

Koperasi Berkah Jaya Lestari
Dusun Negla Sari
Kel. Air Naningan, Kec. Air Naningan
35679 Lampung
Indonesia

Our reference

Date	10.10.2025
Contact	Joachim Essig
Direct	+497141705245
E-Mail	joachim.essig@buhlergroup.com

Your reference

Your enquiry	11.06.2020
Contact	Mr. Tomy Adrianto
Direct	
E-Mail	tomy_adt@mitra-agro.com
Customer no.	301340244

Dear Mr. Adrianto

We thank you sincerely for your enquiry. It is our pleasure to offer you today with reference to our general delivery conditions:

COCOA BEAN PROCESSING LINE 2 T/H

We trust we have correctly interpreted your needs and that the information provided is sufficient to meet your requirements at this stage. Should you have any queries please do not hesitate to contact us.

Making the best out of cocoa – the most valuable ingredient for confectionery

From cocoa beans to cocoa liquor, powder and butter. The Bühler processing technologies, either full cocoa bean roasting or cocoa nibs roasting, offering full control on product quality and on processing efficiency, reliable and proven high end process technology, sophisticated automation and service solutions, as well as customizing according to project requirements.



Over 150 years of experience in the field of cocoa processing technology allows us to offer you exceptionally economical solutions for your cocoa project. Our cocoa nibs alkalinizing and roasting process has set an international benchmark in a wide range of cocoa liquor and powder production, either as a B2B commodity, or as an ingredient for in-house confectionery production. Our cocoa bean roasting technology is offering the traditional cocoa process on highest efficiency, quality and food safety. Regardless if highest capacity processing or small scale production, if commodity producer or chocolate manufacturer, we are the technology partner for the full value chain from cocoa bean to finished cocoa products, finding the best set-up for your specific requirements in aroma and texture development.



Reliability as a fact and shown in figures – our installed base proves a partnership based on best value for money.



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Design Base

Reference Documents

Process Flow Diagram 0301340244-PFD-0001-00, 0301340244-PFD-0002-00,
0301340244-PFD-0003-00, 0301342044-PFD-0004-00
Automation Concept FC-25-00643-AUT-01

Performance Data

Log reduction: > 5-log pathogen reduction

Standard quality cocoa beans

Number of beans : 100 beans +/-20 /100g
Double beans : < 2% relative to flow rate mass
Fermentation index : 1 +/-0.2
Hull content : 12...15 %
Fat content : 50...55%
Moisture content : 6,5...8%

Cocoa nibs specifications for cocoa grinding

Fat content : > 50% (if fat content < 50% cocoa butter addition recommended)
Moisture content : < 2%
Hull content : < 2%
Sand content : < 0.05%
Inlet temperature : < 70°C

Technical Specifications

For all machine parts, unless otherwise specified, the following main voltage is valid:

Type of net TN-C-System
3 phase voltage 400 VAC
1 phase voltage 230 VAC with separate transformer
Frequency 50 Hz
Control voltage 24 VDC
Maximum voltage fluctuation +/- 10 %
Maximum frequency fluctuation +/- 1 %

Norms & Regulations

All plants, machines and parts are manufactured and supplied according to the applicable European Directives and Regulations (Machine Directive 2006/42/EC) in force today.

All components and materials in contact with the product are suitable for that purpose in accordance with EU regulation 1935/2004/EC.

Electrical standards EN/IEC-60204-1
Customer specific norms none

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Item	Qty.	Description
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Scope of Supply

Item	Qty.	Description
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I	1	MECHANICAL SUPPLY
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A	1	DUMPING, CLEANING AND STORAGE SECTION
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A-1	1	DUMPING SECTION
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1	1	BIG-BAG DISCHARGE STATION
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Customer

2	1	DUMPING STATION
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Dumping station with grate.

Dimensions:	approx. 3 x 2 m
Height:	approx. 1.5 m
Load:	4 Tons (fork lift)
Material:	mild steel S235
Surface:	outside: RAL or NCS standard colour inside: clear lacquer food safe gratings galvanized
Grate design:	splitted

Including:

- dust retention slats under the grate
- aspiration connection DN 150
- flange connection to the chain conveyor

Platforms by customer.

3	1	TROUGH CHAIN CONVEYOR AHKG 100
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Technical data

Product	Cocoa beans
Mass flow	6 t/h
Bulk density	0.55 t/m ³
Conveying velocity	0.3 m/s
Overall length	5 m
Inclination	-
Surface	powder-coated for indoor installation
Certification	CE

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Item	Qty.	Description
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Driving station

Design Standard

Drive motor

Motor type Shaft mounted parallel helical geared
 Motor manufacturer GBN (Nord)
 Output 1.5 kW
 Starting relief without

Tail station

Design Standard
 Speed monitoring Yes

Trough types

Design **Wear protection**
 0.1 m single trough with return rail, dust-tight
 3 m hopper trough, 45° with dosing element

Conveying chain

Chain type T2
 With bolted plastic scrapers.

Accessories

Outlet hopper 1 piece(s), circular, 35°
 Support Yes (every 6m)
 Inspection window with probe 1 piece(s)

4 1 GRAVITY SPOUTING

Completely closed gravity spouting system.

Made out of mild steel and in heavy execution.
 Comprising all necessary elements as straight spouts,
 elbow segments, hand hole spouts with cover,
 connection-and transition- pieces, Y-spouts,
 quantity slides, dead boxes, sleeves and clamping rings.

Please note that cocoa beans are very abrasive.
 Our gravity spouting is designed to reduce wear to
 a minimum. However, it cannot be avoided that the pipe
 construction has to be changed during operation of
 the plant due to wear.

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Item	Qty.	Description	
A-2	1	PRE-CLEANING	Option
5	1	ELEVATOR LBEB 360/170	Option
Technical data			
Product		Cocoa beans	
Mass flow		6 t/h	
Bulk density		0.55 t/m³	
Conveying velocity		1.3 m/s	
Overall height		13 m	
Surface		powder-coated for indoor installation	
Certification		CE	
Elevator head			
Wear protection		Yes	
Belt pulley		bright	
Drive motor			
Motor type		Shaft mounted parallel helical geared	
Motor manufacturer		GBN (Nord)	
Output		1.5 kW	
Starting relief		without	
Elevator boot			
Execution		angular	
Type of tensioning device		Spindle tensioning	
Elevator casing			
Execution		angular	
Maintenance casing		1 piece(s)	
Conveyor belt			
Color		white	
FDA certified		Yes	

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Item	Qty.	Description
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Elevator buckets

Material Stainless steel

Accessories

Leveling frame

Inlet nose 1 piece(s)
 with aspiration connection

Inlet connection 1 piece(s)

Connection for central aspiration system 1 piece(s)

Impact pot

Leg supporting floor frame 2 piece(s)

Speed monitor 1 piece(s)

Belt track monitor 4 piece(s)

external evaluation unit 1 piece(s)

6 1 WIND SIFTER V 500 Option

Wind sifter V 500 with expansion chamber

Capacity: until 6 t/h

Feeding P= 0.18 KW

Discharge airlock P= 0.25 KW

Air quantity: approx. 70 m³/min.

Execution:

- Inlet Ø 200 mm
- With weight-loaded feed gate
- With inclinable feed chute
- With 5 adjustable wind guide rails
- Workroom with lighting and 2 observation windows
- With vertically adjustable air compartment and external setting scale
- Expansion chamber with continuous discharge airlock
- With 500 mm extended exhaust air cabinet
- With transition Ø 250 mm, with bead, without throttle slide
- Inlet and outlets with counter flange
- With wear resistant lining, Hardox/ Cryptane
- Standard steel design, prime coated and lacquered
- Coat of lacquer in food-compatible quality
- RAL 7035- light grey

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Item	Qty.	Description
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Sturdy welded steel construction, with inlet buffer
 Feeding shaft including gear motor
 Capacity regulating unit and adjustable feeding chute. This area can be observed via a large window with lighting. The downstream, adjustable air slide facilitates fine adjustment. The position of the slide is documented by means of an indicator and a scale.

In the expansion chamber, heavy particles are separated and discharged via an airlock with a gear motor working over the complete machine width. This area can also be observed via a large window. The fine particles are conveyed to a separator or a filter installation on the suction side.

Electrical Equipment according to ATEX-Zone 22

consisting of:

- Motor
- Cabling approx. 300 mm
- Terminal box
- Ground cable

7	1	GRAVITY SPOUTING	Option
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Completely closed gravity spouting system.

Made out of mild steel and in heavy execution.
 Comprising all necessary elements as straight spouts, elbow segments, hand hole spouts with cover, connection-and transition- pieces, Y-spouts, quantity slides, dead boxes, sleeves and clamping rings.

Please note that cocoa beans are very abrasive.
 Our gravity spouting is designed to reduce wear to a minimum. However, it cannot be avoided that the pipe construction has to be changed during operation of the plant due to wear.

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Item	Qty.	Description
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A-3	1	CLEANING SECTION
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8	1	ELEVATOR LBEB 360/170
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Technical data

Product	Cocoa beans
Mass flow	6 t/h
Bulk density	0.55 t/m ³
Conveying velocity	1.3 m/s
Overall height	13 m
Surface	powder-coated for indoor installation
Certification	CE

Elevator head

Wear protection	Yes
Belt pulley	bright

Drive motor

Motor type	Shaft mounted parallel helical geared
Motor manufacturer	GBN (Nord)
Output	1.5 kW
Starting relief	without

Elevator boot

Execution	angular
Type of tensioning device	Spindle tensioning

Elevator casing

Execution	angular
Maintenance casing	1 piece(s)

Conveyor belt

Color	white
FDA certified	Yes

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Item	Qty.	Description
		Elevator buckets
		Material Stainless steel
		Accessories
		Leveling frame
		Inlet nose 1 piece(s) with aspiration connection
		Inlet connection 1 piece(s)
		Connection for central aspiration system 1 piece(s)
		Impact pot
		Leg supporting floor frame 2 piece(s)
		Belt tensioning device
		Belt installation device
		Hoisting device for legs
		Speed monitor 1 piece(s)
		Belt track monitor 4 piece(s)
		external evaluation unit 1 piece(s)
9	1	SEPARATOR CLASSIFIER MTRC-100/100 Oscillating sieve box with 2 sieve decks, frames covered with perforated steel sheets, ball-type sieve cleaners and withdrawable boxes. Sieve box supported by rubber elements on the frame structure. Inlet with product distributor and outlet box including re-scalping sieve. Drive through 2 out of balance motors with adjustable stroke and throw. Set of metallic sieve frame complete for perforated sheet. 1 Discharge hopper ø120 mm to aspiration channel. Outlet spouts. ATEX Category 3D.
10	1	ASPIRATION CHANNEL MVSL-100 As attachment to a grain separator. Rigid construction with exact adjustable acrylic wall for reproducible extraction results. Intake-aspiration air is distributed to the full depth of the machine with removal- and post- extraction section. Including wear resistant material where necessary. Air quantity regulating valve and connection prepared for central aspiration system. Inside illumination. Protecting grid complete. 1 Discharge hopper ø150 mm to aspiration channel. Aspiration hood.

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Item	Qty.	Description
11	1	DRUM MAGNET DFRT-200/500 For separation of iron particles from the product. With integrated product distributor and product brake. Mild steel construction, housing with inspection doors, product inlet with adjustable radial gate for constant feed of product flow on to the magnetic drum. Rotating drum in stainless steel execution, with incorporated permanent magnet over 1400 Gauss. Drive with hollow shaft gear motor 0.37 kW. Outlets for product and ferrous particles. With floor support.
12	1	DESTONER MTSD-65/120 Operates on the vacuum principle. Steel base frame with sieve slope adjustment. Steel sieve box is supported by special hollow rubber blocks. Withdrawable sieves, Metal frame, covered with steel screens, aluminium diffuse baffles and cube-shaped cleaners. Integrated product distribution system. Plexiglas inspection window, access openings closed with rubber covers. Final separation device for the stones is adjustable. Common board accommodates the product inlet, aspiration connection and air regulating valve with vacuum indicator. Outlets equipped with rubber sleeve valves. Collection funnel for product outlet. Electrical junction box. Oscillating 1 drive. Motor size 0.3 kW ATEX Category 3D. 1 Control glass ø120 mm, with acrylic glass. Hopper with cover plate. Stone container.

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Item	Qty.	Description
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13	1	LOW PRESSURE RADIAL-FAN MHTF
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Air volume: 70 m³/min.

Standard execution:

Compact steel plate construction.

Motor base directly welded to the housing.

The fan wheel is keyed to the motor shaft.

With vibration absorbers.

Squirrel cage motor.

Motor size 5.5 kW

14	1	GRAVITY SPOUTING
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Completely closed gravity spouting system.

Made out of mild steel and in heavy execution.

Comprising all necessary elements as straight spouts,
elbow segments, hand hole spouts with cover,
connection-and transition- pieces, Y-spouts,
quantity slides, dead boxes, sleeves and clamping rings.

Please note that cocoa beans are very abrasive.

Our gravity spouting is designed to reduce wear to
a minimum. However, it cannot be avoided that the pipe
construction has to be changed during operation of
the plant due to wear.

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Item	Qty.	Description
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A-4	1	COCOA BEANS STORAGE
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15	1	ELEVATOR LBEB 360/170
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Technical data

Product	Cocoa beans
Mass flow	6 t/h
Bulk density	0.55 t/m ³
Conveying velocity	1.3 m/s
Overall height	18 m
Surface	powder-coated for indoor installation
Certification	CE

Elevator head

Wear protection	Yes
Belt pulley	bright

Drive motor

Motor type	Shaft mounted parallel helical geared
Motor manufacturer	GBN (Nord)
Output	1.5 kW
Starting relief	without

Elevator boot

Execution	angular
Type of tensioning device	Spindle tensioning

Elevator casing

Execution	angular
Maintenance casing	1 piece(s)

Conveyor belt

Color	white
FDA certified	Yes

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Item	Qty.	Description
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Elevator buckets

Material	Stainless steel
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Accessories

Leveling frame	
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Inlet nose	1 piece(s) with aspiration connection
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Inlet connection	1 piece(s)
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Connection for central aspiration system	1 piece(s)
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Impact pot	
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Leg supporting floor frame	2 piece(s)
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Speed monitor	1 piece(s)
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Belt track monitor	4 piece(s)
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external evaluation unit	1 piece(s)
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16 1 TROUGH CHAIN CONVEYOR AHKG 100**Technical data**

Product	Cocoa beans
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Mass flow	6 t/h
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Bulk density	0.55 t/m ³
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Conveying velocity	0.3 m/s
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Overall length	10 m
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Inclination	-
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Surface	powder-coated for indoor installation
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Certification	CE
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Driving station

Design	Sanitation
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Drive motor

Motor type	Shaft mounted parallel helical geared
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Motor manufacturer	GBN (Nord)
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Output	1.5 kW
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Starting relief	without
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Item	Qty.	Description
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Tail station

Design	Sanitation
Speed monitoring	Yes

Trough types

Design	Heavy-Duty wear protection
1.2 m single trough with return rail, dust-tight	
6.9 m single trough with intermediate bottom, dust-tight	

Conveying chain

Chain type	T3
With bolted plastic scrapers.	

Accessories

Discharge flap	1 piece(s), pneumatical
Outlet hopper	2 piece(s), circular, 35°
Support	Yes (every 6m)
Inspection window with probe	1 piece(s)

- | | | |
|----|---|---|
| 17 | 2 | GRAVITY FLOW DIVERTER
1 gravity flow diverter(s) 35°, ø200 mm, symmetrical. Pneumatic piston with impulse solenoid valve 24 VDC and 2 proximity switches. |
| 18 | 4 | COCOA BEAN SILO WITH CASCADE 25 m³ |
| 19 | 4 | HIGH-LEVEL INDICATOR
Capacitive rope-type probe, 6 m long with sensor, ATEX Category 3D. |
| 20 | 4 | SET OF LOAD CELLS FOR COCOA BEAN SILO 25 m³ |

Customer

Consisting of:
 4 Electronic load cells made of stainless steel, IP68.
 Pressure load: 5000 kg

Universal cable junction box for 1-4 load cells.
 Standard cable 10 m.

Excluded:
 Mounting bracket and further required material (like calibration weights) as well as adjustment work on silo bins.

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Item	Qty.	Description
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21	4	UNIVERSAL CONTROL UNIT MEAG
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For DMS-Transmitter application in painted mild steel housing IP 65.
For application without calibration approval.
Parameters fully controllable by 8 buttons and 14 digit alpha-numerical display.
Comprising:
8 digital inputs 24 VDC,
12 digital outputs 24 VDC,
2 analogue in- and outputs 0/4 ... 20 mA (2 each)
1 interface Profibus-DP slave with max. 12 MBaud
1 serial interface RS 485 with BÜHLER standard scale report.
ProfiNet field bus interface.
DMS Junction box with 10 m cable.

22	4	SLIDE GATE (MAN)
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Slide gate manually operated with hand wheel. Compact design to be mounted between two flanges, DIN PN10. Housing made of grey cast iron, plate made of stainless steel.

Slide gate seal made of PTFE instead of metal on metal.

With 2 proximity switches. With adjustable set points. Wired to terminals.

23	4	TUBULAR SCREW CONVEYOR AHSG-200
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Technical data

Product:	Cocoa beans
Capacity:	10 t/h
Bulk density	0.55 t/m ³
Length:	3 m
Inclination:	0 Degree

Executed in mild steel.

Tubular housing. Shaft with continuous flights.

Necessary intermediate bearings.

1 end outlet.

1 high capacity inlet(s).

Spout connection to outlet DNHIN.Wert, øHIN_2.Wert mm without spout for tension ring connection.

Set of required conveyor supports.

With helical geared motor, manufactured by NORD.

Motor size 0.75 kW

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Item	Qty.	Description
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24	1	TROUGH CHAIN CONVEYOR AHKG 100
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Technical data

Product	Cocoa beans
Mass flow	10 t/h
Bulk density	0.55 t/m ³
Conveying velocity	0.3 m/s
Overall length	8 m
Inclination	-
Surface	powder-coated for indoor installation
Certification	CE

Driving station

Design	Sanitation
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Drive motor

Motor type	Shaft mounted parallel helical geared
Motor manufacturer	GBN (Nord)
Output	1.5 kW
Starting relief	without

Tail station

Design	Sanitation
Speed monitoring	Yes

Trough types

Design	Heavy-Duty wear protection
6.1 m single trough with intermediate bottom, dust-tight	

Conveying chain

Chain type	T3
With bolted plastic scrapers.	

Accessories

Outlet hopper	1 piece(s), circular, 35°
Support	Yes (every 6m)
Inspection window with probe	1 piece(s)

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Item	Qty.	Description
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25 1 GRAVITY SPOUTING

Completely closed gravity spouting system.

Made out of mild steel and in heavy execution.
Comprising all necessary elements as straight spouts,
elbow segments, hand hole spouts with cover,
connection-and transition- pieces, Y-spouts,
quantity slides, dead boxes, sleeves and clamping rings.

Please note that cocoa beans are very abrasive.
Our gravity spouting is designed to reduce wear to
a minimum. However, it cannot be avoided that the pipe
construction has to be changed during operation of
the plant due to wear.

26 1 PRESSURE SWITCH FOR COMPLETE PROCESS LINE

Pressure switch for 230 VAC or 24 VDC.

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Item	Qty.	Description	
A-5	1	ASPIRATION SECTION	
27	1	ASPIRATION DUCTING FOR DUMPING, CLEANING AND STORAGE SECTION	Customer
28	1	PRESSURE SWITCH FOR ASPIRATION DUCTING Pressure switch for 230 VAC or 24 VDC.	
29	1	CYCLONE SEPARATOR MGXG-175 Cone in steel sheet construction. Tangential air intake spout and central air exhaust pipe. Matching counter flanges, gaskets and bolts. 4 Supports	
30	1	HIGH-LEVEL INDICATOR Capacitive probe with rod-type detector, ATEX Category 3D.	
31	1	LOW PRESSURE RADIAL-FAN MHTF Air volume: 292 m ³ /min. Standard execution: Compact steel plate construction. Motor base directly welded to the housing. The fan wheel is keyed to the motor shaft. With vibration absorbers. Squirrel cage motor. Motor size 30 kW	
32	1	DISCHARGE AIRLOCK GZS Discharge airlock with rubber tongues Type GZS 300/300 Steel sheet construction with 4-chamber rotor and screwed rubber lips Gear motor: 0,55 kW with speed monitor according to ATEX Z 22 Material: mild steel Primed and painted according to RAL 7035 light gray without wear lining	
33	1	BAGGING TABLE	Customer

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Item	Qty.	Description	
34	1	ASPIRATION DUCTING FOR WINDSIFTER	Customer
35	1	PRESSURE SWITCH FOR ASPIRATION DUCTING Pressure switch for 230 VAC or 24 VDC.	Option
36	1	CYCLONE SEPARATOR MGXG-95 Cone in steel sheet construction. Tangential air intake spout and central air exhaust pipe. Matching counter flanges, gaskets and bolts. 4 Supports	Option
37	1	HIGH-LEVEL INDICATOR Capacitive probe with rod-type detector, ATEX Category 3D.	Option
38	1	LOW PRESSURE RADIAL-FAN MHTF Air volume: 83 m ³ /min. Standard execution: Compact steel plate construction. Motor base directly welded to the housing. The fan wheel is keyed to the motor shaft. With vibration absorbers. Squirrel cage motor. Motor size 11 kW	Option
39	1	DISCHARGE AIRLOCK GZS Discharge airlock with rubber tongues Type GZS 300/300 Steel sheet construction with 4-chamber rotor and screwed rubber lips Gear motor: 0,55 kW with speed monitor according to ATEX Z 22 Material: mild steel Primed and painted according to RAL 7035 light gray without wear lining	Option
40	1	BAGGING TABLE	Customer

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Item	Qty.	Description
B	1	DEBACTERIZATION, ROASTING, WINNOWING, SHELL AND NIBS HANDLING SYSTEM
B-1	1	DEBACTERIZING AND ROASTING
41	1	ELEVATOR LBEB 360/170
Technical data		
Product		Cocoa beans
Mass flow		10 t/h
Bulk density		0.55 t/m ³
Conveying velocity		1.3 m/s
Overall height		18 m
Surface		powder-coated for indoor installation
Certification		CE
Elevator head		
Wear protection		Yes
Belt pulley		bright
Drive motor		
Motor type		Shaft mounted parallel helical geared
Motor manufacturer		GBN (Nord)
Output		1.5 kW
Starting relief		without
Elevator boot		
Execution		angular
Type of tensioning device		Spindle tensioning
Elevator casing		
Execution		angular
Maintenance casing		1 piece(s)
Conveyor belt		
Color		white
FDA certified		Yes

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Item	Qty.	Description
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Elevator buckets

Material	Stainless steel
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Accessories

Leveling frame

Inlet nose	1 piece(s) with aspiration connection
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Inlet connection	1 piece(s)
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Connection for central aspiration system	1 piece(s)
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Impact pot

Leg supporting floor frame	2 piece(s)
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Speed monitor	1 piece(s)
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Belt track monitor	4 piece(s)
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external evaluation unit	1 piece(s)
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42 1 TUBULAR SCREW CONVEYOR AHSG-200**Technical data**

Product:	Cocoa beans
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Capacity:	10 t/h
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Bulk density	0.55 t/m ³
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Length:	3.5 m
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Inclination:	0 Degree
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Executed in mild steel.

Tubular housing. Shaft with continuous flights.

Necessary intermediate bearings.

1 end outlet.

1 high capacity inlet(s).

Spout connection to outlet DNHIN.Wert, øHIN_2.Wert mm without
spout for tension ring connection.

Set of required conveyor supports.

With helical geared motor, manufactured by NORD.

Motor size 1.1 kW

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Item	Qty.	Description	
43	1	INLET HOPPER 1 T	Customer
44	1	HIGH-LEVEL INDICATOR Capacitive probe with rod-type detector, ATEX Category 3D.	
45	1	LOW-LEVEL INDICATOR Capacitive probe with rod-type detector, ATEX Category 3D.	
46	1	SLIDE GATE (MAN) Slide gate manually operated with hand wheel. Compact design to be mounted between two flanges, DIN PN10. Housing made of grey cast iron, plate made of stainless steel. Slide gate seal made of PTFE instead of metal on metal. With 2 proximity switches. With adjustable set points. Wired to terminals.	
47	1	DEBACTERISATION SYSTEM SLSA 2 STEAM BATTERY Complete steam control with all the necessary valve etc. Assembled on a frame and tested. STEAM ENTRANCE Flanged spout with: – 2 pressure switches – 1 thermometer – 1 manometer The steam control is mounted directly to the vessel. The piping between the steam control and the steam entrance is not included.	

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Item	Qty.	Description
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FEED HOPPER

For the volumetric dosing of the product into the pressure vessel.
Feedhopper in Stainless steel; content 85 kg.

With inlet and outlet stainless steel slide valve ND 200, PN 10
pneumatically actuated. With one high level sensor.

DEBACTERISATION REACTOR

holding capacity: 85 kg
operating pressure: max. 6 bar
test pressure: 9 bar
operating temperature: 165°C (300°C max.)

Hopper consisting of cone and dome cover, connected with flanges.

On dome cover:

- 1 Agitator with flanged gear and electrical motor, output 1.5 kW.
- 1 Flange piece with sight glass
- 1 Product inlet flange with:
 - Inter fitting valve NW150/PN10
 - Housing made of PTFE coated gray iron casting.
 - Plate made of stainless steel.
 - With double acting pneumatic drive.
 - With 2 attached limit switches.
- 1 Connection piece for PT 100.

Steam connected to cone.

Acceptance according to TÜV-PED.

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Item	Qty.	Description
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REACTOR CONNECTION

1 butterfly valve ND 200 / PN10
Housing made of PTFE coated grey cast iron.
Plate made of stainless steel.
With double actuated pneumatic drive and limit switches.

1 special ball valve ND 250, ball made of stainless steel. Mounted at the reactor outlet.

CONNECTION ELEMENTS FROM THE SLSA TO THE SCRO ROASTER

Pneumatically actuated flap box with solenoid valve, to deviate the condensed water and the product remains during the start up phase.

CHECK BIN

Check bin made completely of stainless steel. complete with sieve basket and high level probe.

Furthermore the connection to drain is not included.

STEAM EXPANSION

Steam expansion and water connection.
G1" 6bar water connection.
1 water meter
The connection from the water supply to the steam expansion is not supplied.

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Item	Qty.	Description
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STEAM EXIT

regulation of the steam exit from the reactor.
With 1 butterfly valve and 1 regulated ball valve.

Torque wrench with 2 sets of socket spanners.

Technical data

Electrical power installed:	1.5 kW
Electrical power consumption:	1 kW
Steam consumption (averaged):	420 kg/h
Steam consumption (peak):	2160 kg/h
min. Steam pressure on steam contr.:	8 bar
Compressed air requirement:	0.44 Nm ³ /Charge
Cooling water supply:	600 l/h to 20°C
Cooling water consumption:	530 l/h

It is necessary to install a steam pipe ND 80 from steam boiler to the steam control unit.

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Item	Qty.	Description
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47.1 1 CONTROL SYSTEM SLSA 2 - Siemens TIA

CONTROL UNIT TO DEBACTERISATION SLSA

The operating and visualizing is achieved by means of a touch-industrial-PC, which is placed in the control-/power cabinet.

Functions:

Up to 50 recipes may be apprehended and fed with parameters on the operating panel. Actual values of temperature, throughput, reactor steam pressure and alarm messages, plant flow scheme of process lines etc are shown on the panel. Also available are:

Alarm reporting memory
Manual- automatic- operation
Language switching German - English
Remote maintenance on the PLC
etc.

Scope of delivery:

One combined control-/power- cabinet IP 54 approx. W= 1200 mm, H= 2200 mm, D= 600 mm, completely wired and tested.

In cabinet installed are:

A touch-industrial-PC IPC 477 E
Cabinet fan with filter for ambient temperature of up to 30°C
PLC Siemens TIA
3-pol main switch
Power output connections (Star/Delta as from 7,5 kW)
Control voltage supply 230VAC/24VDC
Automatic circuit breaker
Relays, auxiliary contactors
Terminal strips.

Data recording system

Process data such as temperatures, pressures etc. are recorded in the integrated touch-industrial-PC for reasons of quality management. The hard disk size is predetermining the recording duration. The oldest date will be deleted (after warning- and alarm- notification) if the size should not be sufficient anymore. Hence, the data should be periodically saved on an external hard drive / server (this may be arranged automatically).

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Item	Qty.	Description
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Standards / Guidelines:

The control system is executed according to the European Norms EN 60204-1 respectively according to the "International Electro-technical Commission" - Guidelines IEC 60204-1.

Deviations from this must be communicated with Bühler.

The implementation of the electrical circuit diagrams, the PLC- and visualizing- software as well as the marking of electrical equipment is according to Bühler Standards.

Digital interface for upstream and down stream plant equipment.

48	1	STEAM AND CONDENSATION PIPEWORKS DEBACTERIZATION	Customer
49	1	ASPIRATION DUCTING FOR DEBACTERIZATION SYSTEM	Customer
50	1	INSULATION STEAM PIPEWORK DEBACTERIZATION	Customer

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Item	Qty.	Description
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51	1	CONTINUOUS ROASTER RoaStar™ SCRO 2000
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Continuous roaster for processing Cocoa beans. The concept of the airflow over several roasting zones enables homogeneous roasting of the highest quality. With the RoaStar, Cocoa beans can be roasted while maintaining the original aroma and taste in the desired quality.

Product specification:

Cocoa beans from 6 to 2% H₂O - 2000 kg/h, depending on recipe, raw material and properties of the end product.

System consisting of:

Pre-hopper:

A pre-hopper with an inlet and pneumatic butterfly valve.

Hot-air drying:

Air drying of the pre-hopper to avoid condensation by means of a bypass pipe and control valve from the first heating zone to the pre-hopper.

Heating zones:

Three sequential heating zones made of special steel, with racks made of perforated sheet metal. The large maintenance door with a wide opening offers easy access.

To ensure safety, each heating zone includes a safety water system with spray nozzles that automatically activates when the permissible exhaust-air temperature is exceeded and extinguishes at intervals until the temperature is within the limit values. The second and third heating zones include a bypass pipe with a control valve to condition the exhaust air from the upstream heating zone to prevent condensation.

Heating is steam-powered with one air heater per heating zone. The air heaters are connected to the steam supply unit, which continuously regulates the temperatures. The steam flow is automatically stopped in case of power failure.

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Item	Qty.	Description
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A fan is installed in each heating zone to move the air, and a cyclone separator is installed to separate dust particles from the exhaust air. Each cyclone separator contains an extinguishing device and a collection container for dust.

Cooling zone:

A cooling zone made of special steel with racks made of perforated sheet metal is located after the last heating zone.

Die grosszügige Wartungstür mit breiter Öffnung bietet einen einfachen Zugang.

To ensure safety, the cooling zone includes a safety water system with spray nozzles that automatically activates when the permissible exhaust-air temperature is exceeded and extinguishes at intervals until the temperature is within the limit values.

A control valve at the bottom area of the zone minimizes the amount of untreated product when starting.

A fan is installed in the cooling zone to move the air and a cyclone separator is used to separate dust particles from the exhaust air. The cyclone separator contains an extinguishing device and a collection container for dust.

Discharge:

Discharge airlock across the entire width of the last zone, with gearmotor controlled by a frequency converter for stepless regulation of the discharge and thus also of the throughput.

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Item	Qty.	Description
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RoaStar machine control system:

Siemens design version

The touch panel in the control cabinet visualizes the process and gives the operator full control over the process parameters and influence on the end product.

Functions:

- Recipe management
- Display of current values, e.g. temperature and throughput
- Visualization of analog sensor data as trends
- Alarm management
- Schematics of the process lines
- Maintenance mode/automatic mode
- Remote maintenance on the PLC (Internet connection to the control system required)
- Full-fledged interface to Mercury MES

Consisting of:

- Switch cabinets (control cabinet and power cabinet)
- PLC program TIA
- SCADA process visualization VisiWin
- Licenses

Excluded are:

- Steam and condensate pipe between steam supply rack and heat exchangers
- Hot and cold air piping for connecting the heating and cooling zones and discharging the waste air to the outside
- Insulation of the air piping
- Insulation of the heating zones

52	1	STEAM AND CONDENSATION PIPEWORKS ROASTING	Customer
53	1	ASPIRATION DUCTING ROASTING	Customer
54	1	INSULATION ASPIRATION DUCTING ROASTING	Customer

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Item	Qty.	Description
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55 1 TROUGH SCREW CONVEYOR NFAT-200

Technical data

Product	Cocoa beans
Capacity	2 t/h
Bulk density	0.55 t/m ³
Overall length	4 m
Inclination	0 degree

Conveying trough U-shaped with vertical side walls.
 Front shields with flange bearings and shaft seals.
 Troughs with bolted flange connections.

Certification CE
 Food conformity EU (1935/2004/EG + 10/2011/EG)
 Made of stainless steel (parts in contact with the product).
 1 x End discharge square 255x255.
 Intermediate plain bearing on trough joint.
 Continuously stepped pitch for conveying.
 Trough cover.
 Level measuring instrument with diaphragm.
 Parallel shaft helical geared motor.
 Number of motors: 1
 Motor size 1.1 kW

56 1 GRAVITY SPOUTING

Gravity spouting	stainless sand blasted
Wall thickness	2 mm
Diameter	200 mm
Product break	Yes
Regulating slider	Yes

Modular spouting system, standard components manufactured on the basis of the modular design principle, with beaded edges, are provided with a bead seal ring and fixed in place with a clamp ring. The scope of supply includes straight pipes, elbow segments, flanged nozzles, gaskets, clamping rings, branch pipes and control nozzles.

Seals for barrel construction:

Silicone blue, U-shaped, metal detectable, non-electrostatic conductive
 Food quality: EG 1935/2004 plus FDA
 Earthing bridges made of chrome-nickel steel for electrostatic equipotential bonding, can also be retrofitted without welding or drilling.

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Item	Qty.	Description
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B-2 1 WINNOWING

57 1 WINNOWING SYSTEM WINNOWER SCWI

The winnowing system W 2400 offers an exact degree of separation according to the specific customer needs. The three in one system allows a safe and efficient separation of nibs and shells. The two stage breaking system is optimized to the product properties in order to minimize the amount of fines respectively optimal yield. The optimized separation can be guaranteed by the stage specific aspiration volume as well as the stage adjusted grid and the vibration frequency.

Product:	Cocoa beans
Capacity:	max. 2400 kg/h (infrared pretreated or roasted cocoa beans)
Shell content in nibs:	1.75 ± 0.25 % referred to nibs weight
Nibs content in shells:	1.00 ± 0.25 % referred to shell weight

Machine consisting of:

- 2x BIC 638 bean impact breakers to break the cocoa beans
- Base frame with integrated sieve box with 5 decks. Frame BID21
Standard colour Frame height: 1200 mm
- Vibrator to oscillate the sieve box to ensure an optimal product distribution on the sieve decks
- Drive system to actuate the movable racks on each sieve deck
- 5 screens made out of stainless steel
- Return flow elevator (made of stainless steel) with chute to feed the 2. last bean impact breaker and sampler
- Lifting aspiration hood with integrated and adjustable aspiration system to aspirate the shells
- Automatic pressure regulation system for the adjustable shell separation on each sieve deck
- Internal wiring of all control cables connected to clamping box
- Pneumatic, main cabinet and connection box Mild steel BID21
Standard colour in execution IP54
- Execution W 2400 : normal
- Feeding elevator; height 6 m
- 2 flap boxes for sampling

connection / utility data:

- Electrical power 40 kW
- Compressed air 1.5 m³/h at 6 bar
- Efficiency class: IE3

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Item	Qty.	Description
		Control System W 2400 Functions of machine control: – Intuitive operation via touch screen – Automatic mode with control of the essential parameters – Long-term recording and visualization of process value, all relevant target and actual values can be shown in trend curves and exported – Recipe management for maximum reproducibility – Management of configuration parameters – Display of alarm and message records – User management with configurable rights – Interfaces to plant control panel – Remote support Scope of delivery – Industrial PC 15" (IPC 477E) / Touch Screen / IP65 – Power and control section in one combined electrical cabinet (IP 54, RAL7035) – Cabinet cooling with air conditioning – PLC: Siemens S7-1512 – Panel position: Main Cabinet The control elements have to be integrated in the plant control (extern) : – Feeding elevator – Aspiration fan – Main cyclone with fan, sluice, chipping hammer and level sensor – Control system spark detection Excluded are: – Commissioning for the spark detection and spark suppression – Off-air ducting & roof penetration for off-air ducting

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Item	Qty.	Description
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58 1 GRAVITY SPOUTING

Completely closed gravity spouting system.

Made out of mild and stainless steel in heavy execution.
Comprising all necessary elements as straight spouts,
elbow segments, hand hole spouts with cover,
connection-and transition- pieces, Y-spouts,
quantity slides, dead boxes, sleeves and clamping rings.

Please note that cocoa beans are very abrasive.
Our gravity spouting is designed to reduce wear to
a minimum. However, it cannot be avoided that the pipe
construction has to be changed during operation of
the plant due to wear.

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Item	Qty.	Description
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B-3 1 SHELL HANDLING SYSTEM

59 1 TUBULAR SCREW CONVEYOR AHSG-100

Technical data

Product:	Cocoa shells
Capacity:	1 t/h
Bulk density	0.2 t/m ³
Length:	3 m
Inclination:	20 Degree

Executed in mild steel.

Tubular housing. Shaft with continuous flights.

Necessary intermediate bearings.

1 end outlet.

2 circular inlet.

Spout connection to outlet DNHIN.Wert, øHIN_2.Wert mm without spout for tension ring connection.

Spout connection to outlet DNHIN.Wert, øHIN_2.Wert mm without spout for tension ring connection.

Set of required conveyor supports.

With helical geared motor, manufactured by NORD.

Motor size 0.75 kW

60 1 BLOWER

Execution for pressure operation and belt drive.

Consisting of:

Base carrier with integrated pressure silencer. With hinged motor base as tightening device and mounted on elastic foot pads. Connection casing with integrated check valve. Intake suction side with filter type G4, silencer and monitor for indicating maintenance needs. Ready for connection to pipeline and secured by elastic sleeve and clamps.

Digital pressure sensor with display for pressure- and suction conveying.

With sound absorbing hood for blower unit and motor.

Monitor to check intake-filter condition.

Three-phase motor, ATEX Category 3D.

Motor size 3 kW

Lubrication with food-grade oil according to NSF-H1

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Item	Qty.	Description
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61	1	PRESSURE PNEUMATIC CONVEYING LINE
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Product: Cocoa shells
Capacity: 0.5 t/h
Length: 50 m

Clean-air line, comprising:
10 meter straight pipes, connection and installation parts.
Pipes are prime coated only.

Conveying line, comprising:
50 meter straight pipe(s), 1 cone, 1 expansion pipe, connection and installation parts.
Pipes are prime coated only.
1 sight glass(es) with acrylic tube.
5 wear-resistant bend(s) 90°, with basalt liner.

62	1	IMPACT CRUSHER BIC 638
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For gentle crushing of pre-treated or roasted cocoa beans. Product capacity approx. 0,5 tons of cocoa shells per hour.

Consisting of:
Housing with centrally placed product inlet, integrated centrifugal wheel and impact plates as well as discharge cone and motor mounting plate. A smooth, flared connection piece for joining to the gravity spouting system. The power is transmitted through two narrow type vee-belts to the centrifugal wheel. The wearing parts may be easily exchanged.

Main motor: 1.5 kW
Operating Voltage: 3 x 400 VAC / 50 Hz

63	1	BLOW THROUGH AIRLOCK MPSH-22/22
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Gray cast iron housing with airlock side covers inside and housing bore hard chrome-plated.
Adjustable tapered roller outboard bearings.
Welded mild steel rotor construction with open sides and beveled blades for minimal air leakage.
Additional airlock support.
Check valve.
Parallel shaft geared motor, ATEX Category 3D.
Motor size 0.55 kW

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Item	Qty.	Description	
64	1	HIGH PRESSURE-SMALL FILTER MVRW-6/8 Top-mounted filter Filter fabricated of stainless steel quality class I, consisting of a round shaped steel plate housing. Shock pressure proof up to 0.7 bar. The sleeve cleaning is controlled by the electronic filter control unit FIL-Tronic. VERTICAL placed, easily replaceable filter sleeves with supporting cages. Protection grate. Flange with gasket for floor mounting.	
65	1	SMALL CENTRIFUGAL FAN Capacity: 5 m ³ /min. Cast housing and impeller are made of aluminium. Outlet is equipped with flange and protection grid. With flanged squirrel-cage motor. Motor size 0.37 kW – ATEX Category 3D	
66	1	ASPIRATION DUCTING FOR COCOA SHELL SILO	Customer
67	1	COCOA SHELL SILO 75 m³	Customer
68	1	HIGH-LEVEL INDICATOR Capacitive rope-type probe, 6 m long with sensor, ATEX Category 3D.	

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Item	Qty.	Description
69	1	VIBRO-DISCHARGER MFVH-180 M.S. The Vibro Discharger MFVH is suitable for uniform discharge of free flowing bulk products from concrete-, steel- and synthetic- silos. This guarantees segregation – free and complete emptying out of circular, square or rectangular silo bins. All parts in contact with products streamline designed. This way, a segregation of the product is avoided. Residues are effectively eliminated by closed cavities and by a minimum of dust deposits. The complete set of bearings is lubricated for lifetime. The robust construction parts in contact with product are made of mild steel. With circular inlet flange for installation on a bin outlet of ø180 cm, including sealing material. Flexible suspending elements and flexible connecting sleeve. Freely moving discharge bottom, transition hopper with welded outlet spout, flexible outlet cuff for connection to gravity spouting including clamps. Vibro – motor mounted and grounded on a bracket.
70	1	SILO BIN DISCHARGE SLIDE GATE RVB Type 600 Welded sheet steel housing. Actuation of slide plate manual via screw and handwheel. Limit switch for position indication. Surface treatment for outside installation with multi coat application for corrosion prevention. Connecting part on top: Bin discharge round. Connecting part below: Discharge hopper tapered, large, circular.
71	1	LOW-LEVEL INDICATOR Capacitive probe with rod-type detector, ATEX Category 3D.

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Item	Qty.	Description
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72	1	TUBULAR SCREW CONVEYOR AHSG-315
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Technical data

Product:	Cocoa Shells
Capacity:	10 t/h
Bulk density	0.2 t/m ³
Length:	3.5 m
Inclination:	0 Degree

Executed in mild steel.
Tubular housing. Shaft with continuous flights.
Necessary intermediate bearings.
1 end outlet.
1 high capacity inlet(s).
Set of required conveyor supports.

Surface treatment for outside installation with multi coat application for corrosion prevention.
With helical geared motor, manufactured by NORD.
Motor size 1.5 kW

73	1	GRAVITY SPOUTING
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Completely closed gravity spouting system.

Made out of mild steel and in heavy execution.
Comprising all necessary elements as straight spouts, elbow segments, hand hole spouts with cover, connection-and transition- pieces, Y-spouts, quantity slides, dead boxes, sleeves and clamping rings.

Please note that cocoa beans are very abrasive.
Our gravity spouting is designed to reduce wear to a minimum. However, it cannot be avoided that the pipe construction has to be changed during operation of the plant due to wear.

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Item	Qty.	Description
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B-4	1	PNEUMATIC TRANSPORT FOR NIBS
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74	1	ADDITIONAL FILTER FOR BLOWER
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Additional filter for blower consists of:

- Filter housing FCBS-A 0510 T = 292 mm
399 x 744 x 500 mm
painted sheet steel, white grey RAL9002
- Reduction
RE-RU-308.5 x 613.5 x 400, Ø = 89 mm, shelf 50 mm
painted sheet steel, white grey RAL9002
- FCB housing parts
Filter protection grille FCB-0510/1005
304 x 606 x 12 mm
Material: galvanized steel
- Absolute V Prosafe
VGXL14- 610 x 305 x 292 -P-PS3
EN 1822:2009 H14

75	1	BLOWER
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Execution for pressure operation and belt drive.

Consisting of:

Base carrier with integrated pressure silencer. With hinged motor base as tightening device and mounted on elastic foot pads. Connection casing with integrated check valve. Intake suction side with filter type G4, silencer and monitor for indicating maintenance needs. Ready for connection to pipeline and secured by elastic sleeve and clamps.

Digital pressure sensor with display for pressure- and suction conveying.

With sound absorbing hood for blower unit and motor.

Monitor to check intake-filter condition.

Three-phase motor, ATEX Category 3D.

Motor size 5.5 kW

Lubrication with food-grade oil according to NSF-H1

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Item	Qty.	Description
76	1	PRESSURE PNEUMATIC CONVEYING LINE Product: Cocoa nibs Capacity: 2 t/h Length: 50 m Clean-air line, comprising: 10 meter straight pipes, connection and installation parts. Pipes are prime coated only. Conveying line, comprising: 50 Meter stainless steel straight pipe(s), 1 cone, 1 expansion pipe, connection and installation parts. 1 sight glass(es) with acrylic tube. 5 elbow(s) 90°.
77	1	BLOW THROUGH AIRLOCK MPSS-25/23 QC2 / TK1 Hygienic blow-through airlock made of stainless steel, designed for operating pressures of 0.6 bar (Maximum pressure 0.8 bar). Air leakage is at a minimum because of the precise manufacturing of rotor and housing. Additional airlock support (at the time of order please indicate height). With sight glass. Set of rinsing air parts to airlock. Check valve. Parallel shaft geared motor, ATEX Category 3D. Motor size 0.37 kW
78	1	HIGH PRESSURE-SMALL FILTER MVRW-6/8 Top-mounted filter Filter fabricated of stainless steel quality class I, consisting of a round shaped steel plate housing. Shock pressure proof up to 0.7 bar. The sleeve cleaning is controlled by the electronic filter control unit FIL-Tronic. VERTICAL placed, easily replaceable filter sleeves with supporting cages. Protection grate. Flange with gasket for floor mounting.

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Item	Qty.	Description	
79	1	SMALL CENTRIFUGAL FAN Capacity: 6 m ³ /min. Cast housing and impeller are made of aluminium. Outlet is equipped with flange and protection grid. With flanged squirrel-cage motor. Motor size 0.25 kW – ATEX Category 3D	
80	1	ASPIRATION DUCTING FOR NIBS BUFFER SILO 4 m³	Customer

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Item	Qty.	Description	
B-5	1	ASPIRATION SECTION	
81	1	ASPIRATION DUCTING FOR DEBACTERIZATION, ROASTING AND WINNOWING	Customer
82	1	PRESSURE SWITCH FOR ASPIRATION DUCTING Pressure switch for 230 VAC or 24 VDC.	
83	1	CYCLONE SEPARATOR MGXG-150 Cone in steel sheet construction. Tangential air intake spout and central air exhaust pipe. Matching counter flanges, gaskets and bolts. 4 Supports	
84	1	HIGH-LEVEL INDICATOR Capacitive probe with rod-type detector, ATEX Category 3D.	
85	1	LOW PRESSURE RADIAL-FAN MHTF Air volume: 187 m ³ /min. Standard execution: Compact steel plate construction. Motor base directly welded to the housing. The fan wheel is keyed to the motor shaft. With vibration absorbers. Squirrel cage motor. Motor size 18.5 kW	
86	1	AIRLOCK MPSN-25/23 Gray cast iron housing with 8-pocket rotor, ball bearings, final painting coat. For suction and/or pressure operation to max. +/- 500 mbar. With sight glass. Parallel shaft helical geared motor. Motor size 0.75 kW	
87	1	BAGGING TABLE	Customer

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Item	Qty.	Description	
C	1	COCOA NIBS GRINDING	
C-1	1	COCOA NIBS GRINDING	
88	1	NIBS BUFFER SILO 4 m³ (approx. 2 t)	Customer
89	1	HIGH-LEVEL INDICATOR Capacitive rope-type probe, 6 m long with sensor, ATEX Category 3D.	
90	1	SET OF LOAD CELLS FOR NIBS BUFFER SILO 4 m³ Consisting of: 4 Electronic load cells made of stainless steel, IP68. Pressure load: 2000 kg Universal cable junction box for 1-4 load cells. Standard cable 10 m. Excluded: Mounting bracket and further required material (like calibration weights) as well as adjustment work on silo bins.	
91	1	UNIVERSAL CONTROL UNIT MEAG For DMS-Transmitter application in painted mild steel housing IP 65. For application without calibration approval. Parameters fully controllable by 8 buttons and 14 digit alpha-numerical display. Comprising: 8 digital inputs 24 VDC, 12 digital outputs 24 VDC, 2 analogue in- and outputs 0/4 ... 20 mA (2 each) 1 interface Profibus-DP slave with max. 12 MBaud 1 serial interface RS 485 with BÜHLER standard scale report. ProfiNet field bus interface. DMS Junction box with 10 m cable.	

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Item	Qty.	Description
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92 2 KNOCKING HAMMER

Knocking hammer for weighing hopper.
Pneumatically operated.

93 1 TUBULAR SCREW CONVEYOR AHSG-160

Technical data

Product:	Cocoa nibs
Capacity:	2 t/h
Bulk density	0.55 t/m ³
Length:	2 m
Inclination:	0 Degree

Executed in stainless steel.

Tubular housing, shaft with full flights and necessary intermediate bearings.

Also all parts in contact with the product are of stainless steel.

1 end outlet.

1 high capacity inlet(s).

With helical geared motor, manufactured by NORD.

Motor size 1.5 kW

94 1 METAL SEPARATOR

Compact unit with integrated metal detector, remote control unit - inlet and outlets designed according to the "Jacob" system - for the inspection of coarse-grained, flaky, light, fibrous, crumbly and moist bulk materials.

Product:	Cocoa nibs
Bulk weightt:	600
Product temperature:	-10°C to +80°C

Nominal passage:	ø 100 mm
Throughput:	12000 l/h

Basic scan sensitivity (without product)

ø Fe-ball (1.3505):	0,5 mm
ø VA-ball (1.4301):	0,9 mm
ø NonFe-Ball (brass):	0,6 mm

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Item	Qty.	Description
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Sensitivities

- The detectability of metal particles (sensitivities) is determined by their type, shape and position.
- The basic sensitivities listed above (without product) are achieved with the sensitivities listed in the first column:
- The achievable sensitivity in the product depends on its product effect (intrinsic conductivity of the product e.g. due to moisture, salt content etc.),
the temperature of the material to be tested and influences at the location of the device.

Design

- Frame, detection coil and control enclosure, stainless steel 1.4301 (AISI 304); glass bead blasted
- Separation unit; stainless steel 1.4301 (AISI 304)
- Parts in touch with material; stainless steel 1.4301 (AISI 304), PP, PE-UHMW
- Scanning pipe; PP
- Falling height max. 500 mm above top edge
- Connecting cable device / control unit 3 m

Performance characteristics

- Separates magnetic (iron, nickel, ...) and non-magnetic metals (high-grade steel, aluminium, brass,)
- The separating unit is dust-and water-proof, and can be easily cleaned
- Compact stainless steel design
- Separation by means of a swivel funnel, without interrupting the flow of conveyed material

Control Unit

- Multiprocessor electronics and consequent digital signal processing with DSP-technology
- Quartz-stabilized operating frequency (optional with multi-frequency technology)
- 8-line graphics display with multi language menu-guided dialog
- Automatic teach-in for product effect compensation
- Multi-product memory for 240 different products
- Password-protected system logbook and menu-guided validation system

Including commissioning and certification.

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Item	Qty.	Description
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95 1 GRAVITY SPOUTING

Gravity spouting	stainless sand blasted
Wall thickness	2 mm
Diameter	200 mm
Product break	Yes
Regulating slider	Yes

Modular spouting system, standard components manufactured on the basis of the modular design principle, with beaded edges, are provided with a bead seal ring and fixed in place with a clamp ring. The scope of supply includes straight pipes, elbow segments, flanged nozzles, gaskets, clamping rings, branch pipes and control nozzles.

Seals for barrel construction:

Silicone blue, U-shaped, metal detectable, non-electrostatic conductive
Food quality: EG 1935/2004 plus FDA

Earthing bridges made of chrome-nickel steel for electrostatic equipotential bonding, can also be retrofitted without welding or drilling.

96 1 PreGrind 2000

The PreGrind blade beater mill for the optimum pre-grinding of cocoa nibs. The nibs are ground by means of the blade beater which rotates at high speeds and thereby setting the contained fat free. The ground cocoa mass remains in the grinding zone until the size of the particles are reduced to smaller than the sieve gap dimension. The machine is characterised by its robust construction, the guaranteed production quality due to the rugged gap sieve and the very service friendly design.

Stable machine frame in welded steel construction, with attenuator elements on feet, transitions to all peripheral assembly units as well as to the tempering system.

Heated process zone made of stainless material with dead space free drain.

Changeable wear plate for the bottom of the process zone.

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Item	Qty.	Description
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High precision bearing of the drive shaft.

Holder for the beater blades in stainless execution.

Complete internal piping system for the tempering of the machine.

Product feeding through screw conveyor.

Drive through highly efficient 110 kW IE3motor 24 V / 50Hz.
Transmission with a highly efficient and very robust v-ribbed belt.

Wear resistant, hardened beater blades, simple to exchange for highly efficient grinding at minimal wear.

Highly stable, stainless slotted sieve with an gap width of 0.2 mm.

Discharge pump group for cocoa mass with highly efficient IE3 gear motor, in wear resistant and heated pump housing. Throughput between. 1000-2000 kg/h. With flexible connection to the machine, pre-assembled tempering hoses and pipe sections for the installation of electronic pressure- and temperature sensors.

Electrically heated hose with connection to the process zone and G 3/8" attachment for linking-up on the customers cocoa butter unit.

Open tempering unit for direct connection to a centralized cooling- and hot water system.

Hot water: 45-60 °C, 1.5-2.5 m³/h @ 0.5 bar pressure loss, min. 1.0 - max. 2.50 bar (over pressure).
Cold water: 5-20 °C, ~3 m³/h @ 1.0 bar pressure loss, min. 1.0 - max. 2.50 bar (over pressure).

Electrically operated lifting device for the pivoted cover.

Compact rack unit to place the bearing grease dosing unit, the electrical terminal box and further optional elements.

Aspiration system with fan and manual control flap for achieving an optimal product dehumidification.

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Item	Qty.	Description
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Maintenance set consisting of:
Withdrawing device for the knife holder, for the knife head and for the
gap sieve as well as selected tools and wrenches.

Traverse for the machine transport.

Grease for the shaft bearings.

The machine is controlled by the plant control system. All electrical
equipment is installed in the control- and power cabinet of the plant.

Optical pre-tension measuring instrument for checking and adjusting
the belt tension.

Compact unit complete with integrated electronics and with oscillator,
tuning fork with stainless steel extension tube, stainless steel G1½"
thread. IP 66 plastic housing.

Capacitive pressure sensor. Flush membrane made of Ceraphire. Long
term stable and resistant to overload. Maximum process temperature
125°C.

Process connection G1½".

97	1	ASPIRATION DUCTING FOR COCOA NIBS GRINDING
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Customer

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Item	Qty.	Description
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98	1	TANK 1000 I
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Option

Execution:

Cylindrical standing form

15° conical lid and tank bottom

All corners that are in contact with the product are fabricated with large radius.

Material:

Completely fabricated in stainless steel 1.4301

Finish:

Weld beads ground flush min. K180 and pickled on the outside.

Inside and outside surfaces rolled blank IIIC

Main tank dimensions:

Content :1000 l

Diameter :1000 mm

Cylindrical height :1400 mm

Approx. total height :2200 mm

Base execution:

Manhole on the top with diameter 420 mm.

Safety switch on manhole (applies only to tanks that are delivered with agitator).

Hinged safety grating in manhole (applies only to tanks that are delivered with agitator).

Inlet spouts.

Outlet spouts.

Vent.

Sockets for probes.

Product temperature display at a height of 1.5 m.

Welded grounding tab.

The tank is on 3 supports, with adjustable dome shaped plates or adapter plates for weigh cells.

Transport eyebolts.

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Item	Qty.	Description
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Water jacket:

Warm water jacket with forced circulation. The outlet, bottom and approx. 1/3 of the wall are heated with one circuit. Operating pressure is 3 bar, test pressure is 5 bar.

Insulation:

Welded insulation jacket on the bottom and the walls.
75 mm thick mineral wool insulation.

Anchor agitator: (for masses)

Single stage anchor agitator with wall scrapers
Space between wall and scraper approx. 10 mm
Function: maintain homogeneity, support the heat transfer
Agitator is completely welded
Agitator has an opposing bearing on the tank floor
Rotational speed max. 1.5 m/s
Leak proof tank shaft seal.
Food grade gear lubrication.

Discharge spout: ND 80

"T" spout with 2 manual slide gates according to DIN 11851.

Discharge valve with cap.

Including water jacket for warm water heating with connections.

Ladder stirrup:

3/4" x 500 mm pipe, welded below the top of the cylindrical part of the tank, by the man hole. Intended to be used to hang a ladder onto.

Connecting / Utility Data

Electrical:

– Gearmotor of anchor agitator: 3.0 kW

Hot water consumption data:

- Water flow rate: 0.12 m³/h
- Water temperature: 45-50°C
- Operating pressure: 3bar
- Test pressure: 5bar

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Item	Qty.	Description	
99	1	HIGH-LEVEL INDICATOR With level indicator.	Option
100	1	LEVEL SENSOR Pressure gauge converter in tanks Capacitive pressure sensor. Flush membrane made of Ceraphire. Long term stable and resistant to overload. Maximum process temperature 130°C. Process connection G1½".	Option
101	1	SET OF LOAD CELLS FOR SLURRY TANK 1000 l Consisting of: 2 Electronic load cells made of stainless steel, IP68. Pressure load: 1000 kg Excluded: Mounting bracket and further required material as well as adjustment work on slo bins. Universal cable junction box for 1-4 load cells. Standard cable 10 m. Excluded: Mounting bracket and further required material (like calibration weights) as well as adjustment work on silo bins.	Option
102	1	UNIVERSAL CONTROL UNIT MEAG For DMS-Transmitter application in painted mild steel housing IP 65. For application without calibration approval. Parameters fully controllable by 8 buttons and 14 digit alpha-numerical display. Comprising: 8 digital inputs 24 VDC, 12 digital outputs 24 VDC, 2 analogue in- and outputs 0/4 ... 20 mA (2 each) 1 interface Profibus-DP slave with max. 12 MBaud 1 serial interface RS 485 with BÜHLER standard scale report. ProfiNet field bus interface. DMS Junction box with 10 m cable.	Option
103	1	DOUBLE WALL PRODUCT PIPE FOR SLURRY ADDITION	Customer

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Item	Qty.	Description	
104	1	SLURRY PUMP 2 T/H	Option
		Hygienic rotary lobe pump LEWA with housing heating. Fully assembled on a stainless base plate with coupling and stainless steel coupling guard.	
		Type:	PLP 2-2
		Product:	Slurry
		Viscosity:	3000 - 4000 mPas
		Capacity:	2 t/h
		Speed:	109 rpm @ 50 Hz
		Capacity per rotation:	0.3 l
		Specific gravity:	1.1 kg / dm ³
		Pressure:	3 bar
		Suction side:	positive feed
		Temperature:	40 - 50°C, Cleaning 85°C
		Heating:	Heating jacket on pump cover
		Material - Wetted parts:	1.4404 (< = 0.8 micron)
		Material - Gear box:	1.4301
		Material - Shafts:	Duplex 1.4462
		Material - Lobes:	1.4404
		Shaft seals:	Bi-Wing Clearance class C2 M3; single mechanical seal TC/TC (knife-edge seal rotor product side) coated with foodgrade grease
		Material - elastomers:	FKM acc. to FDA & EC1935 / 2004
		Oil:	Food grade EP220 (0-30°C)
		Connections:	DN 50; DIN 11851 incl. counter parts horizontal connections, shaft top
		Drive:	Nord SK 372.1 - 90LP/4 position term.box: right as seen from shaft end M1 - IP55 50 Hz - 230/400V - IE3 n= 109 rpm - p= 1.5 kW NCS S-0502B - PTC Non-food grade oil

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Item	Qty.	Description
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Execution:	Pump and gear motor mounted on a stainless steel base plate (omega profile horizontal 1.4301) with stainless steel coupling guard.
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Coupling:	J-Flex 150 92° Shore 25-25
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Certificates:	FDA-Certificate for O-Rings
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Important note:	The operator of the pump must ensure that there is product pending at the shaft sealing at any time.
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Documentation:

- operating manual including start-up, operating and maintenance instructions
- sectional drawings with detailed parts list
- Dimensional drawings
- CE certificate and CE label, if applicable
- Declaration of conformity 2006/42/EG, if applicable, 2014/34/EU

Painting:

- Drive: NCS S 0502B
- Stainless steel parts will not painted

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Item	Qty.	Description	
105	1	SLURRY PUMP 40 - 150 KG/H	Option
		Hygienic rotary lobe pump LEWA with housing heating. Fully assembled on a stainless base plate with coupling and stainless steel coupling guard.	
		Type:	PLP 1-1,5
		Product:	Slurry
		Viscosity:	3000 - 4000 mPas
		Capacity:	40 - 150 kg/h
		Speed:	32 rpm @ 50 Hz
		Capacity per rotation:	0.1 l
		Specific gravity:	1.1 kg / dm ³
		Pressure:	3 bar
		Suction side:	positive feed
		Temperature:	40 - 50°C, Cleaning 85°C
		Heating:	Heating jacket on pump cover
		Material - Wetted parts:	1.4404 (< = 0.8 micron)
		Material - Gear box:	1.4301
		Material - Shafts:	Duplex 1.4462
		Material - Lobes:	1.4404
		Shaft seals:	Bi-Wing Clearence class C2 M3; single mechanical seal TC/TC (knife-edge seal rotor product side) coated with foodgrade grease
		Material - elastomers:	FKM acc. to FDA & EC1935 / 2004
		Oil:	Foot grade EP220 (0-30°C)
		Connections:	DN 40; DIN 11851 incl. counter parts horizontal connections, shaft top
		Drive:	Nord SK 372.1 - 71LP/4 position term.box: right as seen from shaft end M1 - IP55 50 Hz - 230/400V - IE3 n= 32 rpm - p= 0.37 kW NCS S-0502B - PTC Non-food grade oil

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Item	Qty.	Description
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Execution: Pump and gear motor mounted on a stainless steel base plate (omega profile horizontal 1.4301) with stainless steel coupling guard.

Coupling: J-Flex 110 92° Shore 15-25

Certificates: FDA-Certificate for O-Rings

Important note: The operator of the pump must ensure that there is product pending at the shaft sealing at any time.

Documentation:

- operating manual including start-up, operating and maintenance instructions
- sectional drawings with detailed parts list
- Dimensional drawings
- CE certificate and CE label, if applicable
- Declaration of conformity 2006/42/EG, if applicable, 2014/34/EU

Painting:

- Drive: NCS S 0502B
- Stainless steel parts will not painted

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Item	Qty.	Description	
106	1	FITTINGS	Option
		The base for a robust and accurate conveying and dosing system of liquid products is the correct application of cut-off devices, pressure sustaining and venting valves. The complete set of armatures is executed for integration into the pipeline system and is of sanitation design. All parts in contact with products are certified by the food industry. Consisting of: – 2 cut-off valve(s), manual operation, with two limit switches – 1 cut-off valve(s), pneumatic operation, with two limit switches – 1 check valve(s) Execution: Cut-off valves: – Housing: <DN50 GG25 / ≥DN50 light alloy casting – Butterfly/shaft: stainless steel 1.4408 Check valves: – Housing: stainless steel 1.4581 – Valve plate/spring: stainless steel 1.4571	
107	1	DOUBLE WALL PRODUCT PIPES FOR NIBS GRINDING	Customer

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Item	Qty.	Description
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108 1 TANK 1000 I

Execution:

Cylindrical standing form

15° conical lid and tank bottom

All corners that are in contact with the product are fabricated with large radius.

Material:

Completely fabricated in stainless steel 1.4301

Finish:

Weld beads ground flush min. K180 and pickled on the outside.

Inside and outside surfaces rolled blank IIIC

Main tank dimensions:

Content :1000 l

Diameter :1000 mm

Cylindrical height :1400 mm

Approx. total height :2200 mm

Base execution:

Manhole on the top with diameter 420 mm.

Safety switch on manhole (applies only to tanks that are delivered with agitator).

Hinged safety grating in manhole (applies only to tanks that are delivered with agitator).

Inlet spouts.

Outlet spouts.

Vent.

Sockets for probes.

Product temperature display at a height of 1.5 m.

Welded grounding tab.

The tank is on 3 supports, with adjustable dome shaped plates or adapter plates for weigh cells.

Transport eyebolts.

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Item	Qty.	Description
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Water jacket:

Warm water jacket with forced circulation. The outlet, bottom and approx. 1/3 of the wall are heated with one circuit. Operating pressure is 3 bar, test pressure is 5 bar.

Insulation:

Welded insulation jacket on the bottom and the walls.
75 mm thick mineral wool insulation.

Anchor agitator: (for masses)

Single stage anchor agitator with wall scrapers
Space between wall and scraper approx. 10 mm
Function: maintain homogeneity, support the heat transfer
Agitator is completely welded
Agitator has an opposing bearing on the tank floor
Rotational speed max. 1.5 m/s
Leak proof tank shaft seal.
Food grade gear lubrication.

Discharge spout: ND 100

"T" spout with 2 manual slide gates according to DIN 11851.

Discharge valve with cap.

Including water jacket for warm water heating with connections.

Ladder stirrup:

3/4" x 500 mm pipe, welded below the top of the cylindrical part of the tank, by the man hole. Intended to be used to hang a ladder onto.

Connecting / Utility Data

Electrical:

– Gearmotor of anchor agitator: 3.0 kW

Hot water consumption data:

- Water flow rate: 0.12 m³/h
- Water temperature: 45-50°C
- Operating pressure: 3bar
- Test pressure: 5bar

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Item	Qty.	Description
109	1	HIGH-LEVEL INDICATOR With level indicator.
110	1	LEVEL SENSOR Pressure gauge converter in tanks Capacitive pressure sensor. Flush membrane made of Ceraphire. Long term stable and resistant to overload. Maximum process temperature 130°C. Process connection G1½".
111	1	BALL MILL NOVA™ S 1000 NOVA™ S ball mill, for fine-grinding of cocoa masses. Its single-pass grinding process, its uniform product fineness quality, and its low product temperatures make the NOVA™ S unique and allow energy-efficient production of a wide variety of top-quality masses. The process is fully automatically controlled. An intuitive operating system enables process parameters to be modified whenever required. The extensive recipe management system allows different recipes to be selected, adjusted, and created. Main benefits at a glance: – NOVA™ Power Control: Consistent product fineness quality thanks to the pressure unit (cylinder/piston unit) at the lower end of the grinding zone. The ball volume in the grinding zone and therefore the motor power are automatically controlled. – NOVA™ Temperature Control: Water cooling for stator, rotor, and gearing (motor is air-cooled) – NOVA™ Flow Control: Lower surge (buffer) hopper with integrated product level sensor, for continuous throughput measurement and pump control Machine consisting of: – Machine support frame of hygienic design, with door at the rear for high ease of servicing – 85 kg product mass – Stator of hardened steel, with hardened pins – Ball charge, 1000 kg, 6 mm or 8 mm balls

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Item	Qty.	Description
		– Closed water system consisting of water pump, water heating, heat exchanger, control valve, expansion vessel, and various sensors and fittings for direct connection to a central cooling water system. Incorporated in the machine support frame – product feeding with FIP gear pump, in mild steel execution – Outlet channel for lower surge hopper – Lower surge hopper, with integrated lid, overflow protection device, and a pressure sensor for flow rate measurement. Jacketed (double walled) for connection to hot water system – Product discharge with FIP gear pump, in mild steel execution – Pressure sensor for overpressure detection after discharge pump – PLC: Siemens TIA S7-1500 Power and control units installed combined in a stand-alone cabinet

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Item	Qty.	Description
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111.1	1	COOLING UNIT OF THE CONTROL CABINET
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Control cabinet ventilation with cooling unit for ambient temperature above 30°C.

Connection and consumption data

Electrical:

- | | |
|-------------------------|------------------|
| – Main motor | 132 kW (at 73Hz) |
| – Feed pump | 2.2 kW |
| – Closed tempering unit | 6 kW |
| – Discharge pump | 2.2 kW |

Water:

- Hot water (for conveying lines, pumps, and lower surge (buffer) hoppers): 40-50°C, 0.5-1 m³/h @ 1.0 bar pressure loss, min. pressure 2 bar
- Cooling water: 5-15°C, 7-15 m³/h @ 1.5 bar pressure loss, min. pressure 2.5 bar

Compressed air:

- Min. 6 bar, 0.2 m³/h

111.2	1	SAMPLER NOVA 1000
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- Sampler integrated in lower surge hopper, for representative and hygienic sampling

111.3	1	STATOR INSULATION NOVA 1000
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- Stator insulation, for prevention of condensate on stator and for noise control

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Item	Qty.	Description
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112	1	ROUND SIEVE FOR 2 T/H COCOA LIQUOR
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The round sieve is invented and customized especially for screening cocoa liquor. The clearance below the unit allows easy cleaning of the floor. The conical bottom of the table frame ensures better flowability of the product and double jacket frame ensures constant, stable temperature of the product. The double tangential spout ensures no blinded spots where the product can built up.

Capacity: 2 t/h
Model: ZS40S128SFCALXB
Diameter: 1016 mm

Including:

- one motor 0,37kW, 50 Hz, 1450 rpm, IP65, with bearings lubricated with FDA grease (H1).
- one blank spacing frame 8" high with no spout in SS304
- one table frame 12" high with 8" rounded spout in SS304 equipped with vent pipe
- one extended base in carbon steel, painted (standard RAL) constuction on three legs (CAL version)
- one lower table in carbon steel, painted (standard RAL)
- one screen, supramagnetic, supertaut 0,4 mm (46TBC), sanitary finished with back-up cloth. Cloth detected by the magnet
- one silicone gaskets
- cover assembly in SS304 with one centered inlet of 8" diameter and one inspection opening of 6" diameter covered by silicone cap
- flexible connector, silicone 8" ID x 1 m long, for inlet cover and spout on table frame
- heating scalping frame (circuit for hot water circulation under the table frame)

The dimensions of the packed unit on the pallet:

Dimension: approx. 157 x 157 x 160 cm
(heigh depends on base extension)
Weight: approx. 450 kg

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Item	Qty.	Description
113	1	HIGH-LEVEL INDICATOR With level indicator.
114	1	INLET HOPPER 15 l FOR PUMP Double jacketed inlet hopper for pump Capacity: 15 Liter cocoa liquor Material: 1.4301 Heating medium: Water Including: - Insulation
115	1	HIGH-LEVEL INDICATOR With level indicator.
116	1	CONVEYING AND METERING PUMPS The FIP internal gear pumps based on hygienic design principles are suitable for gentle conveying of liquid media in the chocolate industry. Various shaft sealing systems consider different product specifications. The FIP pumps are designed for continuous service. The long service life and short exchange times of the seal in conjunction with an easy-to-remove pump head plate minimize the maintenance requirements. Thanks to their constant and non-pulsating conveying action and their high metering precision, these pumps are also suitable for feeding weighing systems. Design: Operating pressure max. 10 bar Shaft seal - Stuffing box packings for chocolate and coating masses, nut pastes, and cocoa masses - Mechanical (axial face) seals for fats, oils, cocoa butter, and emulsifiers Material of - casing GG-25 - shaft St 60.2 - impeller GG-25

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Item	Qty.	Description
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With gear motor for frequency-converter operation, clutch, and pump base plate.

Connection flanges according to DIN 2533.

Pump protection - electrical pressure monitoring system with sensor

- connection size R $\frac{1}{2}$ "

Includes the following types and sizes:

PUMP, TYPE FIP CAMA

for conveying cocoa liquor

density approx. 1.05 kg/dm³

viscosity approx. 2 - 5 Pas

Pump in special wear resistant execution

Bearing bush - hard metal

Pinion pivot - hard metal

Heating jacket - front cover and shaft packing

1x DN 80 PN 16

Conveying capacity range 0.5 - 3.0 t/h

Motor power 2.20 kW

117 2 MAGNET 8000 GAUSS

Quick cleaning magnet filter with double walled housing.

General:

Max. particle size: 16 mm

Min. Fe-particle size that can

be caught: 30µm

Cleaning/Fe-particles disposal: Manual quick cleaning by means of extractor tubes

Built-in length: 400 mm

Interface/connection dim`s: DN80 acc. EN 1092-1 PN16

Interface/connection dimensions-

Flange type: EN 1092-1 PN16

Connections: G 1/2 BSP socket connector opening for warm water (3x)

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Item	Qty.	Description
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Materials - in contact with product:	AISI316L (SS 1.4404)
Material housing:	AISI316L (SS 1.4404)
Version housing:	Doubled walles
Magnetic bar joint:	Extractor tubes and not brazed in plate (welded only)
Material gaskets/seals:	Silicone VMQ O-ring, 60° Shore, red, "Food Safe" acc. EC 1935/2004 and FDA
Surface treatment/finishing:	S1: Grit blastet, Ra 3 µm (interior/exterior)
Welds:	Wm: product chute: completely welded outside the product chute: completely welded where possible
Max. operating/ product temperature:	140°C
Min. Max. ambient temperature:	- 20 to 40°C
Operating pressure:	Max. 10 bar

Magnet info:

Magnetic system:	Magnetic bar in extractor tube
Magnetic bar size:	Tube diameter 25 mm, bar diameter 23 mm
Number of bars:	7
Magnetic quality:	Neodym GSN-42SH, Br 13,300 gauss (at 20°C), Tmax. 150°C
Fiels strength (flux density) on magnetic bar (+/- 10%):	10700 gauss
Fiels strength (flux density) on extractor tube (+/- 10%):	8000 gauss (average value)

Pysical info:

Length:	400 mm
Width:	293 mm
Height:	250 mm
Weight:	33 kg

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Item	Qty.	Description
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118 1 FITTINGS

The base for a robust and accurate conveying and dosing system of liquid products is the correct application of cut-off devices, pressure sustaining and venting valves. The complete set of armatures is executed for integration into the pipeline system and is of sanitation design.

All parts in contact with products are certified by the food industry.

Consisting of:

- 10 cut-off valve(s), manual operation, with two limit switches
- 4 cut-off valve(s), pneumatic operation, with two limit switches
- 3 check valve(s)
- 2 venting valve(s)

Execution:

Cut-off valves:

- Housing: <DN50 GG25 / ≥DN50 light alloy casting
- Butterfly/shaft: stainless steel 1.4408

Check valves:

- Housing: stainless steel 1.4581
- Valve plate/spring: stainless steel 1.4571

Venting valves:

- Housing: Brass
- Ball: Brass

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Item	Qty.	Description	
D	1	COCOA LIQUOR STORAGE, TEMPERING AND BLOCKING-OFF	
D-1	1	COCOA LIQUOR STORAGE	
119	1	DOUBLE WALL PRODUCT PIPES FOR COCOA LIQUOR STORAGE	Customer
120	3	TANK 20000 I Execution: Cylindrical standing form 15° conical lid and tank bottom All corners that are in contact with the product are fabricated with large radius. Material: Completely fabricated in stainless steel 1.4301 Finish: Weld beads ground flush min. K180 and pickled on the outside. Inside and outside surfaces rolled blank IIIC Main tank dimensions: Content :20000 I Diameter :2400 mm Cylindrical height :4500 mm Approx. total height :5600 mm Base execution: Manhole on the top with diameter 420 mm. Safety switch on manhole (applies only to tanks that are delivered with agitator). Hinged safety grating in manhole (applies only to tanks that are delivered with agitator). Inlet spouts. Outlet spouts. Vent. Sockets for probes. Product temperature display at a height of 1.5 m. Welded grounding tab. The tank is on 4 supports, with adjustable dome shaped plates or adapter plates for weigh cells. Transport eyebolts.	

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Item	Qty.	Description
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Water jacket:

Warm water jacket with forced circulation. The outlet, bottom and approx. 1/3 of the wall are heated with one circuit. Operating pressure is 3 bar, test pressure is 5 bar.

Insulation:

Welded insulation jacket on the bottom and the walls.
75 mm thick mineral wool insulation.

Anchor agitator: (for masses)

Single stage anchor agitator with wall scrapers
Space between wall and scraper approx. 10 mm
Function: maintain homogeneity, support the heat transfer
Agitator is completely welded
Agitator has an opposing bearing on the tank floor
Rotational speed max. 1.5 m/s
Leak proof tank shaft seal.
Food grade gear lubrication.

Discharge spout: ND 125

"T" spout with 2 manual slide gates according to DIN 11851.

Discharge valve with cap.

Including water jacket for warm water heating with connections.

Ladder stirrup:

3/4" x 500 mm pipe, welded below the top of the cylindrical part of the tank, by the man hole. Intended to be used to hang a ladder onto.
With safety rail.

Connecting / Utility Data

Electrical:

– Gearmotor of anchor agitator: 5.5 kW

Hot water consumption data:

- Water flow rate: 0.75 m³/h
- Water temperature: 45-50°C
- Operating pressure: 3bar
- Test pressure: 5bar

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Item	Qty.	Description	
121	3	HIGH-LEVEL INDICATOR With level indicator.	
122	3	LEVEL SENSOR Pressure gauge converter in tanks Capacitive pressure sensor. Flush membrane made of Ceraphire. Long term stable and resistant to overload. Maximum process temperature 130°C. Process connection G1½".	
123	3	SET OF LOAD CELLS FOR COCOA LIQUOR TANK 20000 l Consisting of: 2 Electronic load cells made of stainless steel, IP68. Pressure load: 10000 kg Excluded: Mounting bracket and further required material as well as adjustment work on slo bins. Universal cable junction box for 1-4 load cells. Standard cable 10 m. Excluded: Mounting bracket and further required material (like calibration weights) as well as adjustment work on silo bins.	Option
124	3	UNIVERSAL CONTROL UNIT MEAG For DMS-Transmitter application in painted mild steel housing IP 65. For application without calibration approval. Parameters fully controllable by 8 buttons and 14 digit alpha-numerical display. Comprising: 8 digital inputs 24 VDC, 12 digital outputs 24 VDC, 2 analogue in- and outputs 0/4 ... 20 mA (2 each) 1 interface Profibus-DP slave with max. 12 MBaud 1 serial interface RS 485 with BÜHLER standard scale report. ProfiNet field bus interface. DMS Junction box with 10 m cable.	Option

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Item	Qty.	Description
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125 1 CONVEYING AND METERING PUMPS

The FIP internal gear pumps based on hygienic design principles are suitable for gentle conveying of liquid media in the chocolate industry. Various shaft sealing systems consider different product specifications. The FIP pumps are designed for continuous service. The long service life and short exchange times of the seal in conjunction with an easy-to-remove pump head plate minimize the maintenance requirements. Thanks to their constant and non-pulsating conveying action and their high metering precision, these pumps are also suitable for feeding weighing systems.

Design:

Operating pressure max. 10 bar

Shaft seal

- Stuffing box packings for chocolate and coating masses, nut pastes, and cocoa masses
- Mechanical (axial face) seals for fats, oils, cocoa butter, and emulsifiers

Material of

- casing GG-25
- shaft St 60.2
- impeller GG-25

With gear motor for frequency-converter operation, clutch, and pump base plate.

Connection flanges according to DIN 2533.

Pump protection

- electrical pressure monitoring system with sensor
- connection size R½"

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Item	Qty.	Description
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Includes the following types and sizes:

PUMP, TYPE FIP CAMA

for conveying cocoa liquor

density approx. 1.05 kg/dm³

viscosity approx. 2 - 5 Pas

Pump in special wear resistant execution

Bearing bush - hard metal

Pinion pivot - hard metal

Heating jacket - front cover and shaft packing

1x DN 150 PN 10

Conveying capacity range 5.0 - 20.0 t/h

Motor power 9.20 kW

126 1 FITTINGS

The base for a robust and accurate conveying and dosing system of liquid products is the correct application of cut-off devices, pressure sustaining and venting valves. The complete set of armatures is executed for integration into the pipeline system and is of sanitation design.

All parts in contact with products are certified by the food industry.

Consisting of:

- 8 cut-off valve(s), manual operation, with two limit switches
- 9 cut-off valve(s), pneumatic operation, with two limit switches
- 1 check valve(s)
- 3 venting valve(s)

Execution:

Cut-off valves:

- Housing: <DN50 GG25 / ≥DN50 light alloy casting
- Butterfly/shaft: stainless steel 1.4408

Check valves:

- Housing: stainless steel 1.4581
- Valve plate/spring: stainless steel 1.4571

Venting valves:

- Housing: Brass
- Ball: Brass

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Item	Qty.	Description	
D-2	1	COCOA LIQUOR TEMPERING AND BLOCKING-OFF	
127	1	DOUBLE WALL PRODUCT PIPES FOR COCOA LIQUOR TEMPERING & BLOCKING - OFF	Customer
128	1	TANK 5000 I	Option
		Execution: Cylindrical standing form 15° conical lid and tank bottom All corners that are in contact with the product are fabricated with large radius.	
		Material: Completely fabricated in stainless steel 1.4301	
		Finish: Weld beads ground flush min. K180 and pickled on the outside. Inside and outside surfaces rolled blank IIIC	
		Main tank dimensions: Content :5000 I Diameter :1700 mm Cylindrical height :2400 mm Approx. total height :3300 mm	
		Base execution: Manhole on the top with diameter 420 mm. Safety switch on manhole (applies only to tanks that are delivered with agitator). Hinged safety grating in manhole (applies only to tanks that are delivered with agitator). Inlet spouts. Outlet spouts. Vent. Sockets for probes. Product temperature display at a height of 1.5 m. Welded grounding tab. The tank is on 3 supports, with adjustable dome shaped plates or adapter plates for weigh cells. Transport eyebolts.	

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Item	Qty.	Description
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Water jacket:

Warm water jacket with forced circulation. The outlet, bottom and approx. 1/3 of the wall are heated with one circuit. Operating pressure is 3 bar, test pressure is 5 bar.

Insulation:

Welded insulation jacket on the bottom and the walls.
75 mm thick mineral wool insulation.

Anchor agitator: (for masses)

Single stage anchor agitator with wall scrapers
Space between wall and scraper approx. 10 mm
Function: maintain homogeneity, support the heat transfer
Agitator is completely welded
Agitator has an opposing bearing on the tank floor
Rotational speed max. 1.5 m/s
Leak proof tank shaft seal.
Food grade gear lubrication.

Discharge spout: ND 80

"T" spout with 2 manual slide gates according to DIN 11851.

Discharge valve with cap.

Including water jacket for warm water heating with connections.

Ladder stirrup:

3/4" x 500 mm pipe, welded below the top of the cylindrical part of the tank, by the man hole. Intended to be used to hang a ladder onto.
With safety rail.

Connecting / Utility Data

Electrical:

– Gearmotor of anchor agitator: 4.0 kW

Hot water consumption data:

- Water flow rate: 0.32 m³/h
- Water temperature: 45-50°C
- Operating pressure: 3bar
- Test pressure: 5bar

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Item	Qty.	Description	
129	1	HIGH-LEVEL INDICATOR With level indicator.	Option
130	1	LEVEL SENSOR Pressure gauge converter in tanks Capacitive pressure sensor. Flush membrane made of Ceraphire. Long term stable and resistant to overload. Maximum process temperature 130°C. Process connection G1½".	Option
131	1	SET OF LOAD CELLS FOR COCOA LIQUOR TANK 5000 l Consisting of: 2 Electronic load cells made of stainless steel, IP68. Pressure load: 3000 kg Excluded: Mounting bracket and further required material as well as adjustment work on slo bins. Universal cable junction box for 1-4 load cells. Standard cable 10 m. Excluded: Mounting bracket and further required material (like calibration weights) as well as adjustment work on silo bins.	Option
132	1	UNIVERSAL CONTROL UNIT MEAG For DMS-Transmitter application in painted mild steel housing IP 65. For application without calibration approval. Parameters fully controllable by 8 buttons and 14 digit alpha-numerical display. Comprising: 8 digital inputs 24 VDC, 12 digital outputs 24 VDC, 2 analogue in- and outputs 0/4 ... 20 mA (2 each) 1 interface Profibus-DP slave with max. 12 MBaud 1 serial interface RS 485 with BÜHLER standard scale report. ProfiNet field bus interface. DMS Junction box with 10 m cable.	Option

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Item	Qty.	Description
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133

1 CONVEYING AND METERING PUMPS

Option

The FIP internal gear pumps based on hygienic design principles are suitable for gentle conveying of liquid media in the chocolate industry. Various shaft sealing systems consider different product specifications. The FIP pumps are designed for continuous service. The long service life and short exchange times of the seal in conjunction with an easy-to-remove pump head plate minimize the maintenance requirements. Thanks to their constant and non-pulsating conveying action and their high metering precision, these pumps are also suitable for feeding weighing systems.

Design:

Operating pressure max. 10 bar

Shaft seal

- Stuffing box packings for chocolate and coating masses, nut pastes, and cocoa masses
- Mechanical (axial face) seals for fats, oils, cocoa butter, and emulsifiers

Material of

- casing GG-25
- shaft St 60.2
- impeller GG-25

With gear motor for frequency-converter operation, clutch, and pump base plate.

Connection flanges according to DIN 2533.

Pump protection

- electrical pressure monitoring system with sensor
- connection size R½"

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Item	Qty.	Description
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Includes the following types and sizes:

PUMP, TYPE FIP CAMA

for conveying cocoa liquor

density approx. 1.05 kg/dm³

viscosity approx. 2 - 5 Pas

Pump in special wear resistant execution

Bearing bush - hard metal

Pinion pivot - hard metal

Heating jacket - front cover and shaft packing

1x DN 80 PN 16

Conveying capacity range 0.5 - 3.0 t/h

Motor power 2.20 kW

134	1	TEMPERING MACHINE FOR COCOA LIQUOR	Option
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Pre-Cooler Type 9 m²

For cooling of cocoa liquor arriving to the machine at a temperature of 80°C, leaving the machine at approx. 50°C, using cooling water at maximum 30°C.

Capacity in cocoa liquor up to 2000 kg/h

Tempering machine for cocoa liquor

This tempering machine is an integrated solution suitable for all types of high-fat and low-fat pumpable masses, i.e. for chocolate with or without ingredients, cocoa liquor, cocoa butter and fillings. It is designed for supplying moulding lines and enrobers with well-tempered chocolate. The compact construction only requires very little floor space. The tempering machine is mounted on a stainless frame.

General data

For tempering of cocoa liquor being detempered prior to entering the tempering machine. Inlet temperature of 50°C and outlet temperature of 28°C.

Capacity
 max. 2000 kg/h

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Item	Qty.	Description
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Tempering elements each zone:

Zone P	N/A
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Zone 1	12
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Zone 2:	<u>4</u>
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Elements total:	16
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Dimensions

Height	2220 mm
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Width	1040 mm
-------	---------

Depth	1690 mm
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Weight of machine

approx. 2700 kg

Power supply

3 x 400V, 50 Hz

PLC

Allen Bradley or Siemens

PLC and touch screen

Ethernet Interface

Hardware components

Schneider Soft Starter and

Motor Starter

Inclusive of tray and a drain pipe for collection of condensed water, if any. The tray will be placed on consoles under the water systems above the gear motor. The drip tray can be removed for service of the machine.

Gear Motor (HDM)	15 kW
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The capacity indicated is calculated for a cocoa liquor inlet temperature to the first tempering zone of 50°C and a cocoa liquor outlet temperature of 28°C, with the supply of cooling water of max. temperature 8°C.

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Item	Qty.	Description
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Cooling capacity

The machine needs min. 4000 litre/hour of cooling water (approx. 74 kW) during start-up and min. 2100 litre/hour of cooling water (approx. 49 kW) during production.

IP Protection Class IP 54

One Water Heat Exchanger

One plate heat exchanger, expansion tank and circulation pump built into the piping system of the tempering unit, changing the water system of the tempering unit from an open into a closed system; the advantages being reduced corrosion and sediment build-up in elements, blocking of valves, etc.

One Built-In Water System

One additional water system can be built into the tempering unit for heating the cocoa liquor pipe from the tempering machine to filling head.

This unit comprises a water tank, centrifugal pump, thermostat, 4.5 kW immersion heater and solenoid valve.

Electric controls are built into the control panel.

One Explosion Disc

To protect the machine against possible over-pressure, an explosion disc with max. pressure 10 bar, will be installed in the pipe at the inlet to the tempering machine.

Includes system for alarm signal, when the disc is open.

Including start up from a specialist during max. 6 days.

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Item	Qty.	Description
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Detempering Unit, Type 9 m²

For deseeding tempered cocoa liquor, it is vital to re-melt all crystals by reheating the cocoa liquor to 45°C, before it is returned to the tank. We therefore recommend the installation of our detempering unit.

The detempering unit is specially designed with a shell and piping system. Product is pumped through many small diameter pipes contained in an outer shell or jacket containing hot water. Hot water at 50°C to 55°C has to be supplied.

The unit is made completely from stainless steel.

The detemper can be supplied for either horizontal mounting or vertical mounting, according to your preference.

Capacity of cocoa liquor up to 2000 kg/h

Weight of unit

Net weight approx. 390 kg

Insulation of Tempering Elements with Armaflex

In order to prevent issues following condensation of water on the tempering machine, dripping water to the floor etc, we can offer to insulate with Armaflex the "cold zones/areas" on the tempering machine, such as the stack of tempering elements, water pipes, valves, heat exchanger etc.

1 *Tempering system commissioning specialist*

over a period of each 6 day(s)

for the commissioning of the tempering system for cocoa liquor and cocoa butter.

Bühler is solely responsible for supervision of installation and/or supervision of commissioning and/or the supervision of the performance test works for the equipment delivered by Bühler (the "Equipment"). Its tasks and responsibilities are set forth in detail hereinafter. The installation, commissioning and the performance test works as such are carried out by the Buyer in accordance with the stipulations of the technical documentation provided by Bühler.

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Item	Qty.	Description
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135	1	SAMPLER FOR COCOA LIQUOR
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Option

IsoControl M4KSA sampler Silicon

This Model is designed for sampling cocoa liquor up to 1000cP out of a pipe. The sampler is powered by a double effect compressed air plunger piston. The extremity of the plunger takes directly out of the flowing product a sample that drops into the bottle when plunger withdraws. The features ferrule clamp mounting suitable for any line size, quick disconnect to expose all wetted parts for cleaning, FDA and 3A listed seal materials and built-in mount for 24mm thread sample bottles. A bottle sampler adapter suitable to customer's bottle can be provided as an option.

This M4KSA sampler complies with 3A, CE, ATEX, FDA specifications.

Technical specification:

Pressured requested:	3-5 Bar
Max. process pressure:	7 Bar
Process connection:	Clamp DN38 (1" ½)
Sample volume per cycle:	3 ml.
Max. cycle frequency:	30 strokes/min.
Max. capacity:	90 ml / min.

Material:

Body:	SS316
Roughness:	0.4 µm (inside) 0.75 µm (outside)
Seals:	Silicon FDA approved and CE 1935/2004

General: After installation, the sampler should be periodically checked for proper operation.

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Item	Qty.	Description
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Bottle adaptor for Liquid - Size 1

Adaptor for sampling bottles suitable for liquid application.

Specifications:

Size:	T1 (neck Ø33 to Ø70 mm)
Connection:	Thread
Material:	SS316L

Control unit EP 2 actions

This control unit allows to operate a pneumatic sampler from the Labocontrolle range.

Before the first use and through a simple setting mode, all timers has to be adjusted to define a sample taking phase or sequence. This way, the representativity and the repeatability of each sample are seriously improved. The required air pressure could be adjusted with an air pressure regulator installed inside the enclosure.

Operations can be done by using the touch screen of the controller or can be driven by a specific external signal. A manual mode allows anyone to operate the sampler at any time.

Specifications:

Number of actions:	2
Material:	Inox 304 L
Dimensions:	300 x 200 x 120 mm
Protecting rating:	IP 65
Max. pressure inlet:	10 bar
Voltage supply:	24 Vcc

Suitable software for the control of the sampler included.

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Item	Qty.	Description
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Set of 6 HDPE bottles

Set of 6 sampling bottles 1000ml HDPE with threaded cap for in line or portable samplers.

Dimensions:

Diameter:	Ø 90 mm
Height:	200 mm
Neck diameter:	Ø 63 mm

Delivered with CE1935/2004 and FDA certificate

Technical file for IsoControl Sampler

Included:

- User manual (installation, operating, maintenance)
- Technical data sheets
- Certificates (CE, Quality and Material Certificates)
- Drawing

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Item	Qty.	Description
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136	1	FILLING STATION
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Option

The container filling station type Typ 24.2-0-00n in stainless steel design

for automatic and calibrated filling of containers with cocoa liquor with up to 2 .

The filling station is designed for use in normal environments for filling non-flammable liquids.

Basic construction:

Stable column construction on a base frame with machine feet for level compensation.

Filling type:

Overhead level for non-critical products that are filled into open-top containers.

Base height:

With stepless base height adjustment for conversion to different pack heights with an adjustment range of 100 mm to 600 mm (spindle with hand crank).

Machine control:

For intuitive operation of the filling machine via the FEIGE Web HMI, a

15.6" full HD multi-touch screen in combination with a PLC and weighing electronics.

LED display integrated into the control cabinet door for the visual display of machine machine statuses and the fill level in the container.

The filling process is started at the control panel. Further operation, such as activation of the cleaning mode with opening of the filling valve is carried out exclusively via the touchscreen.

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Item	Qty.	Description
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Control cabinet:

Installation location: Mounted on the machine or next to the machine
on a

stand frame

Width: 780 mm

Depth: 350 mm

Height: 1100 mm

Material: Stainless steel housing,
Design element stainless steel acid-resistant
lacquered RAL 5011

Weighing system:

Accuracy class X (1), legal for trade according to OIML R61 consisting
of:

Single-point underframe, stainless steel

- strain gauge load cell (IP68), integrated in the frame construction
- 2 weighing ranges (resolution 3000d each)
 - 2.5 kg to 15 kg, numerical increment 5 g / ...
 - ... up to 30 kg, digit step 10 g / ...
 - ... up to 60 kg, digit step 20 g / ...
- Stainless steel bridge plate 400 x 500mm
- Weighing electronics with "Feige FS-T10E" display and control terminal
- Self-optimising dosing programme
- Control of machine functions
- Cyclic check for safety-relevant states of the filling unit, e.g. of the filling unit, e.g. execution of tolerance checks, Possibility of tare controls, flow control and filling time monitoring
- Product file for saving 1000 data records
- Bright TFT colour display with wide viewing angle viewing angle, large weight display and visualisation of operating states
- Fast, complete and error-free entries thanks to dialogue guidance and alpha-numeric keyboard
- Integrated in the control cabinet
- USB host interface for printer, scanner or USB stick

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Item	Qty.	Description
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Filling unit:

Consisting of

Opening cylinder for pneumatic opening and closing of the filling valve

- Two-stage winding, for coarse flow and adjustable fine flow
- Quick-change filling valve holder made of stainless steel

Heatable 3-way filling valve

Type FVZ-50-RB-WH-3W with short trunk, wind-up, spring-closing
Outer/outlet diameter of short trunk 65/34 mm, suitable for connecting
connection of perforated discs for medium viscosity products.

- Operating pressure up to 10 bar, temperature up to 130°C
Pressure-relieved (no influence of product pressure on closing
behaviour) by the product pressure)

- Without gap and dead spaces
- Self-adjusting sealing bush

G1/8 connection for flushing the sealing chamber or for leakage
monitoring Leakage monitoring

- Product connection Feige flange PN10 for connection of a of a Feige
product hose
- Material 1.4571 (parts in contact with product), seals PTFE

Electro-pneumatic control:

- Design according to VDE / IEC
- Protection class IP55
- Stainless steel control cabinet with operating elements
- Weighing terminal integrated
- PLC function from weighing terminal, control voltage 24 V DC
- Solenoid valve control with valve terminal
- Required operating air pressure 5 bar g dynamic
- Compressed air quality according to DIN ISO 8573-1 class 4
- Compressed air connection G½"
- Compressed air temperature: min. +5°C, max. +40°C
- Electrical connection 230V, 50 Hz

Surface treatment materials:

- Stainless steel design according to Feige design standard VK-03.

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Item	Qty.	Description
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Operating principle of the pail filler type 24.2

- The filling unit is set up to the bucket height.
- One pail at a time is fed to the filling unit
- The pail is positioned manually to the filling unit.
- After pressing the start button, taring takes place automatically.
- The filling nozzle opens for coarse-flow filling.
- The system switches to fine-flow filling when approx. 90 - 95% of the target weight has been reached. The switchover point is adjustable (depending on the filling speed).filling speed).
- The filling nozzle closes when the target weight is reached.
- The drop trap selected as an additional feature comes into operation.
- Cleaning: The cleaning mode is switched on with the key switch "Fill/Clean". With the keys "fine flow" or "coarse flow" buttons, the filling valve can be opened for rinsing (jog mode). (jog mode).

Note:

At a pump capacity of 17m³/h, an output of 166 cartons/h can be achieved. With a filling weight of 29 kg (cocoa mass with a density of 1.27 kg/dm³), this results in a capacity of 4.814 t/h. The pump capacity must be checked or specified by the customer.

Drip collecting device type TF-E

When filling is finished, the drip collecting device is moved below the filling valve by means of a compressed air cylinder. The collected product is fed into a once-way bowl.

Reusable bowl made of stainless steel 1.4301

Container roller conveyor / collecting basin type RBB-Wg-n-50-AW-ss

The collecting basin with inserted, non-driven rollers is installed on the scale platform. For cleaning purposes basin and rollers dismantle easily without tools.

Nominal width:	390 mm
Roller:	diameter 30 mm, stainless steel
Roller pitch:	approx. 50 mm
	Collecting basin made of stainless steel
Total length:	approx. 450 mm

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Item	Qty.	Description
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Conveyor system in the in- and out-feed area

Two container roller conveyor type RBB-n-50 in stainless steel execution

non-driven

Nominal width: 390 mm

Roller: Diameter 32 mm, stainless steel

Roller pitch: approx. 50 mm

Side parts designed as C-bar made of stainless steel

Length: approx. 2000 mm

Including:

Side guide for container roller conveyor type SF-B in stainless steel

The side guide can manually be adjusted to different container sizes.

Support for container conveyor type STR-B in stainless steel execution

stable construction made of stainless steel, height-adjustable by approx. ± 50 mm.

Stationary limit stop type EA-B in stainless steel

at the front end of the container roller conveyor.

Standard data interface Profinet IO type IF-PLC-Ethn

PLC interface for connection to a process control system

Defined data for controlling the filling process is transmitted.

The interface is designed as a Profinet IO device.

controller assigns the Profinet name and IP address to the IO device.

Including:

- Extension data interface type IF-PLC2 / Optional extension to the data interface
- Extension data interface type IF-PLC4 / Optional extension to the data interface
- "IF-Scale-Reader" software package / For simple logging of the dosing results on a PC provided by the customer
- Data interface type IF-FST-Ethn / Ethernet interface for transferring weighing data for PC connection, cable length up to 100 m
- Electrical installation and parameterisation of the filling system before delivery

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Item	Qty.	Description
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Ethernet Interface

Ethernet-Schnittstelle für PC-Anschluss, Kabellänge bis 100 m

Valve flange FVD/Z-50

For connecting the customer's connection to the Feige filling valve.
Block flange DN 50

Note: The valve flange must be welded to the customer's product supply line. This is to implement on site.

Material: 1.4571

Commissioning support of accessories and machines listed in the scope of delivery documents (without optional and alternative equipment) with the following personnel:

1 *Specialist for filling station*

over a period of each 7 day(s)
for commissioning of the filling station.

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Item	Qty.	Description	
137	1	CONVEYOR WITH BOX CLOSING SYSTEM	Customer
138	1	CONVEYOR WITH METAL DETECTOR	Customer
139	1	FITTINGS	Option
The base for a robust and accurate conveying and dosing system of liquid products is the correct application of cut-off devices, pressure sustaining and venting valves. The complete set of armatures is executed for integration into the pipeline system and is of sanitation design. All parts in contact with products are certified by the food industry. Consisting of: – 10 cut-off valve(s), manual operation, with two limit switches – 6 cut-off valve(s), pneumatic operation, with two limit switches – 1 check valve(s) – 1 venting valve(s) Execution: Cut-off valves: – Housing: <DN50 GG25 / ≥DN50 light alloy casting – Butterfly/shaft: stainless steel 1.4408 Check valves: – Housing: stainless steel 1.4581 – Valve plate/spring: stainless steel 1.4571 Venting valves: – Housing: Brass – Ball: Brass			

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Item	Qty.	Description
E	1	MECHANICAL INSTALLATION MATERIAL
E-1	1	MECHANICAL INSTALLATION MATERIAL
140	1	CONSUMABLES & PRIMARY SUPPORTS Consisting of: loose supply of material i. e. steel profiles, steel bars, galvanized- and/or blank sheets of steel for the support and installation of machines and conveying elements. All parts if necessary are prime coated only. Bolts and nuts, screws, studs, washers, spring-washers, wall plugs, dowels, threaded rods. Gaskets, sealing material, rags and fluids for cleaning the equipment only. Welding rods for the initiating period of the installation. Small amounts of touch-up paint for machines only.

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Item	Qty.	Description	
F	1	SPARE PARTS	Option
F-1	1	SPARE PARTS	Option
141	1	SPARE PARTS Recommended strategic spare parts to have on stock, evaluation based on high maintenance experience for plants and machines specified according the final customer installation. (most critical and important spare parts to maintain production, facilities and system to keep down time as short as possible.)	Option

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Item	Qty.	Description	
II	1	STEEL STRUCTURE AND PLATFORMS	Customer
G	1	STEEL STRUCTURE AND PLATFORMS	Customer
G-1	1	STEEL STRUCTURE AND PLATFORMS	Customer
142	1	STEEL STRUCTURE AND PLATFORMS	Customer

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Item	Qty.	Description
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III 1 ELECTRICAL EQUIPMENT

H 1 PLANT CONTROL

To control the plant the automation system Mercury is used, which is optimally adapted for the process. Mercury integrates technology, production and automation and supports continuous production with maximum efficiency and highest quality. Our highly efficient solution can integrate upstream (e.g. ERP) and downstream systems (e.g. machine data). Mercury is web-based, thus adaptive and accessible on any device. It is the backbone for digital services which increase your profitability through new insights and intelligent algorithms. In this system, Bühler combines decades of experience in automation and engineering solutions.

OPERATING CONCEPT

The new processing line is controlled via an operating stations in the control room. The browser of the operating stations is used to access the web server of the plant control system, which runs on a box-IPC in the control cabinet. With this solution, the operator interface can be accessed from any device (tablet, PC) with a browser and the necessary user credentials from anywhere in the production network.

Further details are available on the automation concept drawing (CM-22-00007-AUT-01).

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Item	Qty.	Description
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To control the plant the automation system Mercury MES is used, which is optimally adapted for the process.

Mercury MES integrates technology, production and automation and supports continuous production with maximum efficiency and highest quality. In this system, Bühler combines decades of experience in automation and engineering solutions.

The automation system from Bühler includes the following basic features:

- Process control
- Process monitoring
- Interactive web-based process visualization with status display of plant processes
- Animated product flow
- Alarm display and registration
- Sorting and filter functions in summary tables
- Possibility of remote maintenance
- Password protection against unauthorized access
- User management
- Product management
- Bin management

Operating concept

Industrial PC

The proposed system design is shown in the concept drawing FC-25-00643-AUT-01, which is an integral part of this quotation.

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Item	Qty.	Description
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H-1	1	FUNCTIONS OF THE PLANT CONTROL SYSTEM
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143	1	AUTOMATION ENGINEERING
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Developing the system specifications (SYS) including interface definitions.

PLC software engineering:

User software for the PLC for interlocking, control and monitoring of the process.

Process control software engineering:

User software for the process control system for operation and visualization of the plant.

The plant is displayed in simplified form on the screen (without future machines).

Hardware Engineering:

The planning services for the preparing of the electrical circuit diagrams and the control cabinets of the plant, are included and based on <the specified components>.

The following plant functions and process line elements are made under the consideration of the customers desire:

Plant functions

– Bin management

Convenient overview of all bins and rapid navigation of the system is possible because all silo bins and tanks are defined and registered. If fillings and withdrawals are registered by weight, is the system calculating the filling level in the bins and displays it in a graph. The calculated and measured bin levels are shown if the bins are equipped with level indicators.

– Product management

Definition of raw material, semi-finished- and finished- products, as well as it's allocation to silo bins or tanks, to prevent product cross-contaminations.

– User management

Administration of access rights for different users.

– Event log

Registration of operation messages, warnings and alarms.

– Recipes management

Preparation and management of all recipes for the long-term management and monitoring of the production, to guarantee a repeatable product quality and an improved utilization of raw materials and therefore to secure a higher productivity.

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Item	Qty.	Description
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- Production statistic
- Call wiring diagram directly from visu
It is possible to open and show over the displayed window directly the selected element in the correct digital electronic sheet to support maintenance.
- Job related parameter trends
Job relevant recordings and graphic displays of parameters, such as temperatures, throughputs and pressures. The trend data can be printed out and exported.

Process lines and line functions

- 1x Intake bulk hopper
 - On site operating box
 - Job for process line
 - Job queue for process line
 - Job report for process line
- 1x Cocoa processing beans roasting
 - Integration agitator mill NOVA (1x)
 - 1x Interface RoaStar (STT)
 - 1x Interface DebaC SLSA
 - 1x Interface Winnower
 - 1x Interface PreGrind
 - Automatic weight registering
 - Product interlocking
 - Recipe
 - Sub recipe
 - Job for process line
 - Job queue for process line
 - Job report for process line
 - Event report per job

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Item	Qty.	Description
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- 1x Shell discharge
 - On site operating box
 - Jobfor process line
 - Job report for process line

- 1x Blocking-off
 - 1x Interface "Depositing machine"
 - Automatic weight registering
 - Job for process line
 - Job report for process line

- 1 x Auxiliary 1

ENGINEERING FOR THE CONTROL AND POWER SECTION

Included is the engineering service for preparing the circuit diagrams of the offered process, all based on Bühler standardized CE or IEC conform operating equipment.

Executed according to:

- EN/IEC-60204-1 Safety of machinery
- EN/IEC-61439-1 (-2) Low Voltage- switching device combinations
- EN ISO 13849-1 Safety related parts of controls (PL)
- see technical supplementary sheet (Black box Parameter)

The following documents are rendered:

Electrical scheme with circuit diagram, terminal connection layout, I/O and devices list.

The language for technical documentation is **English**

The language for the user interface is **English**

The language for the general user manual is **English**

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Item	Qty.	Description
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H-2 1 HARDWARE FOR OPERATION AND VISUALISATION

144 1 HARDWARE AND LICENCES

The hardware supplied by Bühler reflects the commonly available in the market at the time of project implementation (technical specifications are minimum values). Overview:

Network hardware components
For connection of the process control system to the network.

Switch
There is (are) 1 switch provided.

1 x Flat screen monitor 38" Curved

OPERATING DEVICES IN THE PLANT

3 x Onsite operation
With illuminated push buttons.

VARIOUS DEVICES

1 x Wall cabinet for IPC
Wall cabinet in stainless steel.

PC LICENSES

1 x Mercury MES runtime (max. 10 connections for operating access)

1 x Software Simatic TIA Portal

2 x Windows server operating system

1 x SQL Server Express

1 x OPC communication software

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Item	Qty.	Description
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1 x VMware workstation player

1 x Microsoft Defender Antivirus

Protect your computers against malware and data loss by Microsoft Defender Antivirus software.

The Microsoft end-user licensing agreement (EULA) shall be accepted by the purchaser.

All rights and obligations of the purchaser in connection with the software refer solely to the purchaser and Microsoft.

Exclusion of liability of Microsoft Defender Antivirus

The software shall be utilized at the purchaser's own risk. The software shall be installed "as it is" without any warranty and liability, specifically yet not restricted to freedom from defects, incompatibility with hardware components or other software products and their components, suitability for a given purpose, etc. In no event shall the purchaser be entitled to claim damages from Bühler for direct or indirect damage, specifically yet not restricted to the software, its installation, or its utilization.

General license rights

If the supply includes software, all rights to this software shall be retained by the supplier. However, the supplier shall grant the purchaser a non-exclusive right to utilize the software in the area covered by the contract, it not being allowed to award any sublicenses. This right shall be limited to utilization according to the contract. Without the supplier's prior consent, the software shall neither be copied nor be made accessible to any third parties or be modified.

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Item	Qty.	Description
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H-3

1 HARDWARE FOR THE CONTROL SYSTEM

All electronic components, such as the main switch, frequency converters, soft starters, fuses, PLC, I/O nodes, are mounted in the combined control- and power panel.

145

1 EQUIPMENT CONTROL- POWER- SECTION

Combined control and power panel

One combined control- and power panel in mild steel IP 54 for installation in +5 to +30°C ambient temperature.

Cabinet color: grey RAL 7035, cable entry: bottom/top

Spare space approx.: 10%

To reduce electrical installation efforts distributed field boxes with I/O-nodes are previewed.

The hardware applied shall be wired up to the terminals as far as required.

Motor starting conditions:

Motors < 15kW with direct online starting

Motors ≥ 15kW with softstarter

All components will be delivered by Bühler's standard suppliers. No customer specifications considered in the quotation.

Supplier list:

PLC-Systems	Siemens S7-1500 family
Remote-I/O	Siemens ET200SP
PC and Monitor	Dell
Touch IPC	Siemens
IT Switches:	Netgear
OT Switches	Siemens, Westermo
Printer	Brother, Hewlett Packard
Cabinets, control boxes	Rittal
Switches, selectors	Schneider Electric
Motor protection, circuit breakers	Schneider Electric
Relays, contactors	Schneider Electric
Safety relay	Pilz
Push-buttons, lamps	Schneider Electric
24 V DC Power supplies	Puls, Phoenix

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Item	Qty.	Description
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Variable frequency drives (VFD)	Schneider Electric Altivar
Soft Starters	Schneider Electric Altistart
Terminals	Weidmüller, Wago
Uninterruptible power supply (UPS)	APC (230/110), Siemens (24)
Scale evaluation unit	Bühler MEAG
Cable glands	Agro

The switching device combinations comprises the offered process, with Bühler standardized CE or IEC conform operating equipment.

Executed according to:

EN/IEC-60204-1	Safety of machinery
EN/IEC-61439-1 (-2)	Low Voltage- switching device combinations
EN ISO 13849-1	Safety related parts of controls (PL)

see technical supplementary sheet (Black box Parameter)

Starting conditions of motors:

Motors < 15 kW with direct-on-line starting.

Motors ≥ 15 kW with soft start.

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Item	Qty.	Description	
H-4	1	PLANT CONTROL OPTIONS	Option
146	1	DATA BACKUP SYSTEM The fully configured backup system of the operating station allows you to recover the whole Mercury including production data in the worst-case scenario of a broken operating station.	Option
147	1	PRINTER The network printer will be fully configured and ready to use for the Mercury.	Option

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Item	Qty.	Description	
IV	1	ELECTRICAL INSTALLATION	Customer
I	1	ELECTRICAL INSTALLATION	Customer
I-1	1	ELECTRICAL INSTALLATION	Customer
148	1	ELECTRICAL INSTALLATION	Customer

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Item	Qty.	Description
V	1	SERVICES
J	1	SERVICES
J-1	1	ENGINEERING
149	1	ENGINEERING SERVICES & DOCUMENTATION
		Below listed engineering services form an integral part of this quotation / contract and are to be provided only in connection with the supplies listed:
		– Process flowsheet(s) – Energy list indicating compressed air, water, steam and electrical requirements – Where applicable aspiration schematics, indicating air quantities, duct diameters and air speeds, typically as part of the process flowsheet(s) – Building indication drawings are limited to the process area and include where applicable: – Main dimensions of the building – Floor height, beam spacing & column locations – Major wall & floor openings – Size of required openings for equipment installation – General floor loads & major, concentrated static and dynamic equipment loads – Drain locations for processing water and/or condensate – Required openings for process related fresh air intake (make-up air) – Machine layout- & installation drawings, indicating equipment, bins, tanks, mechanical and pneumatic conveying elements as well as steel structures for equipment support and/or service access (conveying lines for dry and liquid components only as center line) – Motor- and device flowsheet(s) with corresponding lists – Electrical motor and device location drawings – Sizing and specifying of pumps, pipe diameters, and fittings for liquid components within our scope of supply – Technological and technical advice (No development of recipes, unless noted otherwise) – "As built" flow sheet(s) – Project documentation

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Item	Qty.	Description
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General remarks:

- In respect to content, design and layout, all documents and drawings are done according to our internal standards unless noted otherwise
- Documents other than operating manuals are in English or alternatively German language.

Calculated travel expenses:

2x site visit(s) for kick – off meeting and inquiries, Design Meeting, installation- and commissioning assistance.

The project documentation shall be handed out as follows:

1x English in electronic form

Operation manuals are supplied as standard in an electronic form.

Exempt from this rule are operation manuals that must be handed out additionally in the form of 1 hard copies in order to comply with legal requirements.

Language(s) of the operation manuals:

- English

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Item	Qty.	Description
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149.1	1	FAT
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Option

The customer factory acceptance test (FAT) takes place in the factory where the machines are manufactured. The date will be announced per machine by Bühler approx. 1-2 months before the FAT. Due to flow assembly process, the date will be defined by Bühler factory, i.e. there is no option to choose the FAT date. The FAT usually takes place in Europe, max. 1 day per machine and without product (dry run). In case having more than one machine of the same type, only one of them is going to be configured and tested.

A FAT date is foreseen for the following machines:

- RoaStar
- PreGrind
- Winnower
- Nova S

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Item	Qty.	Description	
150	1	ENGINEERING SERVICES & DOCUMENTATION	Option
		Below listed engineering services form an integral part of this quotation / contract and are to be provided only in connection with the supplies listed:	
		– Process flowsheet(s) – Energy list indicating compressed air, water, steam and electrical requirements – Where applicable aspiration schematics, indicating air quantities, duct diameters and air speeds, typically as part of the process flowsheet(s) – Building indication drawings are limited to the process area and include where applicable: – Main dimensions of the building – Floor height, beam spacing & column locations – Major wall & floor openings – Size of required openings for equipment installation – General floor loads & major, concentrated static and dynamic equipment loads – Drain locations for processing water and/or condensate – Required openings for process related fresh air intake (make-up air) – Machine layout- & installation drawings, indicating equipment, bins, tanks, mechanical and pneumatic conveying elements as well as steel structures for equipment support and/or service access (conveying lines for dry and liquid components only as center line) – Motor- and device flowsheet(s) with corresponding lists – Electrical motor and device location drawings – Sizing and specifying of pumps, pipe diameters, and fittings for liquid components within our scope of supply – Technological and technical advice (No development of recipes, unless noted otherwise) – "As built" flow sheet(s) – Project documentation	

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Item	Qty.	Description
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General remarks:

- In respect to content, design and layout, all documents and drawings are done according to our internal standards unless noted otherwise
- Documents other than operating manuals are in English or alternatively German language.

The project documentation shall be handed out as follows:
1x English in electronic form

Operation manuals are supplied as standard in an electronic form.
Exempt from this rule are operation manuals that must be handed out additionally in the form of 1 hard copies in order to comply with legal requirements.

Language(s) of the operation manuals:

- English

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Item	Qty.	Description
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J-2	1	SUPERVISION OF INSTALLATION
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151	1	INSTALLATION SUPERVISION
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Installation supervision of accessories and machines listed in the scope of delivery documents (without optional and alternative equipment) with the following personnel:

1 *Chief Installation Supervisor mechanical (s)*

on site over a period of each 12 week(s)

Leads the supervisors mechanical site team including a mechanical subcontractor for medium projects.

Customer personnel:

- 15 Mechanic
- 7 Labourer(s) for the mechanical installation
- 1 Site Manager, OHS Coordinator

Included are:

- Salaries based on 50 hours per week
- Travelling costs (e.g. flight, train, car) to place of destination and return
- Daily allowance for:
 - Accommodation (3-Star Hotel European standard or equal quarters)
 - Board with breakfast
 - Sundries
- Cost of possible excess baggage
- Travelling time to site of installation
- Assignment instruction at Bühler AG Uzwil
- Expenses for vaccinations and visas if applicable
- Accident- and health insurance

Excluded are:

- Dues, fees and/or taxes (VAT)
- Local transportation expenses to and from place of installation

The delegation period indicated above is based on the availability of qualified local installation and commissioning personnel, provided by the purchaser and in accordance to the requirements defined by Bühler during the execution of the project.

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Item	Qty.	Description
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The present offer is based on standard prices and conditions not adopted under the pandemic situation. Therefore, any additional costs and/or other expenditures in connection with the coronavirus including but not limited to additional safety measures, quarantine in the country of origin and/or destination, irregular travel / flight routes causing higher prices, and/or exposure to the virus on site are excluded.

Normal working hours

Considered as normal working time are 8.4 hrs. per day or 42 hrs. per week (Monday – Friday)

Overtime

Monday – Friday (06:00 to 23:00 o'clock) normal working time 8.4 hrs.

Monday – Friday (06:00 to 23:00 o'clock) normal overtime > 8.4 hours
= 25%

For installation engineers, commissioning engineers & automation experts

Saturday 25% surcharge

Sunday 50% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

For service engineers

Saturday 50% surcharge

Sunday 75% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

In most cases, commissioning should be planned for Tuesday so that the commissioning engineers can arrive on Monday. Travelling at the weekend is only permitted in exceptional cases.

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Item	Qty.	Description
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The estimation of the above mentioned delegation(s) is based on having a complete building (incl. roof fully enclosed, free from waste materials, cleaned, free from construction equipment and works) and the necessary infrastructure (i.e. access roads, storage area, temp. power, etc.), tooling, local labor as well as locally manufactured items available in adequate quality and quantity.

Moreover, we refer to our General Terms and Conditions of Service, which are an integral part of this document.

Bühler is solely responsible for supervision of installation and/or supervision of commissioning and/or the supervision of the performance test works for the equipment delivered by Bühler (the "Equipment"). Its tasks and responsibilities are set forth in detail hereinafter. The installation, commissioning and the performance test works as such are carried out by the Buyer in accordance with the stipulations of the technical documentation provided by Bühler.

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Item	Qty.	Description
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152	1	INSTALLATION SUPERVISION
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Option

Installation supervision of accessories and machines listed in the scope of delivery documents (without optional and alternative equipment) with the following personnel:

1 Chief Installation Supervisor mechanical (s)

on site over a period of each 2 week(s)

Leads the supervisors mechanical site team including a mechanical subcontractor for medium projects.

Customer personnel:

- 15 Mechanic
- 7 Labourer(s) for the mechanical installation
- 1 Site Manager, OHS Coordinator

The following cost items are included in the quotation:

- Salaries based on 50 hours per week
- Daily allowance of 160 USD for:
 - Accommodation (3-Star Hotel European standard or equal quarters)
 - Board with breakfast
 - Sundries
- Travelling time to site of installation
- Assignment instruction at Bühler office
- Expenses for vaccinations and visas if applicable
- Accident- and health insurance

The following cost items are not included in the quotation and must be provided by the client free of charge:

- Travelling costs (e.g. flight, train, car) to place of destination and return
- Cost of possible excess baggage
- Dues, fees and/or taxes (VAT)
- Local transportation expenses to and from place of installation

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Item	Qty.	Description
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The delegation period indicated above is based on the availability of qualified local installation and commissioning personnel, provided by the purchaser and in accordance to the requirements defined by Bühler during the execution of the project.

The present offer is based on standard prices and conditions not adopted under the pandemic situation. Therefore, any additional costs and/or other expenditures in connection with the coronavirus including but not limited to additional safety measures, quarantine in the country of origin and/or destination, irregular travel / flight routes causing higher prices, and/or exposure to the virus on site are excluded.

Normal working hours

Considered as normal working time are 8.4 hrs. per day or 42 hrs. per week (Monday – Friday)

Overtime

Monday – Friday (06:00 to 23:00 o'clock) normal working time 8.4 hrs.
Monday – Friday (06:00 to 23:00 o'clock) normal overtime > 8.4 hours
= 25%

For installation engineers, commissioning engineers & automation experts

Saturday 25% surcharge

Sunday 50% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

For service engineers

Saturday 50% surcharge

Sunday 75% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

In most cases, commissioning should be planned for Tuesday so that the commissioning engineers can arrive on Monday. Travelling at the weekend is only permitted in exceptional cases.

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Item	Qty.	Description
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The estimation of the above mentioned delegation(s) is based on having a complete building (incl. roof fully enclosed, free from waste materials, cleaned, free from construction equipment and works) and the necessary infrastructure (i.e. access roads, storage area, temp. power, etc.), tooling, local labor as well as locally manufactured items available in adequate quality and quantity.

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153	1	SUPERVISION OF INSTALLATION FOR ASPIRATION DUCTINGS	Customer
154	1	SUPERVISION OF INSTALLATION FOR DOUBLE WALL PRODUCT PIPES	Customer

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Item	Qty.	Description
J-3	1	COMMISSIONING
155	1	COMMISSIONING Commissioning support of accessories and machines listed in the scope of delivery documents (without optional and alternative equipment) with the following personnel: 1 <i>Installation Supervisor mechanical (s)</i> on site over a period of each 3 week(s) Supervise a mechanical site team. Customer personnel: – 1 Site Manager, OHS Coordinator 1 <i>Commissioning Engineer PLC automation(s)</i> on site over a period of each 6 week(s) Commissions and maintains SPS, PLC and safety systems. 1 <i>Commissioning Engineer automation(s)</i> on site over a period of each 4 week(s) Commissions and maintains, panels and hardware including I/O test. 1 <i>Senior Commissioning Engineer Process (s)</i> on site over a period of each 5 week(s) Leads process commissioning engineers and commissions the plant. 1 <i>Commissioning Engineer Process (s)</i> on site over a period of each 3 week(s) Commissions single machines and assists plant commissioning.

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Item	Qty.	Description
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Included are:

- Salaries based on 50 hours per week
- Travelling costs (e.g. flight, train, car) to place of destination and return
- Daily allowance for:
 - Accommodation (3-Star Hotel European standard or equal quarters)
 - Board with breakfast
 - Sundries
- Cost of possible excess baggage
- Travelling time to site of installation
- Assignment instruction at Bühler AG Uzwil
- Expenses for vaccinations and visas if applicable
- Accident- and health insurance

Excluded are:

- Dues, fees and/or taxes (VAT)
- Local transportation expenses to and from place of installation

The delegation period indicated above is based on the availability of qualified local installation and commissioning personnel, provided by the purchaser and in accordance to the requirements defined by Bühler during the execution of the project.

The present offer is based on standard prices and conditions not adopted under the pandemic situation. Therefore, any additional costs and/or other expenditures in connection with the coronavirus including but not limited to additional safety measures, quarantine in the country of origin and/or destination, irregular travel / flight routes causing higher prices, and/or exposure to the virus on site are excluded.

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Item	Qty.	Description
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Normal working hours

Considered as normal working time are 8.4 hrs. per day or 42 hrs. per week (Monday – Friday)

Overtime

Monday – Friday (06:00 to 23:00 o'clock) normal working time 8.4 hrs.

Monday – Friday (06:00 to 23:00 o'clock) normal overtime > 8.4 hours
= 25%

For installation engineers, commissioning engineers & automation experts

Saturday 25% surcharge

Sunday 50% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

For service engineers

Saturday 50% surcharge

Sunday 75% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

In most cases, commissioning should be planned for Tuesday so that the commissioning engineers can arrive on Monday. Travelling at the weekend is only permitted in exceptional cases.

The estimation of the above mentioned delegation(s) is based on having a complete building (incl. roof fully enclosed, free from waste materials, cleaned, free from construction equipment and works) and the necessary infrastructure (i.e. access roads, storage area, temp. power, etc.), tooling, local labor as well as locally manufactured items available in adequate quality and quantity.

Moreover, we refer to our General Terms and Conditions of Service, which are an integral part of this document.

Bühler is solely responsible for supervision of installation and/or supervision of commissioning and/or the supervision of the performance test works for the equipment delivered by Bühler (the "Equipment"). Its tasks and responsibilities are set forth in detail hereinafter. The installation, commissioning and the performance test works as such are carried out by the Buyer in accordance with the stipulations of the technical documentation provided by Bühler.

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Item	Qty.	Description
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156	1	COMMISSIONING
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Option

Commissioning support of accessories and machines listed in the scope of delivery documents (without optional and alternative equipment) with the following personnel:

1 *Installation Supervisor mechanical (s)*

on site over a period of each 1 week(s)

Supervise a mechanical site team.

Customer personnel:

- 1 Site Manager, OHS Coordinator

1 *Senior Commissioning Engineer Process (s)*

on site over a period of each 1 week(s)

Leads process commissioning engineers and commissions the plant.

The following cost items are included in the quotation:

- Salaries based on 50 hours per week
- Daily allowance of 160 USD for:
 - Accommodation (3-Star Hotel European standard or equal quarters)
 - Board with breakfast
 - Sundries
- Travelling time to site of installation
- Assignment instruction at Bühler office
- Expenses for vaccinations and visas if applicable
- Accident- and health insurance

The following cost items are not included in the quotation and must be provided by the client free of charge:

- Travelling costs (e.g. flight, train, car) to place of destination and return
- Cost of possible excess baggage
- Dues, fees and/or taxes (VAT)
- Local transportation expenses to and from place of installation

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Item	Qty.	Description
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The delegation period indicated above is based on the availability of qualified local installation and commissioning personnel, provided by the purchaser and in accordance to the requirements defined by Bühler during the execution of the project.

The present offer is based on standard prices and conditions not adopted under the pandemic situation. Therefore, any additional costs and/or other expenditures in connection with the coronavirus including but not limited to additional safety measures, quarantine in the country of origin and/or destination, irregular travel / flight routes causing higher prices, and/or exposure to the virus on site are excluded.

Normal working hours

Considered as normal working time are 8.4 hrs. per day or 42 hrs. per week (Monday – Friday)

Overtime

Monday – Friday (06:00 to 23:00 o'clock) normal working time 8.4 hrs.

Monday – Friday (06:00 to 23:00 o'clock) normal overtime > 8.4 hours
= 25%

For installation engineers, commissioning engineers & automation experts

Saturday 25% surcharge

Sunday 50% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

For service engineers

Saturday 50% surcharge

Sunday 75% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

In most cases, commissioning should be planned for Tuesday so that the commissioning engineers can arrive on Monday. Travelling at the weekend is only permitted in exceptional cases.

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Item	Qty.	Description
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The estimation of the above mentioned delegation(s) is based on having a complete building (incl. roof fully enclosed, free from waste materials, cleaned, free from construction equipment and works) and the necessary infrastructure (i.e. access roads, storage area, temp. power, etc.), tooling, local labor as well as locally manufactured items available in adequate quality and quantity.

Moreover, we refer to our General Terms and Conditions of Service, which are an integral part of this document.

Bühler is solely responsible for supervision of installation and/or supervision of commissioning and/or the supervision of the performance test works for the equipment delivered by Bühler (the "Equipment"). Its tasks and responsibilities are set forth in detail hereinafter. The installation, commissioning and the performance test works as such are carried out by the Buyer in accordance with the stipulations of the technical documentation provided by Bühler.

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Item	Qty.	Description	
J-4	1	TRAINING	Option
157	1	TRAINING ON CONSTRUCTION SITE Training at the machines/plant site with equipment as described in the scope of supply with the following personnel: 1 <i>Commissioning Engineer PLC automation(s)</i> on site over a period of each 2 week(s) for the training on the plant. 1 <i>Commissioning Engineer Process (s)</i> on site over a period of each 2 week(s) for the training on the plant. Included are: – Salaries based on 50 hours per week – Travelling costs (e.g. flight, train, car) to place of destination and return – Daily allowance for: - Accommodation (3-Star Hotel European standard or equal quarters) - Board with breakfast - Sundries – Cost of possible excess baggage – Travelling time to site of installation – Assignment instruction at Bühler AG Uzwil – Expenses for vaccinations and visas if applicable – Accident- and health insurance Excluded are: – Dues, fees and/or taxes (VAT) – Local transportation expenses to and from place of installation The present offer is based on standard prices and conditions not adopted under the pandemic situation. Therefore, any additional costs and/or other expenditures in connection with the coronavirus including but not limited to additional safety measures, quarantine in the country of origin and/or destination, irregular travel / flight routes causing higher prices, and/or exposure to the virus on site are excluded.	Option

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Item	Qty.	Description
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Normal working hours

Considered as normal working time are 8.4 hrs. per day or 42 hrs. per week (Monday – Friday)

Overtime

Monday – Friday (06:00 to 23:00 o'clock) normal working time 8.4 hrs.

Monday – Friday (06:00 to 23:00 o'clock) normal overtime > 8.4 hours
= 25%

For installation engineers, commissioning engineers & automation experts

Saturday 25% surcharge

Sunday 50% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

For service engineers

Saturday 50% surcharge

Sunday 75% surcharge

For night work between 23:00 and 06:00 o'clock 50% surcharge

In most cases, commissioning should be planned for Tuesday so that the commissioning engineers can arrive on Monday. Travelling at the weekend is only permitted in exceptional cases.

The estimation of the above mentioned delegation(s) is based on having a complete building (incl. roof fully enclosed, free from waste materials, cleaned, free from construction equipment and works) and the necessary infrastructure (i.e. access roads, storage area, temp. power, etc.), tooling, local labor as well as locally manufactured items available in adequate quality and quantity.

Moreover, we refer to our General Terms and Conditions of Service, which are an integral part of this document.

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Price Summary

Item	Qty.	Description	Price	
Delivery by Bühler				
I		MECHANICAL SUPPLY		
A		DUMPING, CLEANING AND STORAGE SECTION		
A-1		DUMPING SECTION	EUR	95'578,-
2	1	DUMPING STATION		
3	1	TROUGH CHAIN CONVEYOR AHKG 100		
4	1	GRAVITY SPOUTING		
A-3		CLEANING SECTION	EUR	120'028,-
8	1	ELEVATOR LBEB 360/170		
9	1	SEPARATOR CLASSIFIER MTRC-100/100		
10	1	ASPIRATION CHANNEL MVSL-100		
11	1	DRUM MAGNET DFRT-200/500		
12	1	DESTONER MTSD-65/120		
13	1	LOW PRESSURE RADIAL-FAN MHTF		
14	1	GRAVITY SPOUTING		
A-4		COCOA BEANS STORAGE	EUR	250'940,-
15	1	ELEVATOR LBEB 360/170		
16	1	TROUGH CHAIN CONVEYOR AHKG 100		
17	2	GRAVITY FLOW DIVERTER		
19	4	HIGH-LEVEL INDICATOR		
20	4	SET OF LOAD CELLS FOR COCOA BEAN SILO 25 m³		
21	4	UNIVERSAL CONTROL UNIT MEAG		
22	4	SLIDE GATE (MAN)		
23	4	TUBULAR SCREW CONVEYOR AHSG-200		
24	1	TROUGH CHAIN CONVEYOR AHKG 100		
25	1	GRAVITY SPOUTING		
26	1	PRESSURE SWITCH FOR COMPLETE PROCESS LINE		

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A-5	ASPIRATION SECTION	EUR	50'336,-
28	1 PRESSURE SWITCH FOR ASPIRATION DUCTING		
29	1 CYCLONE SEPARATOR MGXG-175		
30	1 HIGH-LEVEL INDICATOR		
31	1 LOW PRESSURE RADIAL-FAN MHTF		
32	1 DISCHARGE AIRLOCK GZS		
B	DEBACTERIZATION, ROASTING, WINNOWING, SHELL AND NIBS HANDLING SYSTEM		
B-1	DEBACTERIZING AND ROASTING	EUR	1'842'947,-
41	1 ELEVATOR LBEB 360/170		
42	1 TUBULAR SCREW CONVEYOR AHSG-200		
44	1 HIGH-LEVEL INDICATOR		
45	1 LOW-LEVEL INDICATOR		
46	1 SLIDE GATE (MAN)		
47	1 DEBACTERISATION SYSTEM SLSA 2		
47.1	1 CONTROL SYSTEM SLSA 2 - Siemens TIA		
51	1 CONTINUOUS ROASTER RoaStar™ SCRO 2000		
55	1 TROUGH SCREW CONVEYOR NFAT-200		
56	1 GRAVITY SPOUTING		
B-2	WINNOWING	EUR	701'956,-
57	1 WINNOWING SYSTEM WINNOWER SCWI		
58	1 GRAVITY SPOUTING		

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B-3	SHELL HANDLING SYSTEM	EUR	118'248,-
59	1 TUBULAR SCREW CONVEYOR AHSG-100		
60	1 BLOWER		
61	1 PRESSURE PNEUMATIC CONVEYING LINE		
62	1 IMPACT CRUSHER BIC 638		
63	1 BLOW THROUGH AIRLOCK MPSH-22/22		
64	1 HIGH PRESSURE-SMALL FILTER MVRW-6/8		
65	1 SMALL CENTRIFUGAL FAN		
68	1 HIGH-LEVEL INDICATOR		
69	1 VIBRO-DISCHARGER MFVH-180 M.S.		
70	1 SILO BIN DISCHARGE SLIDE GATE RVB		
71	1 LOW-LEVEL INDICATOR		
72	1 TUBULAR SCREW CONVEYOR AHSG-315		
73	1 GRAVITY SPOUTING		
B-4	PNEUMATIC TRANSPORT FOR NIBS	EUR	50'877,-
74	1 ADDITIONAL FILTER FOR BLOWER		
75	1 BLOWER		
76	1 PRESSURE PNEUMATIC CONVEYING LINE		
77	1 BLOW THROUGH AIRLOCK MPSS-25/23 QC2 / TK1		
78	1 HIGH PRESSURE-SMALL FILTER MVRW-6/8		
79	1 SMALL CENTRIFUGAL FAN		
B-5	ASPIRATION SECTION	EUR	28'746,-
82	1 PRESSURE SWITCH FOR ASPIRATION DUCTING		
83	1 CYCLONE SEPARATOR MGXG-150		
84	1 HIGH-LEVEL INDICATOR		
85	1 LOW PRESSURE RADIAL-FAN MHTF		
86	1 AIRLOCK MPSN-25/23		

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C	COCOA NIBS GRINDING		
C-1	COCOA NIBS GRINDING	EUR	1'246'150,-
89	1 HIGH-LEVEL INDICATOR		
90	1 SET OF LOAD CELLS FOR NIBS BUFFER SILO 4 m³		
91	1 UNIVERSAL CONTROL UNIT MEAG		
92	2 KNOCKING HAMMER		
93	1 TUBULAR SCREW CONVEYOR AHSG-160		
94	1 METAL SEPARATOR		
95	1 GRAVITY SPOUTING		
96	1 PreGrind 2000		
108	1 TANK 1000 l		
109	1 HIGH-LEVEL INDICATOR		
110	1 LEVEL SENSOR Pressure gauge converter in tanks		
111	1 BALL MILL NOVA™ S 1000		
112	1 ROUND SIEVE FOR 2 T/H COCOA LIQUOR		
113	1 HIGH-LEVEL INDICATOR		
114	1 INLET HOPPER 15 l FOR PUMP		
115	1 HIGH-LEVEL INDICATOR		
116	1 CONVEYING AND METERING PUMPS		
117	2 MAGNET 8000 GAUSS		
118	1 FITTINGS		
D	COCOA LIQUOR STORAGE, TEMPERING AND BLOCKING-OFF		
D-1	COCOA LIQUOR STORAGE	EUR	344'730,-
120	3 TANK 20000 l		
121	3 HIGH-LEVEL INDICATOR		
122	3 LEVEL SENSOR Pressure gauge converter in tanks		
125	1 CONVEYING AND METERING PUMPS		
126	1 FITTINGS		
E	MECHANICAL INSTALLATION MATERIAL		
E-1	MECHANICAL INSTALLATION MATERIAL	EUR	50'791,-
140	1 CONSUMABLES & PRIMARY SUPPORTS		

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III		ELECTRICAL EQUIPMENT		
H-1		FUNCTIONS OF THE PLANT CONTROL SYSTEM	EUR	366'725,-
143	1	AUTOMATION ENGINEERING		
H-2		HARDWARE FOR OPERATION AND VISUALISATION	EUR	36'404,-
144	1	HARDWARE AND LICENCES		
H-3		HARDWARE FOR THE CONTROL SYSTEM	EUR	116'826,-
145	1	EQUIPMENT CONTROL- POWER- SECTION		
V		SERVICES		
J		SERVICES		
J-1		ENGINEERING	EUR	808'397,-
149	1	ENGINEERING SERVICES & DOCUMENTATION		
J-2		SUPERVISION OF INSTALLATION	EUR	133'722,-
151	1	INSTALLATION SUPERVISION		
J-3		COMMISSIONING	EUR	262'090,-
155	1	COMMISSIONING		
TOTAL PRICE			EUR	6'625'491,-
For delivery of item(s) 2 - 4, 8 - 17, 19 - 26, 28 - 32, 41 - 42, 44 - 47.1, 51, 55 - 65, 68 - 79, 82 - 86, 89 - 96, 108 - 118, 120 - 122, 125 - 126, 140, 143 - 145, 149, 151, 155				
FOB Named port of boarding according to Incoterms 2020				
CIF - Costs *			EUR	360'908,-
TOTAL PRICE			EUR	6'986'399,-
For delivery of item(s) 2 - 4, 8 - 17, 19 - 26, 28 - 32, 41 - 42, 44 - 47.1, 51, 55 - 65, 68 - 79, 82 - 86, 89 - 96, 108 - 118, 120 - 122, 125 - 126, 140, 143 - 145, 149, 151, 155				
CIF Seaport Indonesia according to Incoterms 2020				

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“Insurance cover is acceptable on basis CIF Indonesian Port of Discharge / CIP Indonesian Airport of Destination only but without cover of any post-carriage somewhere to any site/city in the Indonesian midland.

The scope of cover must not include warehouse-warehouse clauses. The insurance ends at the Indonesian Port of Discharge/Airport of Destination. Customers have to arrange for insurance coverage from Port of Discharge/Airport of Destination up to their warehouse with their local Indonesian insurance companies.”

*Today's supply markets are very uncertain. Prices and availability of many goods and materials are fluctuating in previously unseen manner. Unfortunately, the situation is not only expected to last but most likely get even more challenging. Therefore, Bühler would like to advise that prices will be increased after the expiry of this quote. The following escalation formula is therefore included in this quote to reflect the impact from the market beyond the validity: **Final price = Price shown in this quote x (1 + 0.6% x n)**; n = number of months between expiry and contract getting into force. In case the expiry date is more than 6 months old, the quotations automatically becomes invalid.*

***Actual costs + service charge**

The quoted freight costs are an indicative estimation only and might fluctuate during the execution of order. Bühler will calculate the actual freight costs plus a service charge of 20% after shipment. For the avoidance of doubt, Bühler explicitly reserves the right to adjust the Purchase Price in accordance with the actual freight costs at the time of shipment. Additional costs (if any) will be invoiced separately.

***Shipment and service disruptions**

Given the current circumstances in the Red Sea, Gulf of Aden and Indian Ocean resulting in schedule changes, lack of empty containers, capacity shortages and the recent, partial cancellation of war and political risk coverage by our insurer, unexpected delays in transporting goods may occur. Bühler has no influence on the geopolitical situation and therefore disclaims any liability for such delays. Your patience and understanding are greatly appreciated during this time.

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Options

The prices to be understood CIF Seaport Indonesia according to Incoterms 2020

5	1	ELEVATOR LBEB 360/170	EUR	34'474,-.
6	1	WIND SIFTER V 500	EUR	47'577,-.
7	1	GRAVITY SPOUTING	EUR	3'205,-.
35	1	PRESSURE SWITCH FOR ASPIRATION DUCTING	EUR	976,-.
36	1	CYCLONE SEPARATOR MGXG-95	EUR	10'743,-.
37	1	HIGH-LEVEL INDICATOR	EUR	860,-.
38	1	LOW PRESSURE RADIAL-FAN MHTF	EUR	4'831,-.
39	1	DISCHARGE AIRLOCK GZS	EUR	14'861,-.
98	1	TANK 1000 I	EUR	40'409,-.
99	1	HIGH-LEVEL INDICATOR	EUR	864,-.
100	1	LEVEL SENSOR Pressure gauge converter in tanks	EUR	2'133,-.
101	1	SET OF LOAD CELLS FOR SLURRY TANK 1000 I	EUR	15'058,-.
102	1	UNIVERSAL CONTROL UNIT MEAG	EUR	5'404,-.
104	1	SLURRY PUMP 2 T/H	EUR	26'632,-.
105	1	SLURRY PUMP 40 - 150 KG/H	EUR	20'259,-.
106	1	FITTINGS	EUR	2'645,-.
123	3	SET OF LOAD CELLS FOR COCOA LIQUOR TANK 20000 I	EUR	53'852,-.
124	3	UNIVERSAL CONTROL UNIT MEAG	EUR	16'212,-.
128	1	TANK 5000 I	EUR	71'003,-.
129	1	HIGH-LEVEL INDICATOR	EUR	864,-.
130	1	LEVEL SENSOR Pressure gauge converter in tanks	EUR	2'133,-.
131	1	SET OF LOAD CELLS FOR COCOA LIQUOR TANK 5000 I	EUR	15'058,-.
132	1	UNIVERSAL CONTROL UNIT MEAG	EUR	5'404,-.
133	1	CONVEYING AND METERING PUMPS	EUR	12'700,-.
134	1	TEMPERING MACHINE FOR COCOA LIQUOR	EUR	817'072,-.
135	1	SAMPLER FOR COCOA LIQUOR	EUR	15'964,-.
136	1	FILLING STATION	EUR	261'757,-.
139	1	FITTINGS	EUR	9'109,-.

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141	1	SPARE PARTS	EUR	545'088,-.
146	1	DATA BACKUP SYSTEM	EUR	1'357,-.
147	1	PRINTER	EUR	1'049,-.
149.1	1	FAT	EUR	5'835,-.
150	1	ENGINEERING SERVICES & DOCUMENTATION	EUR	16'857,-.
152	1	INSTALLATION SUPERVISION	EUR	20'186,-.
156	1	COMMISSIONING	EUR	25'910,-.
157	1	TRAINING ON CONSTRUCTION SITE	EUR	53'784,-.

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Customer supply

1	1	BIG-BAG DISCHARGE STATION	.
18	4	COCOA BEAN SILO WITH CASCADE 25 m ³	.
27	1	ASPIRATION DUCTING FOR DUMPING, CLEANING AND STORAGE SECTION	.
33	1	BAGGING TABLE	.
34	1	ASPIRATION DUCTING FOR WINDSIFTER	.
40	1	BAGGING TABLE	.
43	1	INLET HOPPER 1 T	.
48	1	STEAM AND CONDENSATION PIPEWORKS DEBACTERIZATION	.
49	1	ASPIRATION DUCTING FOR DEBACTERIZATION SYSTEM	.
50	1	INSULATION STEAM PIPEWORK DEBACTERIZATION	.
52	1	STEAM AND CONDENSATION PIPEWORKS ROASTING	.
53	1	ASPIRATION DUCTING ROASTING	.
54	1	INSULATION ASPIRATION DUCTING ROASTING	.
66	1	ASPIRATION DUCTING FOR COCOA SHELL SILO	.
67	1	COCOA SHELL SILO 75 m ³	.
80	1	ASPIRATION DUCTING FOR NIBS BUFFER SILO 4 m ³	.
81	1	ASPIRATION DUCTING FOR DEBACTERIZATION, ROASTING AND WINNOWING	.
87	1	BAGGING TABLE	.
88	1	NIBS BUFFER SILO 4 m ³ (approx. 2 t)	.
97	1	ASPIRATION DUCTING FOR COCOA NIBS GRINDING	.
103	1	DOUBLE WALL PRODUCT PIPE FOR SLURRY ADDITION	.
107	1	DOUBLE WALL PRODUCT PIPES FOR NIBS GRINDING	.
119	1	DOUBLE WALL PRODUCT PIPES FOR COCOA LIQUOR STORAGE	.
127	1	DOUBLE WALL PRODUCT PIPES FOR COCOA LIQUOR TEMPERING & BLOCKING - OFF	.
137	1	CONVEYOR WITH BOX CLOSING SYSTEM	.

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138	1	CONVEYOR WITH METAL DETECTOR	.
142	1	STEEL STRUCTURE AND PLATFORMS	.
148	1	ELECTRICAL INSTALLATION	.
153	1	SUPERVISION OF INSTALLATION FOR ASPIRATION DUCTINGS	.
154	1	SUPERVISION OF INSTALLATION FOR DOUBLE WALL PRODUCT PIPES	.

Seamless range of services for our partners. Trust in the service of the market leader.



Bühler offers a seamless range of comprehensive customer services to ensure optimal uptime and performance of your equipment throughout its lifecycle.

Lab Services. With its expertise and laboratory facilities, Bühler offers you a broad range of physical and chemical analyses in order to innovate processes and improve equipment.

Consulting. Strategy, plant performance, or energy consulting are just some of the consulting services that will improve your product quality, production processes and energy efficiency.

Training. At Bühler training centers – or at any site worldwide– specially trained experts pass on their hands-on expertise to customers' employees. Additionally, the newest e-learning platform "ChocoGenius" allows online video-based trainings and troubleshooting for operators.

Maintenance. Adjustment of production cycles to prevent downtime, loss in efficiency or product quality, ranging from individual services to complete outsourcing.

Spare and Wear Parts. Highest standards of reliability apply to original Bühler parts. They are perfectly adapted and ensure performance and production safety.

Repairs. Dedicated to minimizing downtime in the event of an incident: Fast and reliable technical repair on site or service via the Bühler hotline.

Revision. Bühler evaluates, overhauls, adjusts or renews customer installations, including Bühler and third-party machines.

Retrofits. Individual upgrades and conversion kits enable time-worn Bühler machines to be restored to current technology and efficiency standards.

Technical Specifications

Any IEC Ex, EAC Ex, A-Ex (e.g: Class 2 div II, Class II Zone 22), CCCEx or NFPA 70, NFPA 79 execution is excluded.

The machines and control elements may not be operated in an ATEX-Zone. For reasons of standardization, individual devices may be executed in 3D equipment category.

For all machine parts, unless otherwise specified, the following main voltage is valid:

Type of low voltage net	TN-C System
3-phase voltage	400 VAC
1-phase voltage	230 VAC (internal control voltage with separate transformer)
Frequency	50 Hz
Control voltage	24 VDC
Maximum voltage fluctuation	+/- 10 %
Maximum frequency fluctuation	+/- 1 %

Low-voltage switchgear and control gear assemblies

Interface specification according EN 61439 Black-Box-Concept with following switchgear combination (SC) Black-Box Parameter:

Max. allowed short circuit current	
SC plant control	50 kA Ipk on the SC-input terminals, fed by LTD (NSHV)
SC machine control	25 kA Ipk on the SC-input terminals, fed by LTD (NSHV)
Pollution degree for SC:	
Closed control cabinets	3
Operating equipment	2
Design according to EN 61439-2	2
Installation place	Indoors
RDF InA according to EN 61439-1	1
InA max.	630A
EMC classification	Environment A
Incoming unit at SC	Copper
Ambient air temperature limits	Standard EN 60204-1
Maximum SC inside temperature	50°C without cooling system; 40°C with active cooling system
Air humidity conditions	Standard EN 60204-1
Maximum altitude above sea level	Standard EN 60204-1
IP protection for enclosed SC	IP 54 (Indoors IP2X, parts to operate finger safe)
SC request on staff	instructed Persons
Conductors and wiring colors	Standard EN 60204-1
Outgoing unit for drives	Connections takes place directly on the output terminals of the applied operating equipment
Cable entries	
SC of plant control	below
SC of a machine control	below and above

Standard surface treatment of main machines

Due to differences in manufacturing conditions and treatment processes, it may be that nuances are created in terms of color and surface characteristics, for which no complaints can be accepted. The same applies in the general sense to the surfaces of other, non-coated materials such as stainless steel, plastic, etc.

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Exclusions

The scope of supply and services by Bühler is determined and limited by the description of this quotation.

All other required equipment and services not included in this quotation are excluded, such as:

Freight Dues

- Transport to site if not specified by Incoterms
- Insurances, if not specified by Incoterms
- Customs duties, value added tax and all other taxes
- Unloading, storage and transportation into the building

Land and Building

- Property, Soil investigation.
- Earth and landscaping work of any kind, e.g. excavations, trenches, roads, yards etc.
- Piling.
- Any architectural, civil and/or structural work for buildings canopies, foundations, bases and break throughs.
- Foundations, plinths, openings.
- Building connection elements, building renovations and similar.
- Covering of floor/wall- openings, floor and building reinforcements.
- Connecting catwalks and operators platforms.
- Explosion prevention devices, fire protection and fire fighting equipment e.g. fire extinguishers.
- Sprinkler systems.
- Air conditioners.
- Lightning protection grounding.
- Phone/Intercom/video systems
- Emergency lighting system
- Foundation bolting to the floor. Fixing to ceilings, if necessary.
- Separating walls between warm area/cold area. (air conditioned warm area, temperature: 24° C – 26° C, relative air humidity: 50% +/-5%, air conditioned cold area, temperature 18° C – 20° C, relative air humidity: 50% +/-5%)

Auxiliary and Various Equipment

- Labelling of all product conveying lines, aspiration ducting and gravity spouts in respect of transported medium and flow directions.
- Passenger lift, freight elevator, silo winch and hoisting equipment.
- Maintenance workshop with furnishings and equipment e.g. tools, machines, hoists, devices, consumable goods.
- All chocolate tempering machines and systems unless noted in the quotation
- Laboratory with furnishings, testing equipment with accessories.
- Pumps, piping, valves and spouting for product unless noted elsewhere.
- Air conditioning for pneumatic conveying.
- Aspiration ducting unless noted elsewhere.
- Insulation material for heat, cold, sound, weather etc.
- Repairing and supply of replacement parts for re-used machines and accessory parts.
- Spare and wear parts

Utilities, Energy Supply and Distribution

- Fresh, hot/cold industrial and waste water components and systems i.e. preparation, pumps, piping, insulation and aggregates, etc. as well as compressed air and steam supply utilities.
- Entire energy supplies, i.e. electricity, water, compressed air, steam, gas etc. and its piping.
- Operating materials (lubricants and other)

Conformity Licences and ATEX

- Attendance and support for obtaining licenses, necessary assessments from the officials as well as complying with obligations requested from the experts.
- CE Declaration for equipment which is not delivered by Bühler
- Explosion prevention and fire protection measures and equipment unless noted elsewhere.

Electrical process installation

- Any electrical process installation including electrical engineering, installation supervision, material- and cable-lists, material and related installation work. Any cabling from/to machines and from/to control cabinets.

Power Supply

- Any power supply systems including electrical engineering, installation supervision, voltage generator-, distribution- and compensation-systems, installation materials and related installation work.

Earthing System

- Any grounding systems for buildings, machinery and lightning protection including electrical engineering, installation supervision, installation materials and related installation work

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Infrastructure

- Any equipment or systems for infrastructure including communication-, building lighting, power factor correction, fire protection- and monitoring-systems as well as electrical planning, installation supervision, installation materials and related installation work.

Plant Control System

- If plant control system is delivered by 3rd Party: Preparing and inspection of control system descriptions. Interface definition to third suppliers.
- Remodelling of the existing control panel and/or necessary elements.
- Wire numbering (wire labeling is included = connection-number on both sides of the wire)
- Additional measurements or filters to prevent possible annoyances caused by harmonious oscillations from the installation
- Software descriptions
- Flow or logic diagrams
- Any kind of source programs, source codes, etc.

Installation, Commissioning and Training

- Necessary temporary residence and working permits for expatriates, supervisors and administrators.
- Office/storage containers on site (lockable)
- Site security (24/7)
- Temporary power panels for mechanical- and electrical installation
- Welding gas and electrodes for the complete installation etc.
- Provision of scaffolding of any kind necessary for installation of equipment, as specified by Bühler.
- Cost of any Inspection (Pre-Shipment Inspection and/or Destination Inspection and/or Random Inspection etc. by a designated Inspection Company or any third party Company).
- Organizing a safety coordinator.
- Calibration acceptance incl. local calibration costs and calibration weights.
- Supervision of installation, setting to work and commissioning with product unless noted elsewhere.
- Product to cleanout the system
- Crane and lifting devices
- Installation of equipment
- Demounting and disposal of existing equipment as well as packaging materials
- Cleaning works of plant and equipment
- Operator and staff training unless noted elsewhere.

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Engineering Services & Documentation

- Any automation engineering related to the plant control system, such as hardware design, software development for PLC's/PC's or visualization software unless noted elsewhere.
- Any electrical installation engineering drawing(s) with corresponding bill of materials unless noted elsewhere.
- Any electrical infrastructure engineering (everything upstream of MCP'S and MCC's) unless noted elsewhere.
- Any infrastructure engineering related to compressed air, steam, cold, warm or hot water design other than an energy requirement list.
- Any fabrication drawings outside of our scope of supply for bins, tanks, hoppers, and steel structures as a basis for local manufacturing unless noted elsewhere.
- Isometric views of pipes for liquid components unless within our scope of supply.
- "As-built" installation drawings
- Any civil, structural and/or architectural engineering and/or drawings.
- Any professional engineering fees for stamping and approval of drawings or other information of any kind.
- Costs for the coordination with third party suppliers that are not contracted with Bühler
- Translation of documentation into languages other than mentioned.

Operating Supplies

- All necessary moulds.
- All mould cavities have to be volumetrically laid out for different specific filling gravities and properly designed to allow for easy demoulding.
- Cooling media (i.e. Pecosol, glycol etc.)
- Grease and oil

Platforms and Supports

- All platforms, railings, bridges, ladders, fixing, support and suspending constructions as well as chimney holes unless noted elsewhere

Local Regulations Fees

- Local union rules and factory acts for installations, buildings, structures and premises.
- All costs for necessary local/regional acceptance, test/permission for governmental concerns.

Financing

- Any banking costs in regard to the L/C, as well as any financing costs

General

- i.e. all parts and equipment not specifically mentioned in our quotation.

Besides exclusions listed, please refer to the content of the individual quotation items for any additional exclusions which may be mentioned thereunder.

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Plant control

- Mobile control devices (tablets, mobile phones etc.)
- Identification systems (Barcode, Scanner or RFID)
- Hard / software supplies for management systems (e.g. ERP interface, Formulation system interface, etc.).
- Hard- & software supply for network security i.e. Firewall, Virus protection etc.
- Wireless local area networks (design, administration, security, routers, cabling, etc.)
- Internet connection (Provider) for service and support
- Remodelling of the existing control panel and/or necessary elements.
- Emergency Control application in case of fire
- Wire labelling
- Spare Parts
- Power metering equipment for control and motor control panel
- Motor control panels of draw-out type
- Other than to European machine directive 2006/42/EC, low voltage directive 2014/35/EC and EMC directive 2014/30/EC harmonized standards.
- Responsibility for any influences and possible annoyances caused by harmonic oscillations from the Bühler installation.

Power supply

- Total power supply

Earthing system

- Total earthing system

Process installation

- Terminal boxes.
- Service switches/Inspection switch (0-Manual-Remote) next to motor, excepted for elevators, chain conveyers, scales and flow balancer.
- Other than to European machine directive 2006/42/EC, low voltage directive 2014/35/EC and EMC directive 2014/30/EC harmonized standards.

Infrastructure

- Total infrastructure

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Commercial Conditions

Bühler is continuously seeking to develop new products. For this reason, a product offered in this quotation may be replaced by a new one prior to delivery. In such a case, Bühler reserves the right to supply a new product instead.

Enclosed pamphlets of sub-suppliers are for general information only. The actual sources of supplies will depend on the purchase conditions at the date of order.

Terms of Payment

- 30% of the Contract Price within 14 calendar days from receipt of the Order Confirmation by the Customer/effective date of Contract to be paid to the bank account of Bühler AG as notified to the Customer against presentation of the respective invoice.
- 70% of the Contract Price (pro rata) delivery to be paid to the bank account of Bühler AG against presentation of the usual transport document and the respective invoice by an irrevocable letter of credit (divisible, allowing part payments/instalments) which Customer undertakes to have issued by a first class bank acceptable to Bühler AG at Customer's cost and expense in the amount of 70% of the Contract Price under the Uniform Customs and Practices for Documentary Credits of the International Chamber of Commerce (UCP latest version), within 14 days from receipt of the Order Confirmation by the Customer/effective date of Contract in favor of Bühler AG and confirmed by a German bank.

The price of this offer is based on the spot rate EUR to CHF as published in the internet by UBS – Public Version UBS Quotes: <https://www.ubs.com/global/en/quotes.html> as at 10.10.2025.

Bühler solely reserves the right to adjust its offer and/or contract value should the spot rate on the date the contract enters into force and/or the signing of the contract and/or of receipt of your firm order vary by more than 3% from the spot rate as at 10.10.2025, as mentioned here above.

No re-export to Russia / Belarus

(1) The Customer shall not sell, export or re-export, directly or indirectly, to the Russian Federation and/or to Belarus or for use in the Russian Federation and/or in Belarus any goods supplied under or in connection with this Agreement that fall under the scope of Article 14f of the Swiss Ordinance imposing Measures in Connection with the Situation in Ukraine (SR 946.231.176.72) and/or Article 12g of Council Regulation (EU) No 833/2014, and/or Article 10 of the Swiss Ordinance imposing Measures against Belarus (SR 946.231.116.9) and/or Article 8g of Council Regulation (EU) No 765/2006.

(2) The Customer shall undertake its best efforts to ensure that the purpose of paragraph (1) is not frustrated by any third parties further down the commercial chain, including by possible resellers.

(3) The Customer shall set up and maintain an adequate monitoring mechanism to detect conduct by any third parties further down the commercial chain, including by possible resellers, that would frustrate the purpose of paragraph (1).

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(4) Any violation of paragraphs (1), (2) or (3) shall constitute a material breach of an essential element of this Agreement, and Bühler shall be entitled to seek appropriate remedies, including, but not limited to termination of this Agreement.

(5) The Customer shall immediately inform Bühler about any problems in applying paragraphs (1), (2) or (3), including any relevant activities by third parties that could frustrate the purpose of paragraph (1). The Customer shall make available to Bühler information concerning compliance with the obligations under paragraph (1), (2) and (3) within two weeks of the simple request of such information.

Time of Delivery

Approx. 15-17 months ex works after receipt of written order and clarification of all technical issues and receipt of contractual payments and/or Letters of Credit as per above terms of payment.

This quotation is valid until 9th November 2025 for material of our own manufacture. For material from outside suppliers, we can maintain the quoted prices only if these suppliers do not increase their prices in the meantime.

This quotation is subject to confirmation in case of order.

Bühler will undertake its commercial reasonable efforts to deliver the machines / equipment in accordance with the specified dates. The Customer, however, is aware, acknowledges and agrees that currently availability and/or delivery time of particular 3rd party components embedded into the machines / equipment (such as but not limited to electronical components) is critical and might be impacted by factors beyond Bühler's influence. Bühler will timely inform the Customer if it becomes aware of possible difficulties in this respect and make commercially reasonable efforts to mitigate the consequences. In this case, Bühler will inform the customer when the adjusted delivery times will take place. Any hereof resulting delay in the delivery of the machines / equipment (either wholly or partially) shall be understood as an excused delay for which Bühler is not responsible. The delivery times are to be understood as non-binding, but shall be observed by the Parties to the best of their knowledge and belief. In such case, the Customer cannot claim any damages and/or penalties from Bühler and the delivery dates shall be adjusted accordingly.

Origin of the goods

Upon request Bühler will inform you about the continent where the equipment and/or spare parts offered in this quotation are manufactured or sourced by Bühler. However, Bühler will not provide you with an itemized declaration of the country of origin. Any warranties, representations and liability of Bühler with respect to the country of origin of the offered equipment and/or spare parts are expressly waived.

General

All required equipment and services for this particular job, but not included in the scope of supply, have to be free issued by the client or others.

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Safety concept

Control systems supplied by Bühler are an integral part of the safety concept designed to prevent accidents in plants with our supplied equipment. If control systems are in use from other sources without our involvement, Bühler will reject any liability for occurring faults and consequential damages and costs.

Delay caused by the purchaser

The stipulated dates and deadlines depend on the fulfilment of the contractual commitments of the purchaser. The dates and deadlines are accordingly extended, if payment terms and any other contractual terms are not fulfilled on time by the purchaser, this refers particularly to delivery terms, start of installation and commissioning date.

All costs incurred due to any delay by the purchaser's, will be charged in full . This applies especially to:

- Delivery delays and / or storing of equipment: if the purchaser payment commitments are not adhered to, or if due to lack of information to the carrier storage facilities are incomplete or unsuitable and therefore Bühler is not able to make shipments;
- Installation delays or additional time required: if the building or infrastructure is not ready as planned;
- Installation delays or additional time required: if the purchaser does not comply with the contractual commitments in regard to tool availability, staff resources or any other logistic requirements;
- Changes in provision of installation and/or commissioning personnel: if this information is only provided less than 15 days of the delegation's departure and therefore delays or cancellation of the experts travelling is necessary or if installation and/or commissioning procedures are constantly interrupted.

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We refer you to our General Terms and Conditions of Sale, our General Terms and Conditions of Service and our Digital Solution Terms and Conditions of Bühler attached, forming an integral part of this Confirmation of Order.

Uzwil, 10th October 2025

Bühler AG

A handwritten signature in blue ink, appearing to read 'N. Gutsell'.

Niklaus Gutsell
Head of Sales
Chocolate Masses

A handwritten signature in blue ink, appearing to read 'J. Essig'.

Joachim Essig
Head of Sales
Cocoa

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General Terms and Conditions of Sale

1. Entry into force, Application and Modifications

- i The present General Terms and Conditions of Sale („GTC“) shall apply to the agreement made between the customer (“Customer“) and Bühler (“Agreement“) as far as the parties have made no other written agreements in individual cases, irrespective of whether an order is placed in one of Bühler's online-shops or by an offline transaction. Any Customer's conditions shall not be binding, except if expressly accepted by Bühler in writing.
- ii The Agreement shall only be effective upon Bühler's written confirmation of Customer's order (“Order Confirmation“) and shall consist of the Order Confirmation, the present GTC as well as any Annexes attached thereto.
- iii Any addition to or modifications of any provision of the present GTC, shall only be binding upon the parties if executed in writing and duly signed by both parties.

2. Scope of Supply

- i Machines, components, spare parts or other equipment (together “Products“) under the Agreement are exhaustively specified in the Order Confirmation, whereby Bühler is entitled to unilaterally make changes to improve the Products provided such changes do not result in a price increases for the Customer.
- ii For the avoidance of doubt as far as not differently agreed in the Agreement installation and commissioning of the Products is in the sole responsibility of the Customer. If Customer requests supporting services from Bühler a separate agreement shall be concluded.

3. Drawings technical documentation and Software

- i Brochures and catalogues, including but not limited to presentations in the online shops, are not binding. Any information contained in drawings and technical documents are only binding upon Bühler if the Order Confirmation explicitly so provides.
- ii Bühler reserves all rights to the drawings and technical documents delivered by Bühler. They shall not be made available to third parties nor used for purposes other than those for which they were provided. If no order is placed, all drawings and technical documents which have already been handed over are to be returned to Bühler immediately.
- iii As far as software is included in the Products, all rights to that software remain the property of Bühler. Bühler grants to the Customer a non-exclusive right of use of the software together with the Products for the contractually agreed purpose, whereby the right to sublicense shall be excluded. In particular, the Customer may not, subject to mandatory applicable law, disassemble, decompile, decrypt, reverse engineer the software or make the software available to third parties without the prior written consent of Bühler. In case of infringement, Bühler may withdraw the right of use. For third-party software, the conditions of use of the third-party licensor apply, and the third-party licensor, as well as Bühler, may assert a claim in the event of infringement.

4. Prices

- i All prices shall be deemed to be net without any deduction whatsoever. The Incoterm shall be agreed upon in the Order Confirmation. Unless otherwise agreed in writing and signed by both parties, place of performance for all contractual obligations is Uzwil/Switzerland.
- ii Any and all additional charges, such as but not limited to, freight charges, insurance premiums, fees for export, transit, import and other permits, as well as for certifications, shall be borne by the Customer.
- iii Likewise, the Customer shall bear all value-added taxes, sales taxes, profit taxes, income taxes, withholding taxes, social contributions, registration fees, as well as all other kinds of taxes, fees, levies, custom duties and the like which are levied against Bühler, its related companies, its factories or its personnel in connection with the Agreement.

5. Terms of payment

- i The Customer shall make payments in accordance with the terms agreed in the Order Confirmation without any deduction for cash discount, expenses, taxes, levies, fees, duties and the like. The payment obligations of the Customer are fulfilled as soon as the total prices have been credited in full to Bühler's bank account indicated in the respective invoice. Except with prior approval in writing by Bühler the Customer must not retain payments or set-off any payments against counter-claims.

- ii If transport, delivery or taking over is delayed or prevented due to reasons beyond the control of Bühler, or if minor parts are missing or if post-delivery work, which does not prevent the Products from being used, is necessary, Customer shall still observe the agreed dates of payment.

- iii Without prejudice to any other legal right of Bühler, if the Customer, for any reason whatsoever, is overdue with any payment, or if Bühler has valid reasons to expect that the Customer will not make payment on time, Bühler is entitled to suspend the further execution of the Agreement, to stop the fabrication and/or to retain the Products which are ready for dispatch. Bühler is entitled to suspend the further execution of the Agreement until Bühler and the Customer have mutually agreed on new payment and delivery terms and Bühler has received satisfactory securities within a reasonable period of time. If no agreement is found and/or no securities are provided, Bühler is entitled to terminate the Agreement and to claim damages.

- iv With effect from the agreed due date until receipt of the outstanding payment, the Customer shall, without additional reminder, pay interest on the outstanding amount at a rate of 5 % over the 12-months-LIBOR, per annum, for the contractual reference currency valid at the due date plus the agreed late payment interest. In case the relevant LIBOR is negative, a base floor rate of 0.0% shall apply. If the Customer is overdue with a payment or the delivery of an agreed security for more than two weeks, the entire balance of payment shall be due immediately.

6. Retention of title

- i Bühler or its designated representative remains the owner of the Products until the full receipt of the payments in accordance with the terms set forth in the Agreement.
- ii Customer shall cooperate in any measures necessary for the protection of Bühler's title. In particular, to enter or notify the reservation of title in the required form in public registers, books or similar records, all in accordance with the relevant national laws, at the Customer's expense. During the reservation of title, the Customer shall, at his own cost, maintain the Products, insure them for Bühler's benefit against theft, breakdown, fire, water and/or other risks and take all measures to ensure that Bühler's title is in no way compromised or rescinded.

7. Delivery time

- i The delivery time shall start as soon as the Agreement is validly entered into, all official formalities such as, but not limited to, import, export, transit and payment permits have been completed, the agreed payments and securities are given, and the main technical points settled. The delivery time shall be deemed to be observed when by that time Bühler has sent a notice to the Customer that the Products are ready for dispatch. Compliance with the delivery time is conditional upon the Customer's due fulfillment of all of his contractual obligations. Bühler shall explicitly be entitled to partial deliveries.
- ii The delivery time shall be reasonably extended (1) if events of Force Majeure occur; (2) in case of a delay of Customer's contractual obligations, especially if the Customer has delayed the delivery of the necessary documents, the agreed payments and/or securities or the performance of work; (3) Customer subsequently wishes to modify the Products. If delivery is delayed for reasons attributable to Bühler, Bühler shall be granted a reasonable grace period to effect delivery.
- iii Customer shall notify Bühler in good time of any special requirements regarding forwarding, transport and insurance. The benefit and the risk of the Products shall pass to the Customer in accordance with the Incoterms 2020 agreed in the Agreement. If dispatch is delayed for reasons beyond the control of Bühler, the Products are stored and insured at the expense and the risk of the Customer and the risk shall pass to the Customer at the originally foreseen time. As far as not deviating agreed in writing transport shall be executed at the expense and risk of the Customer and Customer is responsible to insure the Products against risks of any kind.

8. Packing

- i Packing will be charged to the Customer and is not returnable.

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9. Examination and Acceptance

- i After delivery Customer shall examine the Products within reasonable time and immediately notify Bühler in writing of any deficiencies. If Customer submits no substantiated written complaint within two weeks for single machines, or within two months for complete plants, calculated from the date of delivery at the place of performance the Products are considered as having been flawlessly accepted. Acceptance is further deemed completed if (1) Customer refuses to participate in an agreed acceptance inspection despite being requested by Bühler to do so; (2) Customer refuses to sign an acceptance certificate without being entitled to do so; (3) Customer productively uses the Products or (4) there are only minor deficiencies that do not jeopardize the contractually agreed operation of the Products.
- ii Subject to a deemed acceptance in accordance with the previous paragraph, the execution of an acceptance is subject to a separate agreement to be concluded between the parties. If the Products fails to pass a performance test due to solely Bühler's fault, Bühler shall be given reasonable opportunity to take remedial steps to pass such performance test.
- iii Bühler shall not later than at the date of delivery, provide information and drawings which are necessary to permit the Customer to install, commission, operate and maintain the Products. Bühler shall not be obliged to provide manufacturing drawings of the Products or of spare parts.

10. Warranty

- i The warranty period is 12 months beginning with Bühler's notification of readiness for dispatch. The warranty performances are subject to Customer's due fulfillment of its payment and examination obligations.
- ii Bühler shall be responsible to repair or replace any parts, which, before the expiry of the warranty period, are proven to be deficient due to bad materials, faulty design or poor workmanship or other circumstances having occurred before the transfer of risk in accordance with the agreed Incoterms. The right to cancel the Agreement or reduce the purchase price shall be excluded. Customer's remedy shall not be deemed to have failed of its essential purpose so long as Bühler is willing and able to repair or replace the non-conforming parts.
- iii If Products are still defective, despite a replacement or repair, Bühler is entitled to take back the defective Products against reimbursement of the received payments. If not explicitly deviating agreed by the parties, replaced parts become Bühler's property. For replaced or repaired parts the warranty period starts anew and lasts 6 months from the replacement or completion of the repair or until the expiry of the original warranty period which ever is later. In no case is the warranty period after replacement or completion longer than double the warranty period stipulated in the preceding paragraph.
- iv Excluded from Bühler's warranty and liability are all deficiencies beyond the control of Bühler, especially if resulting from normal wear and tear, false information from the Customer, improper maintenance, failure to observe the operating instructions, excessive loading, use of any unsuitable material, influence of chemical or electrolytic action, combination with non-original parts, installation, changes and/or repairs by the Customer or third parties, events of Force Majeure, or if Customer does not immediately take all appropriate steps to mitigate the damage and give Bühler the possibility to remedy the defect.
- v Should Bühler explicitly acknowledges an express warranty in the Order Confirmation with regard to performance, energy consumption, or the like, such express warranty shall be only valid if Bühler has the opportunity to prove the warranted parameters are achieved after completed commissioning. The Customer shall at its expense make available the necessary raw materials, energy, etc., as well as qualified personnel for such test. If the express warranties are not or only partially achieved, Customer shall give Bühler the necessary time and possibility to carry out improvements. If Bühler is not given the possibility to prove the warranted parameters within two months from first test, then those parameters are deemed to be proven.
- vi Bühler's liability for adequate advice or breach of any additional obligations is limited to unlawful intent or gross negligence. Unless specifically set forth otherwise in the Agreement, Bühler expressly waives any warranty, representations and liability with respect to the country of origin of the Products.

11. Returns

- i Except for hidden defects, which were not recognizable despite prompt examination, returns are solely accepted if declared within the period set forth in clause 9.i. Hidden defects are solely accepted if a return notification is immediately sent to Bühler after recognition. Bühler is under no obligation to accept returns of flawless Products, returns notified after expiry of the warranty period set forth in clause 10.i, or returns of damaged, contaminated or used Products.
- ii The Customer shall notify returns to Bühler using Bühler's return form. Bühler will thereupon provide the Customer with a return covering note including the return number. After receipt of the return covering note, the Products, including the return covering note, must be sent within 5 working days to Bühler. Any equipment not listed on the return covering note will not be accepted and returned to Customer.
- iii The Customer is solely responsible for the transfer back to Bühler and for any damages, costs and/or expenses, such as but not limited to shipping costs, customs duties or taxes, incurred on Customer and/or Bühler during the return shipping of the Products. Where such costs are directly incurred on Bühler Customers undertakes to indemnify Bühler in full. With the exception of incorrect or defective Products for which Bühler shall bear exclusive responsibility.

12. Exclusion and Limitation of Bühler's Liability

- i Claims for compensation for damages, irrespective of the underlying legal basis and/or ground, out of or in connection with the Agreement, are exhaustively covered by these GTC. Any claim not originating from the scope of the Agreement is expressly precluded.
- ii Irrespective of this, Bühler shall only be liable for typical and foreseeable damage. Liability for damages out of or in relation to reduction of price, termination of or withdrawal from the Agreement, loss of production, loss of use, loss of orders, recall costs, loss of profit and other indirect, direct or consequential damage are expressly excluded.
- iii Furthermore liability out of or in relation to the Agreement is in any case whatsoever limited to the total price paid by the Customer under the Agreement.
- iv Subject to clause 13, liability is also excluded for compensation claims from third parties against the Customer for infringements of intellectual property rights.
- v This exclusion of Bühler's liability does not apply to unlawful intent, gross negligence, negligent harm to life, body, health, or as far as it is otherwise contrary to compulsory law.
- vi The limitation or exclusion of liability extends to Bühler as well as any person employed or appointed by Bühler to perform any contractual obligation under the Agreement.

13. Intellectual Property

- i Bühler shall defend at its expense any action or proceeding initiated by a third party against the Customer to the extent that such action or proceeding is based on a claim that Products infringe a valid and enforceable patent or other intellectual property right of the third party, subject to the following conditions: (i) Bühler is promptly notified by the Customer in writing of the intellectual property infringement claim; (ii) the Products claimed to infringe the third party's right was not modified by the Customer; (iii) the Products are used for the purpose(s) intended by Bühler; (iv) the claim is based upon a valid and enforceable intellectual property right that had been issued as of the date of the delivery under this Agreement; (v) the Customer renders to Bühler whatever information and assistance reasonably required for the defence or settlement of the claim; and (vi) infringement took place within three (3) years from the Products date of delivery. At its own discretion Bühler alternatively has the right to (i) procure for the Customer the right to continue using the Products; (ii) replace the Products with a non-infringing product; (iii) modify the Products so as to avoid infringement; or (iv) remove the Products and refund to the Customer the originally paid price.

14. Compliance

- i In exercising its rights and performing its obligations under this Agreement Customer, its subsidiaries and any of its respective businesses or employees shall comply with all applicable laws, regulations and orders, including but not limited to those related to anti-corruption and export controls (understood as and including but not limited to the transfer of controlled products, technologies or services for the development, production, use or stockpiling of nuclear, chemical, or biological weapons or missiles) imposed by the United States of America (U.S.), European Union (EU), United Kingdom (UK), Switzerland and/or any other countries.

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General Terms and Conditions of Sale

15. Data Protection

- i. The Customer acknowledges and agrees that Bühler may have access to personal data (i.e. information relating to identified or identifiable natural person for example names, functions or contact details) of the Customer's employees, representatives, consultants, agents, contractors and other personnel. Such personal data may be processed by or on behalf of Bühler in accordance with Bühler's privacy policy, available at www.buhlergroup.com/privacy in order to enter into and perform any rights and/or obligation under these GTC as well as related purposes, including but not limited to order and payment processing, tolls and import/export management, customer relationship management, business accounting and general administrative purposes. The parties agree that they act as independent controllers (as defined under applicable law) in relation to personal data processed in accordance with this clause. The Customer further undertakes to inform its personnel of whom Bühler is entitled to process personal data and obtain valid consent (if applicable) thereby especially pointing out Bühler's right to transfer personal data to third parties or third parties abroad while ensuring a comparable data protection level as described in Bühler's privacy policy, in accordance with applicable law.

16. Termination by Bühler

- i. If unforeseen events considerably change the economic effect or the content of the Products or Bühler's activities, or if the performance subsequently becomes impossible, the Agreement shall be adopted appropriately. If such adaption is economically not justifiable, Bühler shall be entitled to terminate the Agreement or the parts affected thereby. If Bühler wishes to terminate the Agreement Bühler shall immediately after recognition of the consequences of the event, inform Customer; this applies even if an extension of the delivery time has been agreed beforehand. In case of termination Bühler shall be entitled to payment of those parts of the Products which have already been carried out. Claims for damages from the Customer because of such termination are excluded.

17. Force Majeure

- i. Any event, which was unforeseeable at the time of conclusion of the Agreement, is beyond either party's control and renders the further performance of contractual obligations impossible or unreasonably onerous, as for instance adverse weather conditions or other acts of nature (including, but not limited to, earthquake, hurricane, landslide, or flood), war (declared or not), acts of terrorism, sabotage, piracy, riot, fire, explosion, epidemics, actions or inactions of governmental or quasi-governmental authorities or shortages of labour, fuel, or raw materials qualifies as a force majeure event ("Force Majeure").

- ii. If a Force Majeure event lasts longer than consecutive 6 months either party shall be entitled to terminate the Agreement at any time by giving ten business days prior written notice. For the avoidance of doubt, in case of a Force Majeure event the Customer shall not be entitled to claim any damages or any other compensation.

18. Severability

- i. If any provision of the Agreement shall be found to be invalid or unenforceable, the invalidity or unenforceability of that provision shall not affect the other provisions of the Agreement. The parties shall substitute any invalid or unenforceable provision with provision that achieves to the greatest extent possible the same legal and commercial objectives.

19. Assignment

- i. Neither party may assign or otherwise transfer any right or obligation under the Agreement without prior written consent of the other party.
- ii. Bühler may, however, transfer and/or delegate the exercise of its rights and/or the performance of its obligations under the Agreement to another affiliated company that it deems fit for the intended purpose.
- iii. All terms, provisions and conditions of the Agreement are binding upon and inure to the benefit of the parties and their respective successors.

20. Jurisdiction/applicable law

- i. The place of jurisdiction shall be exclusively CH-9240 Uzwil. The applicable law shall be exclusively substantive Swiss law to the exclusion of the United Nations Convention on Contracts for the International Sale of Goods (1980).

Bühler AG
CH-9240 Uzwil
Version: 04/2020

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General Terms and Conditions of Service

11. Entry into force, Application and Modifications

- i The present General Terms and Conditions of Service („GTC“) shall apply to the agreement made between the customer (“Customer“) and Bühler (“Agreement“) as far as the parties have made no other written agreements in individual cases, irrespective of whether an order is placed in one of Bühler's online-shops or by an offline transaction. Any Customer's conditions shall not be binding, except if expressly accepted by Bühler in writing.
- ii The Agreement shall only be effective upon Bühler's written confirmation of Customer's order (“Order Confirmation“) and shall consist of the Order Confirmation, the present GTC as well as any Annexes attached thereto.
- iii Any addition to or modifications of any provision of the present GTC, shall only be binding upon the parties if executed in writing and duly signed by both parties.

2. Scope

- i The scope of any service to be provided by Bühler might it be for advisory, supervisory, maintenance, training, consultation or other services are exhaustively specified in these GTC and the Order Confirmation (“Services“). Bühler may unilaterally make changes to improve the Services provided such changes do not result in an increase of the Fee.
- ii For the avoidance of doubt the delivery of any machines, components, spare parts or other equipment is exclusively subject to separate agreement to be concluded between the parties.
- iii Bühler undertakes to provide the Services in the recognized state-of-the-art manner at the time when the Agreement is concluded by qualified personnel or third party subcontractors (“Delegate“) and in accordance with all applicable laws and regulations.
- iv Any further warranty shall be excluded to the extent permitted under applicable law.
- v Bühler may perform the Services at its choice on the premises of the Customer or in its own workshops.

3. Provision of Services

- i The start of the Services shall be subject to the availability of the necessary planned staff, the travel possibilities, the issuance of the necessary visa and permits as well as Customer's due and timely performance of all of its obligations under these GTC.
- ii Bühler will provide a service report to the Customer after completion of the Services, or in time intervals as agreed in the Order Confirmation.
- iii Bühler reserves all rights to the drawings and technical documents delivered in connection with the provision of the Services. The drawings and technical documents shall not be made available to third parties nor used for purposes other than those for which they were provided.
- iv Except if expressly otherwise agreed in writing, the Delegates will solely perform Services assigned by Bühler. If a third party assignment has been approved in writing by Bühler such work shall be at the full responsibility of the Customer and invoiced additionally.

4. Fee

- i Subject to an explicit agreed lump-sum payment the fee for the Services shall be calculated based on time and material spent at the rates set forth in the Order Confirmation (“Fee“).
- ii The Fee shall be deemed to be net without any deduction whatsoever. Any and all additional charges, such as but not limited to, freight charges, insurance premiums, fees for export, transit, import and other permits, as well as for certifications, shall be borne by the Customer.
- iii Likewise, the Customer shall bear all value-added taxes, sales taxes, profit taxes, income taxes, withholding taxes, social contributions, registration fees, as well as all other kinds of taxes, fees, levies, custom duties and the like which are levied against Bühler, its related companies, its factories or its personnel in connection with the Services.
- iv In case of extraordinary conditions (e.g. difficult working conditions, shift work etc), Bühler is entitled to invoice to the Customer appropriate additional compensation to be determined by Bühler on the basis of fair market price.

5. Travel, transport and other Service costs

- i Travelling time as well as a reasonable period for preparation and follow-up time shall also be regarded as working time.

- ii Travelling time is understood to be the entire time the Delegate spends for the journey to and from the place of performance of the Services (e.g. effective travelling, waiting, hand-over/move-in to/move-out of accommodation facilities time including additional costs for indirect travel routes or extraordinary means of transportation to reduce travel and/or waiting risks) and will be invoiced as follows:
 - (a) Rail journeys: a second class railway ticket for delegations within Switzerland; a first class railway ticket for delegations abroad;
 - (b) Car journeys: an appropriate fee on a kilometre basis; and
 - (c) Air travels: an economy class ticket and in exceptional cases (e.g. special circumstances, flying time exceeding eight hours, etc.) a business class ticket.
- iii Bühler invoices to Customer the costs for the transport of luggage, tools and other materials as well as other transportation costs, if any.
- iv Subsistence expenses for the Delegates, including accommodation, board, both as best as possible in conformity with Swiss standard of living and allowances for daily expenses shall be paid additionally by the Customer.
- v Any other extra costs resulting from the delegation such as issuance of visa, vaccinations, pharmaceuticals, special equipment necessary due to climatic conditions, etc. also need to be borne by the Customer.

6. Terms of payment

- i As far as not deviating agreed in the Order Confirmation, the Customer shall make payments without any deduction for cash discounts, expenses, taxes, levies, fees, duties and the like within 30 (thirty) days after the date of the invoice. Customer's payment obligation are fulfilled as soon as the total Fee has been credited in full to Bühler's bank account indicated in the respective invoice. Except with Bühler's prior written approval the Customer must not retain payments or set-off any payments against counter-claims.
- ii The Customer shall, upon Bühler's request, make an advance payment or provide a security (e.g. bank guarantee, letter of credit) in the amount of the estimated costs. In case an advance payment or security is agreed, Bühler is under no obligation to start with the Services until the advance payment or security is received.
- iii Without prejudice to any other legal right of Bühler, if the Customer, for any reason whatsoever, is overdue with any payment, or if Bühler is, due to the circumstances occurring since entering into the Agreement, seriously concerned to not receive payments in total or in due time, Bühler is entitled to suspend the further performance of the Services until new, for Bühler acceptable payment and delivery terms, are agreed and Bühler has received satisfactory securities within a reasonable period of time. If no agreement is found and/or no securities are provided, Bühler is entitled to terminate the Agreement and to claim damages.
- iv With effect from the agreed due date until receipt of the outstanding payment, the Customer shall, without additional reminder, pay interest on the outstanding amount at a rate of 5 % over the 12-months-LIBOR, per annum, for the contractual reference currency valid at the due date plus the agreed late payment interest. In case the relevant LIBOR is negative, a base floor rate of 0.0% shall apply. The foregoing shall be without prejudice to any other right of Bühler described in these GTC. Payment of default interest does not release the Customer from its payment obligation.

7. Working Time

- i Regular working time of the Delegate amounts to 42 (forty-two) hours per week from Monday to Friday between 6.00 a.m. and 11.00 p.m but not more than 10 hours per day. Any Services outside regular working time are subject to Bühler's prior written approval.
- ii Overtime will be charged to the Customer at the rates in accordance with clause 4 and a supplement as defined hereinafter:
 - (a) Supplement of 25% for Services performed within the regular working time but exceeding 42 hours per week or 10 hours per day;
 - (b) Supplement of 50% for Services performed on Saturdays or night work from 11.00 p.m. to 6.00 a.m.; and
 - (c) Supplement of 75% for Services performed on Sundays and public holidays at the place of performance (from midnight to midnight).
- iii In case the Delegate is to work on two consecutive weekends, he shall be allowed to claim a corresponding amount of days off during the following week, for which the normal subsistence expenses shall be paid.
- iv If due to reasons not attributable to Bühler the actual daily working time has to be shorter than the agreed regular working time, Bühler is entitled to charge the full agreed regular working time to the Customer.

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- v If, due to any reason not attributable to Bühler, Services cannot be performed or Delegate is retained on Customers site after termination of the Services, all costs resulting therefrom shall be borne by the Customer. If circumstances are hindering or rendering the performance of the Services impossible at the agreed time, Customer shall immediately inform Bühler in writing and take all reasonable measures to avoid or mitigate any possible damages.
 - vi Both parties are entitled to suspend or cause suspension of the Services in case that events beyond the will and control of the parties, which were unforeseeable at the time of conclusion of the Agreement such as, in particular, fire, explosion, natural catastrophes (e.g. earthquake, flood, etc.), epidemics, mobilization, confiscation, war (declared or not), acts of piracy, riots, embargoes, boycott, lack of means of transportation, etc. render the performance of the Services impossible or unreasonably difficult ("Force Majeure"). Bühler is also entitled to suspend the Services in case one of its subcontractors is affected by Force Majeure. Neither party shall be deemed to be in default of its contractual obligations whilst performance thereof is prevented by Force Majeure and the agreed timetable shall be extended accordingly by a period equal to that during which the Force Majeure contingencies and their effects have occurred. Each party shall fulfill its obligations insofar as they have become due before the occurrence of Force Majeure. In case of Force Majeure lasting for more than 6 (six) consecutive months, each party is entitled to terminate the Agreement by written notice to the other party, whereby the parties shall mutually agree on the allocation of the accrued costs. No liquidated damages or any indemnity shall be claimed by either party in case of Force Majeure. However, the Customer shall pay any part of the Services, which, but for such delay, have already been rendered by Bühler.
- 8. Undertakings of the Customer**
- i As far as the Services are provided on Customer's side the Customer undertakes and is responsible to take all necessary measures to prevent accidents or damages, guarantee safe access to the place where the Services are rendered, while ensuring at all times Bühler's minimum occupational health and safety requirements, which can be found under www.buhlergroup.com. Consequently Customer is fully responsible for all accidents, damages caused or influenced by inadequate safety conditions.
 - ii As far as the Services are not performed on Customer's side the Customer is responsible for the disassembly, transportation and assembly of the products.
 - iii The Customer shall provide, within due time and at its own cost, the necessary assistant staff in the quantity and with the qualifications necessary for the performance of the Services and is responsible to adequately insure them according to the applicable law. The Customer is fully liable for all acts and/or omissions of its assistant staff, or other third persons not engaged by Bühler, even though Bühler might have the overall responsibility for the Services. The Customer's ensures that the assistant staff adheres at all time with Bühler's or Delegate's instructions.
 - iv As far as not deviating agreed in the Order Confirmation the Customer shall provide Bühler in due time and at its costs with the necessary tools and materials necessary for the provision of the Services (e.g. lifting tools, ropes, scaffoldings, workbenches, welding equipment, lubrication, plaster, foundations, power, water, other operating resources).
 - v The Customer shall make available to Bühler suitable storage room for the safe and separate storage of equipment, tools and/or materials.
 - vi As far as not explicitly deviating agreed in the Order Confirmation, the Customer is responsible to procure the needed spare parts in due time and at its cost to enable Bühler to provide the Services.
 - vii Any necessary residence, working and/or other governmental permits (in particular for overtime, night and Sunday work) as well as any required technical documentation shall be organized by the Customer within due time and at its own cost.
 - viii The Customer undertakes to immediately inform Bühler of any exceptional circumstances or special requirements which need to be taken into consideration while providing the Services.
 - ix In case in accordance with local applicable law additional insurance for illness and/or accident for the Delegate compared to the requirement under Swiss law is necessary the Customer undertakes to inform Bühler in advance in writing.
- 9. Customer's delay**
- i If the Customer fails to duly perform any of its obligation under these GTC in time Bühler may (a) suspend in whole or in part its performance, (b) choose at its own discretion to carry out or employ a third party to carry out Customer's obligations, or otherwise take such measures as under the circumstances are appropriate to avoid or alleviate the effect of Customer's delay, and/or (c) inform Customer in writing of such failure and ask it to fulfil its obligation within a reasonable period of time but in no case longer than 14 (fourteen) calendar days.
 - ii Customer shall reimburse Bühler for any damages, costs and/or expenses, which are reasonably incurred by Bühler as a result of Customer's failure to duly fulfil its contractual obligations.
- 10. Exclusion and Limitation of Bühler's Liability**
- i Claims for compensation for damages, irrespective of the underlying legal basis and/or ground, out of or in connection with the Agreement, are exhaustively covered by these GTC. Any claim not originating from the scope of the Agreement is expressly precluded. Bühler shall solely be liable for direct damages to Customer's property intentionally or grossly negligently caused during the preparation or performance of the Services.
 - ii Irrespective of this, Bühler shall only be liable for typical and foreseeable damage. Liability for damages out of or in relation to reduction of price, termination of or withdrawal from the Agreement, loss of production, loss of use, loss of orders, recall costs, loss of profit and other indirect, direct or consequential damage are expressly excluded.
 - iii Furthermore liability out of or in relation to the Agreement is in any case whatsoever limited to the total Fee paid by the Customer under the Agreement.
 - iv Subject to clause 13, liability is also excluded for compensation claims from third parties against the Customer for infringements of intellectual property rights.
 - v This exclusion of Bühler's liability does not apply to unlawful intent, gross negligence, negligent harm to life, body, health, or as far as it is otherwise contrary to compulsory law.
 - vi The limitation or exclusion of liability extends to Bühler as well as any person employed or appointed by Bühler to perform any contractual obligation under the Agreement.
- 11. Complaints**
- i Any complaints concerning the Services or Delegate's behaviour shall be immediately notified to Bühler in writing, stating the reason for such complaint.
- 12. Compliance**
- i In exercising its rights and performing its obligations under this GTC Customer, its subsidiaries and any of its respective businesses or employees shall comply with all applicable laws, regulations and orders, including but not limited those related to anti-corruption and export controls (understood as and including but not limited to the transfer of controlled products, technologies or services for the development, production, use or stockpiling of nuclear, chemical, or biological weapons or missiles) imposed by the United States of America (U.S.), European Union (EU), United Kingdom (UK), Switzerland and/or any other countries.
- 13. Data Protection**
- i The Customer acknowledges and agrees that Bühler may have access to personal data (i.e. information relating to identified or identifiable natural person for example names, functions or contact details) of the Customer's employees, representatives, consultants, agents, contractors and other personnel. Such personal data may be processed by or on behalf of Bühler in accordance with Bühler's privacy policy, available at www.buhlergroup.com/privacy in order to enter into and perform any rights and/or obligation under these GTC as well as related purposes, including but not limited to order and payment processing, tolls and import/export management, Customer relationship management, business accounting and general administrative purposes. The parties agree that they act as independent controllers (as defined under applicable law) in relation to personal data processed in accordance with this clause. The Customer further undertakes to inform its personnel of whom Bühler is entitled to process personal data and obtain valid consent (if applicable) thereby especially pointing out Bühler's right to transfer personal data to third parties or third parties abroad while ensuring a comparable data protection level as described in Bühler's privacy policy, in accordance with applicable law.

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14. Termination by Bühler

- i If unforeseen events considerably change the economic effect or the content of the Services, or if the performance subsequently becomes impossible, the Agreement shall be adopted accordingly. If such adaption is economically not justifiable, Bühler shall be entitled to terminate the Agreement or the parts affected thereby. Bühler shall immediately after recognition of the consequences of the event, inform the Customer. In case of termination Bühler shall be entitled to payment of those parts of the Services which have already been performed. Claims for damages from the Customer because of such termination are excluded.

15. Severability

- i If any provision of the Agreement shall be found to be invalid or unenforceable, the invalidity or unenforceability of that provision shall not affect the other provisions of the Agreement. The parties shall substitute any invalid or unenforceable provision with provision that achieves to the greatest extent possible the same legal and commercial objectives.

16. Assignment

- i Neither party may assign or otherwise transfer any right or obligation under the Agreement without prior written consent of the other party.
- ii Bühler may, however, transfer and/or delegate the exercise of its rights and/or the performance of its obligations under the Agreement to another affiliated company that it deems fit for the intended purpose.
- iii All terms, provisions and conditions of the Agreement are binding upon and inure to the benefit of the parties and their respective successors.

17. Jurisdiction/applicable law

- i The place of jurisdiction shall be exclusively CH-9240 Uzwil. The applicable law shall be exclusively substantive Swiss law to the exclusion of the United Nations Convention on Contracts for the International Sale of Goods (1980).

Bühler AG
CH-9240 Uzwil,
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1. Conclusion of Agreement and Scope.

- i. These present DSTC apply to the Agreement made between the Customer and Bühler for providing the Digital Solution indicated and further specified in the Service Description as included in the Agreement and/or additional contractual documents forming an integral part of the Agreement, irrespective of whether an order is placed by an online or an offline transaction. If the parties expressly agree in writing, the Agreement might provide for rules that deviate from and/or prevail over the present DSTC.
- ii. Subject to the terms of the present DSTC and the terms of the Agreement, including but not limited to Customer's obligations, Bühler grants the Customer a non-exclusive, non-transferable and non-sub licensable right to access and use the Digital Solution during the term of the Agreement via interfaces and system elements as specified in the Service Description for its internal business purposes.
- iii. Any additional services provided by Bühler on request of the Customer, such as, for instance, implementation or installation of the Digital Solution or support, and/or the rent or purchase of equipment is subject to a separate Agreement to be concluded between the parties in accordance with the respective Bühler general terms and conditions as applicable and will be charged additionally.

2. Operation of Digital Solution.

- i. Bühler shall operate the Digital Solutions during the term of the Agreement by using adequately skilled personnel.
- ii. Subject to Customer's due fulfilment of all of its obligations, the availability, if any, of the Digital Solution shall be defined and set forth in the respective Service Description.
- iii. Bühler shall provide and Customer shall use the Digital Solution by using generally accepted, appropriate and applied industry standard information security technologies and techniques. Bühler operates an information security management system (ISMS) and adheres to high business information security standards with the objective to preserve confidentiality, integrity and availability of information processed by Bühler. More details are available under http://www.buhlergroup.com/Information_Security_global_20200301_en.
- iv. The Customer acknowledges and agrees to be solely responsible to continuously ensure and provide as described in the Service Description (a) a secure connection to the public internet with appropriate bandwidth; (b) any hardware, software, or other equipment necessary to establish and maintain the secure connection; and (c) any further requirements specified therein.
- v. Except if explicitly otherwise agreed in the Agreement, Bühler and/or Bühler's subcontractors, are permitted to remotely via including, but not limited to, internet or mobile phone connection access, monitor and/or adapt, in accordance with the terms set forth in the Agreement, the Customer Equipment, as necessary to support and/or provide the Digital Solution.
- vi. Bühler shall not have any obligation under the present Agreement if and to the extent the Customer fails to duly fulfil its co-operation obligations, especially but not limited to those described in clause 5 and 6 of the DSTC.

3. Bühler's Obligation.

- i. Bühler shall safeguard that the Digital Solution and/or Remote Support will be provided (i) with all reasonable skill and care and in accordance with applicable professional standards adhered to by companies offering services of the type offered by Bühler; (ii) in accordance with the applicable laws and, (iii) not to cause intentionally any fault or malfunction in the software or any interruption to the business processes of Customer.
- ii. Further Bühler shall obtain the necessary licenses or rights required for the due operation of the Digital Solution and/or Remote Support and that such licenses and rights will be valid for the duration of the Agreement.
- iii. If the Digital Solution and/or Remote Support do not meet the requirements described in this clause 3 i and ii. above, at the Customer's request, Bühler will use its reasonable best efforts to remedy any defect and to ensure that the Digital Solution and/or the Remote Support will substantially be in compliance with such requirements.
- iv. The remedies as described in clause 3 iii are Customer's sole remedies for the Digital Solution and/or the Remote Support and are exclusive and in lieu of all other remedies or warranties (which are expressly disclaimed), whether express or implied, statutory, common law or otherwise, including warranties for workmanship or performance, merchantability, fitness for a particular purpose.

4. Fee.

- i. In exchange for the performance of the Digital Solution in accordance with the terms of the Agreement, the Customer shall pay the Fee as specified in the Agreement.
- ii. The Fee shall be deemed to be net without any deduction, cash discount whatsoever. Any and all additional charges, such as but not limited to, insurance premiums, fees for export, transit, import and other permits, as well as for certifications, value-added taxes, sales taxes, as well as all other kinds of taxes, fees, levies, custom duties and the like which are levied against Bühler in connection with the Digital Solution are additionally paid by the Customer.
- iii. After the Minimum Term Bühler may, by giving 30 calendar days' prior written notice, at any time change the Fee and/or the basis on which it is calculated. The Fee however may not be adapted more than once every 12-months commencing after the Minimum Term or any anniversary of that date.
- iv. Within 20 calendar days after receipt of Bühler's notice to adapt the Fee, the Customer may give written notice to Bühler to terminate the Agreement as of the date the adaption would enter into force.

5. Terms of Payment.

- i. The Customer shall make payments in accordance with the terms set forth in the Agreement within 30 (thirty) days of the invoice date.
- ii. Customer's payment obligation is fulfilled after the total Fee has been credited in full to Bühler's bank account indicated by Bühler.
- iii. Customer shall not be entitled to retain or set-off any payments against counter-claims.
- iv. Without prejudice to any other contractual or statutory right of Bühler, if, for any reason whatsoever, a payment to be effected by the Customer is overdue, or if Bühler has valid reasons to expect that Customer will not make payment on time, Bühler may suspend the further performance of the Digital Solution and/or Remote Support until Bühler has received satisfactory security within a reasonable period of time. If no agreement is found and/or no securities are provided within such reasonable period of time, Bühler is entitled to terminate the Agreement and reserve any further rights that Bühler may have.
- v. In case of late payment, Bühler is entitled to charge interest for delay of 5 % over the then relevant 12-months-LIBOR (or any corresponding successor exchange rate), per annum, for the agreed contractual currency with effect from the agreed due date for payment until receipt of the outstanding payment plus the agreed late payment interest. In case the relevant LIBOR is negative, a base floor rate of 0.0% shall apply. The foregoing shall be without prejudice to any other right of Bühler under the Agreement. Payment of interest for delay does not release the Customer from its contractual obligations.

6. Customer's Obligations.

- i. The Customer shall, at its expense, promptly provide all resources and documents and take all measures reasonably required for the operation of the Digital Solution and/or Remote Support. Customer further explicitly undertakes to solely use the Digital Solution and/or Remote Support in accordance with the terms and conditions of the Agreement and for the purpose it was designed for as stated in the Service Description.
- ii. Without limiting the generality of the foregoing, the Customer shall (i) co-operate with Bühler in all matters relating to the Agreement, (ii) provide Bühler with such information and materials as Bühler may reasonably require in order to perform the Agreement, and ensure that such information is complete and accurate in all material respects; (iii) obtain and maintain all necessary licenses, permissions and consents which may be required; (iv) comply with all applicable laws, (v) keep all Bühler Materials at the Customer's premises in safe custody at its own risk, (vi) maintain the Bühler Materials in good condition until returned to Bühler, (vii) not dispose of or use the Bühler Materials other than in accordance with Bühler's written instructions, (viii) ensure that the Customer Equipment fulfils the requirements set forth in the Agreement, (ix) immediately inform Bühler of any exceptional circumstances or special requirements influencing the Agreement, and (x) not change the set-up of the Bühler Materials without Bühler's prior written consent.
- iii. The Customer acknowledges that Bühler's performance might be delayed, negatively impacted or completely interrupted if Customer fails to duly or timely fulfil any of its contractual obligations.
- iv. Further specific co-operation obligations of the Customer, as far as necessary for the provision and/or operation of the Digital Solution, may be set forth in the Agreement.

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7. Changes to the Digital Solution.

- i. Bühler may from time to time develop Updates. Bühler will use commercially reasonable efforts to: (a) rollout and install appropriate Updates that require a temporary expected downtime of the Digital Solution, and (b) provide notice to Customer, when reasonable and possible. If an Update requires more than an expected temporary downtime the parties shall mutually agree on the date and time of such Update. Whereby Bühler shall ensure that any Update shall not have material adverse effect to the Customer Equipment.
- ii. The Customer acknowledges and agrees to timely install any Update if so required in the individual situation. Further the Customer acknowledges and agrees that if it prevents installation of an Update (i) the Customer may be deprived of relevant functionality of the Digital Solution and (ii) Bühler has no obligation and liability whatsoever in respect of the Digital Solution, and (iii) Bühler may give notice of early termination in accordance with clause 18 below.
- iii. At any time Bühler may otherwise reasonably change the Digital Solution, and implement such change, as far as such change does not materially adversely affect the quality and/or security of the Digital Solution. By giving 30 calendar days' prior written notice Bühler may, further, implement changes materially adversely affecting the quality and/or security of the Digital Solution if (a) such change is required by mandatory applicable law, (b) a subcontractor implemented material changes or (c) the relevant functionality of the Digital Solution is definitely discontinued. Within 30 calendar days after the date of the notification of the material adverse change Customer may terminate the Agreement by giving written notice.

8. Third Party Products.

- i. The Customer acknowledges that Bühler's performance may rely on and/or interoperate with Third Party Products. Customer further acknowledges and agrees that Third Party Products (i) are beyond Bühler's reasonable control; (ii) are subject to the terms and conditions of the respective license; and (iii) may impact the use and reliability of the Digital Solution.
- ii. Bühler undertakes to apply reasonable care in selecting the Third Party Products and their relevant suppliers. The Customer, however, acknowledges and agrees that Bühler has no obligation, provides no warranty and assumes no liability, in respect of these Third Party Products. Even though Bühler applied due care in choosing any Third Party Products, Bühler does not, and does not have to, ensure that these Third Party Products will operate in a reliable manner and will not impact the way the Digital Solution is being operated. The Customer acknowledges and agrees that if the Third Party Products affect the Digital Solution the Customer shall have, to the extent permitted by applicable law, no right to raise any claim against Bühler. On Customer's written request, Bühler will provide a list of the Third Party Products and their suppliers.

9. Customer's Default.

- i. Without limiting or affecting any other right or remedy available, if performance of any of Bühler's obligations under the Agreement is prevented or delayed by an act of the Customer, or an omission or failure by the Customer to perform any contractual obligation, as set forth, but not limited to, in clause 6 Customer's Obligations, Bühler shall (i) have the right to suspend and/or delay the performance of any of its contractual obligation, especially but not limited to the provision and/or operation of the Digital Solution and/or Remote Support until the Customer remedies the Customer Default; and (ii) have the right to carry out Customer's obligations, or otherwise take appropriate measures to avoid or alleviate the effects of the Customer Default.
- ii. The Customer undertakes to remedy any Customer Default without delay. After remediation of Customer Default Bühler shall, if reasonable, resume its contractual obligation within due course.
- iii. On written demand, Customer will duly reimburse Bühler for any costs and/or losses directly or indirectly sustained by Bühler due to the Customer Default as far as Customer has not remedied such Customer Default within reasonable time. For the avoidance of doubt Bühler is not liable for any direct or indirect costs, expenses and/or losses sustained or incurred due to a suspension or delay based on a Customer Default.

10. Limitation of Liability.

- i. Bühler's liability for damages of any kind (such as, but not limited to, damages to property, damages caused by personal injury, as well as any

other direct or indirect or consequential damages) caused to the Customer by Bühler and/or Bühler's directors, officers, employees and other auxiliary persons, sub-contractors shall be excluded to the fullest extent permitted by applicable law. The remedies expressly set out in the present DSTC are the sole remedies of the Customer in case of a breach of the Agreement by Bühler, and any further liability or obligation of Bühler and any further rights or claims of the Customer, in particular claims not expressly mentioned for damages, reduction of Fee, termination of or withdrawal from the Agreement are expressly excluded.

- ii. The foregoing limitation shall apply regardless of the legal basis of the claim, such as, but not limited to, non-performance, non-proper performance, breach of warranty for defects or in title, default or tort. If, irrespective of the reason, the foregoing limitation of liability does not apply, Bühler's maximum liability out of or in connection with the Agreement is limited to the Fee paid under the Agreement.
- iii. The Customer shall immediately notify Bühler of any complaint it may have in writing, specifying in detail the reasons for its complaint. Only reproducible failures are considered a default of Bühler.

11. Data.

- i. By means of the Digital Solutions, Bühler may receive Third Party Data and/or Customer Data. Third Party Data and Customer Data together "Raw Data". Bühler will assemble, store, analyze, diagnose, combine and otherwise operate such Raw Data in order to create Data.
- ii. The Customer shall ensure, and herewith confirms, the legality, reliability, integrity, and quality of the Customer Data as well as the legality of the use of the Customer Data as contemplated herein. The Customer herewith grants and assigns to Bühler a perpetual, worldwide, royalty-free license and right to use the Customer Data (including, for the avoidance of doubt, Data that is derived from Customer Data) for the provision of the Digital Solution to the Customer as well as the provision of similar digital solutions to other customers and, more generally, for use within any present and/or future digital solution Bühler provides and/or may provide to the Customer and other customers. This right includes, in particular, the sharing of Customer Data and Data with, and its transfer to, Bühler's subcontractors, and/or third parties for providing the Digital Solution to the Customer as well as similar digital solutions to other customer, developing existing or new technologies, products, services, software and/or Digital Solutions as well as anonymized benchmarking. For the avoidance of doubt Bühler will keep the Customer Data confidential in accordance with clause 15 of the present DSTC and not share non-anonymized Customer Data with other customers. All rights in technology, products, software and/or Digital Solution developed by Bühler out of or in connection with the Customer Data and Data exclusively belong to Bühler and Bühler is entitled to use them for free and any purpose Bühler deems fit.
- iii. As far as Third Party Data forms part of, or has been used to create, the Data which Bühler will make available to the Customer, the Customer acknowledges and agrees that legality, reliability, integrity, accuracy and quality of such Third Party Data is the sole responsibility of the third party from which such Third Party Data was received and that to the extent permitted by applicable law Bühler has no obligations and assumes no warranty and no liability, in respect of such Third Party Data.
- iv. Bühler provides all Data "as is". It is Customer's sole responsibility to ensure that its use of the Data is fit for its intended purpose. Bühler has no duty to take any action as to how the Customer may interpret and/or use the Data or Customer's actions based on the receipt of the Data. The Customer further acknowledges and agrees that to the extent permitted by applicable law Bühler makes no and shall make no warranty (express, implied or statutory), including any warranty with respect to accuracy, completeness or fitness for any specific purpose in respect to the Data.

12. Pilot Digital Solutions.

- i. If the Digital Solution is indicated as a pilot, trial and/or beta version or is offered free of charge it qualifies as a Pilot.
- ii. Pilots are subject to additional limitations of the present clause 12, which prevail over the deviating terms of the present DSTC. Pilots are to the extent permitted by applicable law provided "as is" without any warranty and excluding all liability to the fullest extent permitted under applicable law.
- iii. Without limiting the generality of the foregoing Customer especially acknowledges and agrees that Pilots may not have been fully tested or verified and that therefore they may become unavailable, be changed or discontinued, their performance might be negatively affected and/or might not meet industry standards. Therefore, Pilots might negatively affect

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Customer's internal processes and business operations or impair the functionalities of Customer's Equipment.

- iv. Customer may only use Pilots for its internal use for the purpose of reviewing, evaluating and testing Pilots. The Customer acknowledges and agrees that the use of Pilots is at its sole risk.

13. Intellectual Property Rights.

- i. All intellectual property rights in, arising out of or in connection with the Digital Solution and the Data (other than intellectual property rights in any materials and/or Customer Data provided by the Customer) shall be owned by Bühler or its suppliers, as the case may be.
- ii. By entering into the Agreement, the Customer agrees that Bühler has the right to use all such existing and future intellectual property rights that may inure to the benefit of the Customer (or its employees or other auxiliary persons of the Customer) by operation of applicable law to Bühler, and the Customer undertakes to do all acts and sign all documents that may be required to perfect such use. The Customer waives any moral rights in the Digital Solution and/or Data to which it is now or may at any future time be entitled under any applicable law. The Customer shall ensure and guarantee that all of its employees and other auxiliary persons that may have been involved in the creation of such intellectual property rights have validly assigned the rights they may have to the Customer, and have waived any moral rights they may have, so that the Customer can validly and fully provide the right to use such intellectual property rights to Bühler.
- iii. Notwithstanding the foregoing, Customer shall not, directly or indirectly, reverse engineer or aid or assist in the reverse engineering of all or any part of Digital Solution except and only to the extent that such activity is mandatorily permitted by applicable law. Reverse engineering includes, without limitation, decompiling, disassembly, sniffing, peeling semiconductor components, or otherwise deriving the source code.
- iv. The Customer grants Bühler a non-exclusive, royalty-free, non-transferable license to copy and modify Data provided by the Customer to Bühler for the term of the Agreement for the purpose of providing the Digital Solution and Remote Support to the Customer.

14. Force Majeure.

- i. Either party shall be entitled to suspend performance of its contractual obligations if and to the extent its performance or performance by a subcontractor is directly or indirectly prevented by reasons of Force Majeure.
- ii. Neither party shall be deemed to be in default of its contractual obligations and no liquidated damages or indemnity shall be claimed whilst performance thereof is prevented by Force Majeure and the agreed timetable shall be extended accordingly by a period equal to that during which the Force Majeure contingencies and their effects have occurred. Each party shall fulfil its obligations insofar as they have become due before the occurrence of Force Majeure.
- iii. In case of Force Majeure lasting for more than 6 (six) consecutive months, each party is entitled to terminate the Agreement by written notice to the other party, whereby the parties shall mutually agree on the allocation of the accrued costs.

15. Confidentiality.

- i. The parties undertake to keep Confidential Information confidential and to not disclose Confidential Information to any third party other than to their officers, employees or advisors who have a need to know for the performance of the Agreement and are bound by no less stringent confidentiality undertakings.
- ii. The confidentiality obligation shall not apply to information which is (i) trivial or obvious; (ii) in the public domain through no fault of the receiving party; (iii) at the time of disclosure already known to the receiving party other than as a result of a breach of any obligation of confidentiality hereunder; or (iv) required to be disclosed by law based on order of the competent authorities. It is understood, however, that if the receiving party receives an order to disclose the Confidential Information it shall exercise in good faith all applicable legal remedies to prevent, or limit to the largest possible extent, the disclosure of the other party's Confidential Information and, to the extent permitted by applicable law, immediately inform the disclosing party of any such disclosure order.
- iii. The confidentiality obligations shall survive the termination of the Agreement for as long as the disclosing party has a reasonable interest in the confidentiality of the Confidential Information.

16. Data Protection.

- i. The Customer and Bühler acknowledge that the processing of personal data is not the primary purpose of the Agreement. However, if and to the extent processing of personal data as defined under applicable data protection laws (i.e., as a rule, information relating to an identified or identifiable natural person) is required for and/or in connection with the performance of the Agreement, Bühler and the Customer shall comply with the applicable data protection laws.
- ii. The Customer, in particular, acknowledges and agrees that Bühler may have access to Customer Personal Data. Bühler will, and the Customer confirms Bühler may, process (or have, on its behalf, processed) Customer Personal Data in accordance with Bühler's privacy policy, available at www.buhlergroup.com/privacy in order to enter into and perform the rights and obligations under the Agreement. The purposes for which Bühler processes Customer Personal Data include, but are not limited to, order and payment processing, tolls and import/export management, customer relationship management, business accounting and general administrative purposes.
- iii. The Customer further undertakes to inform its personnel and other above mentioned persons of whom Bühler may, in accordance with the previous paragraph, process Customer Personal Data and obtain valid consent (if required under applicable data protection law) thereby especially pointing out Bühler's right to transfer Customer Personal Data to third parties or third parties abroad while ensuring a comparable data protection level as described in Bühler's privacy policy.
- iv. The parties agree that insofar as Bühler processes Customer Personal Data of the Customer for its own business purposes pursuant to the foregoing paragraphs, Bühler acts as independent controller (as defined under applicable data protection law).

17. Compliance.

- i. In exercising its rights and performing its obligations under this Agreement Customer, its subsidiaries and any of its respective businesses or employees shall comply with all applicable laws, regulations and orders, including but not limited to those related to anti-corruption and export controls (understood as and including but not limited to the transfer of controlled products, technologies or services for the development, production, use or stockpiling of nuclear, chemical, or biological weapons or missiles) imposed by the United States of America (U.S.), European Union (EU), United Kingdom (UK), Switzerland and/or any other countries.

18. Term and Termination.

- i. The Agreement enters into effect upon confirmation by Bühler and is valid at least for the Minimum Term, if any, unless early terminated pursuant with the present DSTC.
- ii. Unless otherwise provided in the Agreement, either party may terminate the Agreement by giving one month prior written notice per end of the Minimum Term or any anniversary thereof. Customer may further terminate the Agreement if Customer objects to a change of the DSTC as set forth in clause 19 or to a material adverse change of the Digital Solution as set forth in clause 8 or to a change of the Fee as set forth in clause 5.
- iii. If a party commits any material breach, the non-breaching party may (i) serve written notice on the breaching party describing the breach and requiring the breaching party to remedy the breach within 30 working days from receipt of the written notice, and (ii) if the breach is not remedied within the 30 working days grace period terminate the Agreement by submitting a written notice.
- iv. The Agreement can be terminated by either party with written notice at any time if the other party ceases or threatens to cease to carry on business or becomes unable to pay its debts or if bankruptcy or composition or similar proceedings are opened against that party.
- v. Upon termination of the Agreement for whatever reason (a) Bühler is entitled to receive (at least) payment for the Digital Solutions provided up to the date of termination; (b) Customer shall cease using the Digital Solution; (c) Customer is entitled to obtain subject to the specification set forth in the Service Description a copy of Customer Data against payment of a fee compensating Bühler's time and effort. Termination is without prejudice to any other right or remedy of either party under the Agreement and shall not affect the parties' accrued rights.

19. Miscellaneous.

- i. Nothing in the Agreement shall be read and/or understood to create a partnership or joint venture between the parties.

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- ii. The parties may not assign or otherwise transfer any of their rights or obligations under the Agreement without prior written consent of the other party. Bühler may, however, transfer its rights and obligations hereunder and/or delegate the exercise of its rights and/or the performance of its obligations under the Agreement to another Bühler affiliate.
 - iii. The Agreement shall be binding upon signature and inure to the benefit of the parties and their authorized successors.
 - iv. The Agreement supersedes any and all prior agreements, either oral or written, between the parties hereto with respect to the subject matter of the Agreement and replaces and supersedes any general terms and conditions of either party not expressly included in by reference in the Agreement. No representations, inducements, promises, or agreements, orally or otherwise, have been made, that are not embodied in the Agreement.
 - v. If any provision of the Agreement shall be found to be invalid or unenforceable, the invalidity or unenforceability of that provision shall not affect the other provisions of the Agreement. The parties will substitute the invalid or unenforceable provision with a provision that achieves to the greatest extent possible the same legal and commercial objectives.
 - vi. Bühler reserves the right to change the present DSTC at any time, with prior notice. The most recent version of the DSTC will govern the Digital Solution and Remote Support and will always be available on the respective website the Digital Solution is operated. If the Customer does not contradict in writing within 30 days from Bühler posting the new version of the DSTC on the respective website and continues to use the Digital Solution, the Customer shall be deemed to have agreed to be bound by the revised DSTC. The last update is indicated at the end of the present DSTC.
- 20. Governing law and arbitration/jurisdiction.**
- i. The Agreement and the present DSTC are exclusively governed by the substantive laws at Bühler's place of jurisdiction. Any dispute arising out of or in connection with the Agreement and/or the present DSTC shall be finally settled under the Rules of Arbitration of the International Chamber of Commerce by one arbitrator appointed in accordance with the said rules. The language of arbitration shall be English. The place of arbitration shall be Zurich, Switzerland if Bühler is located in Europe, Africa, Middle East or South America. If Bühler is located in Asia or Oceania the place of arbitration shall be Singapore and if Bühler is located in North America, the place of arbitration shall be New York, USA.

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Country specific terms

As far as the applicable law as defined in clause 20 of the present DSTC in accordance with Bühler's location is one of the countries mentioned just below, the hereinafter paragraphs shall apply and replace the corresponding paragraph of the DSTC above or if there is no corresponding clause in the DSTC shall apply additionally.

Indonesia

Please refer to the bilingual version.

Glossary

Capitalized terms used in this DSTC shall have the meaning as set forth below:

Agreement	the understanding between the Customer and Bühler whereby the present DSTC form an integral part.
Bühler	Party to the Agreement providing the Digital Solution, whereby all Bühler affiliates are included.
Bühler Materials	materials, equipment, documents and other property of Bühler.
Confidential Information	information disclosed to or obtained by a party under the Agreement relating to the other party's intellectual property, know-how, business, customers, as well as other information that by its nature or content reasonably appear to be confidential.
Customer	Party to the Agreement obtaining the Digital Solution as defined in the Agreement.
Customer Data	data received by Bühler from or through Customer Equipment.
Customer Default	an act and/or an omission or failure by the Customer to perform any contractual obligation, as set forth, but not limited to, in clause 6.
Customer Equipment	Customer's machine, device and/or equipment dedicated for the use of the Digital Solution and/or Remote Support.
Customer Personal Data	data of the Customer's officers, employees, representatives, consultants, contractors and other natural persons working for or on behalf of the Customer and by which the natural person can be identified or is identifiable.
Data	Raw Data that has been assembled, stored, analyzed, diagnosed, combined and otherwise operated by Bühler in order to create output data.
Digital Solution	the service and/or solution as set forth in the Service Description and provided over an online connection.
DSTC	the present Digital Solution Terms and Conditions.
Fee	the remuneration in exchange for the performance of the Digital Solution as described in detail in the Service Description and payable by the Customer in accordance with the terms of the Agreement
Force Majeure	events beyond either party's (or its subcontractor's) reasonable control, which are unforeseeable at the time of conclusion of the Agreement and which render the further performance impossible or commercially unreasonable difficult, such as strike, government action, riot, armed conflict, accident, epidemic, extremes of weather or event of nature, unavailability of raw materials or of normal means of transport and cyber-attacks.
Minimum Term	minimum duration as described in the Agreement.
Pilot	Digital Solution indicated as pilot, trial and/or beta version or offered free of charge.
Raw Data	Customer Data and Third Party Data combined.
Remote Support	remote access, monitoring and/or adaption of the Customer Equipment, or as otherwise necessary in order to support and/or provide the Digital Solution.
Service Description	documentation forming an integral part of the Agreement and describing the Digital Solution, collected Raw Data as well as Customer's obligation in detail.
Third Party Data	data received by Bühler from or through equipment used by other Bühler customers.
Third Party Products	products, software (such as proprietary, freeware or open source software) and/or services procured, licensed and/or used by Bühler as well as third party software embedded in Customer Equipment and/or Bühler Materials.
Updates	patches, enhancements and/or modifications to improve the security as well as the performance of the Digital Solution, Remote Support and/or the underlying systems and components.

Bühler Insights Connectivity - IT Requirements

Released on April 12, 2021

The Bühler Insights gateway fetches data from machines in the local factory network and sends them to Bühler Insights. There are specific requirements for both the local network and the outgoing internet connection. **Network and firewall rules** need to be configured according to this document. The gateway requires an **active internet connection** for two communication channels, Data and Remote Access.

WAN: Below mentioned ports must be open in internet firewall:

Port	Remarks
443	Protocol: HTTPS – TCP; Data Channel, Remote Access Channel Must allow WebSockets over HTTPS
53	Protocol: DNS – TCP/UDP; Domain Name Service If no local/internal DNS is used. Important is that the Gateway can reach a DNS via this port
123	Protocol: NTP – UDP; NTP Clock Synchronization If no local/internal NTP is used. Important is that the Gateway can reach a NTP via this port

WAN: Below mentioned Fully Qualified Domain Names (FQDN) must be open in user's internet firewall:

FQDNs to be Whitelisted	Remarks
buhler-prod-telemetry.azure-devices.net global.azure-devices-provisioning.net buhlerprodwestorage.blob.core.windows.net buhlerprodcr.azurecr.io buhlerprodcr.westeurope.data.azurecr.io pki.buhlergroup.io fleet.buhlergroup.io monitoring.buhlergroup.io definitionupdates.microsoft.com	Data Channel Data is sent over a secure connection to Bühler Insights on the Azure cloud. Gateway to Internet must allow Web Sockets over HTTPS. Additionally, it needs to access a set of Bühler and Azure FQDNs.
ubiquityas1.asem.it ubiquityas2.asem.it ubiquityrs1.asem.it ubiquityrs2.asem.it ubiquityrs3.asem.it ubiquityrs4.asem.it ubiquityrs5.asem.it ubiquityrs6.asem.it ubiquityrs7.asem.it	Remote Access Channel To facilitate remote connections for gateway maintenance purpose, the gateway must be able to connect to these addresses. <u>Note:</u> Remote access to gateway does not allow to access OT network
SG-SGP-ANX-C002.teamviewer.com DE-FRA-ANX-C010.teamviewer.com US-DAL-ANX-C001.teamviewer.com	Remote Access Channel Soon, the Ubiquity Remote Access solution will be replaced by TeamViewer Tensor to facilitate remote connections for gateway maintenance purpose, the Gateway must be able to connect to these addresses. <u>Note:</u> Remote access to gateway does not allow to access OT network
Internal DNS server can be used. For global DNS , whitelist 1.1.1.1, 4.4.4.4 and 8.8.8.8	Domain Name Service If no local/internal DNS is used. Important is that the Gateway can reach global DNS
Internal NTP server can be used. For global NTP , whitelist *.pool.ntp.org or time.windows.com	NTP Clock Synchronization If no local/internal NTP is used. Important is that the Gateway can reach global NTP

 **SSL inspection is not supported**

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LAN: The Bühler Insights Gateway reads out values from devices on the network using different protocols. If a firewall is in place between the Bühler Insights Gateway and the factory network, the firewall must be configured to allow the protocols which are used. The **IPv4 network** is supported.

Protocol	Used for (examples)	Port
MQTT	BCube, Autonomous Refining, Molding Lines	TCP ports 1883 and 8883 (TLS)
AMQP	WinCos / Mercury, MYRG	TCP ports 5672 and 5671 (TLS)
MEAG	YMS, Pellet Mill	TCP port 2101
OPC-UA	Plant PLCs, Cookie Lines, GD Machines	According to specific installation